

The Mathematics of Market Impact and Order Execution

Impact Kernels, Feasible Policies, and Execution Certificates

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Abstract

An execution policy must complete a parent mandate while choosing quantities, order instructions and destinations under uncertain liquidity. This monograph connects that operational problem to impact models and conditional bounds on execution cost. A controlled, tagged order-book model preserves the parent ledger, live-order reservations and the trader’s information. Feasible signature policies admit exact quadratic reductions in an exogenous setting; partially observed endogenous books require separate planning and observation-reduction certificates.

The structural analysis covers impact kernels, manipulation, nonlinear size laws, cross-impact, information value and venue allocation. The control analysis separates physical feasibility, approximation error, model error and numerical optimization error. A baseline-relative extension answers a further deployment question: how much of a proposed schedule change can be accepted without increasing cost under any law in a declared uncertainty family? For diagonal impact uncertainty and ellipsoidal linear charges, the worst-case comparison along a feasible schedule segment is an explicit quadratic. Its minimizer gives both an acceptance fraction and a guaranteed saving. A capacity condition preserves completion at each update.

A joint fill-completion calculation shows how dependence alone can reverse a routing decision while preserving both marginal laws. A discrete implementation layer adds reservation-preserving order updates, robust completion masks and exact integer allocation. Finite-sample row coverage and a policy-transfer bound separate calibration uncertainty from the loss of the emitted implementation.

Reproducible synthetic experiments, including exact rational cases, test the identities and certificates. The results concern supplied models and explicit uncertainty bounds. They do not establish market calibration, counterfactual identification or live-market outperformance.

Keywords: optimal execution; market impact; propagator models; manipulation; path signatures; information bounds; Nash equilibrium; NYSE; Nasdaq; stochastic control.

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1 Introduction

Market impact and execution quality are related but distinct objects. A trade can consume available liquidity, reveal information, and change later order flow. Its execution cost also depends on spread, fees, priority, latency, and the completion mandate. A mathematical execution model must identify which of these mechanisms it represents and which it leaves to an additional observation or control model. [Bouchaud \(2009\)](#) reviews the meaning, measurement and mechanisms of price impact; [Webster \(2023\)](#) develops a broad treatment of its modeling and quantitative-trading applications.

1.1 The claim of this monograph, and how to read it

The organising claim is a discipline rather than a formula. A structural statement about execution cost — a kernel, an exponent, a convexity property — is not usable until it is paired with a bound on what is lost when the statement is only approximately true, stated in the same units as the cost being minimised. Sections [12–16](#) supply the structural side: what an execution cost functional can look like, and which size laws and cross-impact matrices are excluded outright. Sections [17](#) and [29](#) supply the certificate side: policy constructions whose approximation gap, model error and optimization tolerance are separately accounted and jointly bounded. For finite observations and binary hidden liquidity, Section [29.7](#) makes the global planning gap computable by lower and upper bounds. Section [29.8](#) then compares the finite controller with policies using richer observations, under verified conditional action-law envelopes. It distinguishes the cost of discarded information from belief rounding and gives a constructive endpoint calculation in the binary case. Its action-comparison refinement can certify that discarded information does not change the optimal instruction, even when a cost interval remains. Parts [V–VI](#) carry both into the instruction sets, reservations and auction mechanisms that an order actually meets.

A reader seeking the main execution contribution can begin with Sections [29.4](#), [29.7](#) and [29.8](#): the primitive-model certificate, Theorem [29.18](#) and Theorem [29.22](#) connect a controlled book to a policy and its information benchmark. Sections [12](#), [13](#) and [17](#) give the structural foundations and the exogenous quadratic specialization. The Taylor and area identities explain model restrictions; the contribution comparison in Section [29.9](#) identifies which control constructions use established order-book, signature and dynamic-programming foundations. A reader who wants only the execution layer can begin at Section [22](#), treating the impact theory as a specification of the cost functional. The numerical sections are checks on the stated models, not evidence about markets; Section [40](#) states what evidence would be required and Section [40.4](#) isolates two conditional restrictions and the data needed to test them. Three of the results below — the finite-grid form of Theorem [13.1](#), the ray form of Corollary [12.8](#), and Theorem [27.1](#) — are accompanied by Lean 4 formalizations; Appendix [E](#) states exactly what that does and does not cover.

Section [37](#) distinguishes six practical model gaps from mechanisms already represented, and gives an exact dependence stress in which unchanged marginal fills and completion prices reverse the routing decision.

1.2 Mechanisms and execution certificates

Two useful axes are response shape and persistence. A response can be linear or nonlinear in a specified input, such as signed trading rate. A displacement can persist, decay, or enter only the current execution premium. Trade size, metaorder size, participation and trading rate are different inputs; an exponent estimated for one is not automatically an exponent for another.

The monograph is primarily theoretical: it develops tractable impact benchmarks and connects them to execution on NYSE and Nasdaq. Its results are conditional on explicit trading and cost

conventions. Splitting invariance gives a Cauchy equation for permanent trade impact. Linear transient cost is an energy form; round-trip nonnegativity is a condition on its zero-mass subspace. Exponential and power-law kernels yield explicit risk-neutral schedules. A local temporary-cost model yields uniform execution, while nonlinear transient models require separate admissibility and existence checks. The latter retain exact size homogeneity when their response is a homogeneous power; nonlinearity alone does not make optimal shape size-dependent. For several assets, [Proposition 13.6](#) and [Corollary 13.9](#) separate transient cost into symmetric energy and weighted area and retain the terminal-state contribution of the exponential lift.

The execution layer then treats schedules, instructions and destinations as a joint decision under uncertain fills. It derives accounting identities, reservation constraints, allocation rules, information-value bounds, and separate NYSE and Nasdaq auction representations. Sections 38 and 39 complete the implementation interface: reservation-preserving order updates, robust completion masks, exact integer allocation, finite-sample row coverage, and a policy-transfer bound for the emitted implementation.

Scope, stated once. Every theorem below is conditional on its displayed hypotheses: a specified cost convention, a specified feasible set and, from Part IV onward, a supplied controlled law or uncertainty set. Every numerical calculation uses stipulated synthetic parameters; none is estimated from NYSE or Nasdaq records, and none is evidence of market calibration, counterfactual identification or live-market outperformance. Section 40 states what such evidence would require. Individual sections repeat a caveat only when it concerns a hypothesis specific to that section.

The central execution construction is stated in [Theorems 17.3, 17.7 and 17.8](#). Exponential kernel coordinates give an exact finite-degree objective; an adapted feature transformation makes completion a pathwise identity; and a common feasible coefficient set makes approximation errors comparable in execution-cost units. Relative to the signature approach of [Kalsi et al. \(2020\)](#), the emphasis is exact kernel augmentation and feasibility-preserving error accounting. The signature policy class remains finite and exogenous, so its certificate explicitly separates its approximation gap from errors within that class. The numerical section includes both a conventional liquidity-learning example and a direct test of this construction.

A second connection is developed in [Section 29](#). Starting from the order-flow/clearing decomposition of [Cont et al. \(2025\)](#), tagged queues produce own trades, cash charges and information in a single update. The resulting controlled generator supplies the joint kernel for execution; its stationary response gives a precise propagator reduction under additional assumptions. A coupling certificate combines primitive event-rate error with same-path impact-cost error, optimization tolerance and policy-class error. A 19-state queue example checks the construction and its decision consequences. This complements the exogenous signature results without assuming that an endogenous book retains their fixed quadratic moment objective.

For the finite controlled model, [Corollary 29.11](#) combines model accuracy, common feasibility, global optimization tolerance and a vanishing signature class gap into a conditional consistency theorem. Its proof retains initial observations and past actions, approximates finite scores on large compact sets, and enforces history-dependent feasibility through an exact mask. It supplies no general computational depth rate or convexity claim. [Proposition 29.12](#) gives an explicit approximation in the fully observed finite-state case, while [Proposition 29.13](#) charges reference interventions and any error in reducing that reference to a quadratic objective. [Section 29.9](#) compares the construction with order-book factorization, signature execution and the control-density results of [Bank et al. \(2025\)](#).

[Section 29.7](#) makes the approximation and optimization gap computable for a finite-observation

model with binary hidden liquidity. A concavity lower relaxation and physical hidden-state evaluation enclose the unrestricted partially observed optimum and a feasible controller’s cost. The controller uses a recursively rounded belief and never observes the hidden state. [Theorem 29.18](#), [Proposition 29.19](#), and [Corollary 29.20](#) give a stopping certificate, an explicit mesh bound, a signature-table implementation, and transfer to a perturbed model. These use classical POMDP planning foundations with explicit execution feasibility and error accounting. The experiment in [Section 34.1](#) includes twelve decisions, seven instructions, action-dependent liquidity transitions and private fill information; its global certificate is obtained by finite calculation, without an assumed optimization oracle.

1.3 From a parent mandate to executable decisions

A decision to acquire a position is incomplete until a sequence of executable instructions has been specified. A requirement to buy 200,000 shares before the close does not say when to trade, whether to cross the spread, how long to wait in a queue, how much to display, or where to route. Classical optimal execution makes the schedule explicit ([Bertsimas and Lo, 1998](#); [Almgren and Chriss, 2001](#)); order-placement models make uncertain liquidity explicit across venues, order types and dark pools ([Cont and Kukanov, 2017](#); [Cartea and Jaimungal, 2015](#); [Kratz and Schöneborn, 2015](#); [Huang et al., 2015](#)). In practice these ingredients are often fitted and optimized separately: a scheduler assumes an average cost, a placement rule maximizes one child’s fill probability, a router chases a rebate, and a hidden-stock estimate is treated as observable supply. Parts V–VI give them one interface: a state containing remaining quantity, deadline, live commitments, public observations, private reports and a belief about unobserved liquidity; an action specifying quantity, venue, instruction, price and lifetime; and one joint transition law used both to optimize and to evaluate. The marginal continuation value of an unexecuted share connects scheduling urgency to willingness to pay at a venue.

Part I

Impact: mechanisms and evidence

2 Five mechanism families

A separable propagator is a useful model class, not a universal representation of market microstructure. [Table 1](#) distinguishes exact examples from reductions that require additional assumptions or approximation.

2.1 Informational: conditional prices

In the one-period Gaussian Kyle equilibrium, the price is linear in aggregate order flow, with coefficient $\lambda = \sigma_v / (2\sigma_u)$ under the model’s normalizations ([Kyle, 1985](#)). Gaussian signal structure and strategic optimization matter. A martingale need not be linear in its observations, and linearity of conditional expectation as an operator on random variables does not imply linearity of its value as a function of observed order flow. Sequential-trade and continuous-time informational models have their own filtering and equilibrium assumptions ([Glosten and Milgrom, 1985](#); [Back, 1992](#)).

A constant-kernel linear propagator is therefore a useful informational benchmark, not a consequence of every martingale-pricing model. Microstructure invariance adds hypotheses about the

scaling of trading activity and costs across assets (Kyle and Obizhaeva, 2016); it does not identify a universal propagator pair.

2.2 Mechanical: book depletion and resilience

A block-shaped book with linear resilience gives the exponential linear-impact model of Obizhaeva and Wang (2013). A nonlinear book shape instead requires a state for consumed volume or price displacement. If $\Phi(x) = \int_0^x \varphi(u) du$, a trade of size q changes volume depletion from E to $E + q$, and its price displacement is $\Phi^{-1}(E + q) - \Phi^{-1}(E)$. This is state dependent. It is generally not obtained by convolving $\Phi^{-1}(v)$ with an exponential kernel.

Alfonsi et al. (2010) and Predoiu et al. (2011) study shaped-book models with explicit resilience and admissibility conditions. A result about nonlinear rate response inside a separable propagator must not be applied to all of these models merely because their book shape is nonlinear.

2.3 Statistical: propagators and correlated flow

Empirical propagators relate prices to a history of signed trades. The response $\mathcal{R}(\ell) = \mathbb{E}[(p_{t+\ell} - p_t)\varepsilon_t]$ mixes the direct kernel with trade-sign correlations. In the idealized stationary linear setting with sign correlation proportional to $\ell^{-\alpha}$, a propagator exponent $\beta = (1 - \alpha)/2$ can produce diffusive scaling (Bouchaud et al., 2004; Lillo and Farmer, 2004). Diffusive variance growth alone is weaker than the full conditional martingale property. A measured response curve must not be read directly as an isolated-trade decay kernel.

2.4 Latent liquidity: an implicit price response

Latent-order-book models distinguish resting displayed orders from intentions that become active as prices change. A locally linear latent density motivates square-root displacement in an appropriate regime (Tóth et al., 2011). The nonlinear latent-book model of Donier et al. (2015) contains the price trajectory inside its response equation; it is not, in general, the separable specification $f(v) = \sqrt{|v|} \text{sgn}(v)$ with a fixed kernel. Its linear-response limit and its large-rate regime must be distinguished. Consequently, its no-manipulation result cannot be inferred from a scalar exponent inequality for a different propagator model.

2.5 Self-exciting activity

Hawkes models describe event clustering and interactions between prices and trades (Bacry and Muzy, 2014). Nearly unstable scaling limits depend on the assumptions on kernel tails (Jaisson and Rosenbaum, 2015). Under a specified Hawkes-based construction, Jusselin and Rosenbaum (2020) connect impact scaling and rough volatility. These are model-specific limit results, not consequences of a branching ratio near one alone. Activity memory, trade-sign memory, and volatility roughness must retain their distinct definitions.

3 Empirical restrictions and their limits

Concavity depends on the conditioning. Many metaorder studies report concave impact, often summarized by a square-root relationship over a range of sizes (Tóth et al., 2011; Bershova and Rakhlin, 2013). Almgren et al. (2005) estimate impact using size and trading-rate variables; Zarinelli et al. (2015) examine departures from a single power law. Peak midpoint response, average

Family	Primitive	Relationship to a separable propagator
Informational	Conditional value given observations	Linear Gaussian equilibria supply examples; martingale pricing alone does not imply linear response
Mechanical	Book shape and resilience state	Block-shaped linear book gives an exponential kernel; nonlinear shapes generally need a state-dependent model
Statistical	Signed-event history and response	A propagator can be fitted conditional on flow dependence and clock conventions
Latent liquidity	Evolving unobserved order density	Nonlinear price equations can have propagator approximations in specified regimes
Self-exciting	Interacting event intensities	Scaling limits require explicit tail, stability and price assumptions

Table 1: Mechanisms and the scope of their propagator representations.

execution premium and total dollar cost are different targets. A square-root peak-response estimate does not by itself identify the exponent of an instantaneous rate function f .

Decay is horizon and model dependent. [Brokmann et al. \(2015\)](#) find slowly decaying impact after controlling for correlated orders and predictive information, with decay towards zero or a small residual. [Bucci et al. \(2018\)](#) find substantial same-day reversion and further longer-horizon decay in a different sample. These results do not establish a universal permanent fraction or a universal two-thirds asymptote. The clock, conditioning variables, order overlap and observation horizon belong in every decay estimate.

Efficiency constrains joint dynamics. Persistent order signs and comparatively weak return predictability constrain the joint price/flow model ([Bouchaud et al., 2004, 2018](#)). They do not uniquely select a kernel without assumptions on innovations, liquidity response and other order-book events.

An admissibility inequality is not an empirical equality. The condition $\beta + \delta \geq 1$ in Theorem 8.3 is a necessary restriction for a specified nonlinear rate-propagator model with unrestricted round trips. It is not sufficient in general. Combining exponents estimated for different observables or clocks does not show that markets satisfy equality. Nor does an inequality alone prove statistical weak identification. These questions require a joint likelihood or other identifying design.

Part II

Impact: theory and closed forms

4 Setup

4.1 Schedules and admissible domains

Fix $T > 0$ and signed quantity X . In the linear theory, a schedule is a finite signed Borel measure μ on $[0, T]$ with mass X . Write $\mathcal{M}_X$ for this affine set, restricted to finite absolute energy whenever a singular kernel is used. The zero-mass elements of $\mathcal{M}_0$ are round trips. For a buy-only problem use $\mathcal{M}_X^+ = \{\mu \geq 0 : \mu([0, T]) = X\}$, $X > 0$; this is a different constraint set. If $\mu(dt) = v_t dt$, remaining inventory is $x_t = X - \int_0^t v_s ds$.

A nonlinear function of a trading rate is defined on rates, not automatically on measures with atoms. In the nonlinear sections use measurable rates for which all displayed integrals are absolutely finite. Bounds on rates and restrictions on selling must be specified separately. The impact benchmarks below use deterministic schedules; later sections introduce adapted execution controls.

Table 2: Notation across the impact and execution layers. Local uses are identified explicitly.

Symbol	Meaning and scope
$X, Q; x_t, q_k$	Total signed impact quantity or parent size; remaining quantity
$G, G_\infty; \gamma_p$	Impact kernel and its long-run component; permanent coefficient in the quadratic benchmark
β, δ	Power-kernel decay exponent and nonlinear rate-response exponent
η, λ, σ	Temporary-cost coefficient, inventory-risk weight and price volatility
$\kappa; \rho, \zeta$	Local benchmark parameter; discrete persistence and impact-state loading
$\mu; b$	Trading measure in impact theory; posterior over latent liquidity in control
c in the potential condition	Constant marginal energy on the optimal support
c_v in routing	Capacity of venue v in the current allocation problem
C in the belief bound	Range of conditional expected total policy cost
c in the simulation bound	Upper bound on one stage's shifted nonnegative loss
$\ \nu\ _{TV}$	Total-variation (mass) norm of a signed measure (Sections 14, 20)
$TV(P, Q)$	Distance between probability laws, $\sup_A P(A) - Q(A) $ (Sections 18, 29, 39)
μ in routing	Common marginal cost of a venue allocation (Theorem 27.1)
$\mu_s^a; \mu$ in §40.3	One-clock primitive mark law (Section 29); logging policy (Section 40.3)

4.2 The propagator model

Definition 4.1 (Separable rate propagator). Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be odd, continuous and nondecreasing, with $f(0) = 0$, and let $G \geq 0$ be a nonincreasing locally integrable kernel. For an admissible rate,

$$S_t = S_0 + \int_0^t G(t-s)f(v_s) ds + Z_t, \quad (4.1)$$

where Z is an integrable zero-mean martingale with $Z_0 = 0$.

The limit $G_\infty = \lim_{t \rightarrow \infty} G(t)$ is the permanent component and $G_{\text{tr}} = G - G_\infty$ the transient component. A purely temporary execution premium is specified separately as $\eta f(v_t)$. Writing a Dirac kernel for that local term is only shorthand: its normalization is fixed by the execution cost convention, not by a one-sided convolution at an endpoint.

4.3 Cost and block conventions

With sufficient integrability to exchange expectation and integration,

$$\mathcal{C}[v] = \int_0^T \int_0^t G(t-s) f(v_s) v_t \, ds \, dt. \quad (4.2)$$

For $f(v) = v$, symmetrization gives

$$\mathcal{C}[\mu] = \frac{1}{2} \mathcal{E}_G[\mu], \quad \mathcal{E}_G[\mu] = \iint G(|t-s|) \mu(ds) \mu(dt). \quad (4.3)$$

For bounded kernels this extends to atoms by charging a block a the self-impact cost $G(0)a^2/2$, corresponding to average execution through a linear book. A strictly pre-impact block quote would charge zero self-impact; a post-impact quote would charge $G(0)a^2$. These are different economic conventions and can change manipulation properties. They are not interchangeable. For $G(t) = t^{-\beta}$, $\beta > 0$, a nonzero atom has infinite self-energy and is excluded. Adapted stochastic strategies additionally require conditions that make the martingale exposure have zero expectation; this does not follow from writing down an arbitrary random measure.

5 Permanent impact and splitting

Definition 5.1 (Splitting invariance). A trade-size impact function h is splitting invariant if

$$h(q_1) + \cdots + h(q_n) = h(q_1 + \cdots + q_n) \quad \text{for every finite signed trade sequence.} \quad (5.1)$$

This is an assumption about additive permanent displacement, not a consequence of martingale pricing or a definition of every informational model.

Theorem 5.2 (Linearity and a specified manipulation model). *If h is measurable and splitting invariant, then $h(q) = \kappa q$; if also nondecreasing, $\kappa \geq 0$. Separately, suppose h is continuous, odd and nonlinear, each trade adds $h(q)$ permanently to the prevailing price, and a trade q executes at the prevailing price plus $h(q)/2$. With no spread or other charge, some finite round trip then has negative cost.*

Proof. Splitting invariance for two trades is Cauchy's equation; its measurable real solutions are linear. Monotonicity fixes the sign of the coefficient. For the second assertion, buying $X > 0$ in n equal slices and selling X in one block gives, relative to the initial price,

$$\mathcal{C}_n = \frac{X}{2} \{h(X) - nh(X/n)\}. \quad (5.2)$$

Indeed, the purchases cost $Xnh(X/n)/2$ and the sale contributes $-X\{nh(X/n) - h(X)/2\}$. Buying in one block and selling in n slices gives $-\mathcal{C}_n$, using oddness. If the bracket vanished for every $X > 0$ and integer $n \geq 1$, integer subdivision and multiplication would give rational homogeneity on the positive half-line. Continuity would then give $h(X) = Xh(1)$ there, and oddness would extend it to all real X , a contradiction. Thus one of these two round trips has strictly negative cost. For strictly concave h with $h(0) = 0$, $h(X/n) > h(X)/n$ for $n > 1$, so the sliced accumulation is the profitable orientation. $\square$

The theorem connects to the permanent-impact restrictions of [Huberman and Stanzl \(2004\)](#), but its converse uses the particular additive trade-size and average-price conventions just stated. It is not a claim about all equilibrium or nonlinear book models.

Proposition 5.3 (Schedule-independent permanent cost). *For linear response and $G = G_\infty + G_{\text{tr}}$,*

$$\mathcal{C}[\mu] = \frac{1}{2}G_\infty X^2 + \frac{1}{2}\mathcal{E}_{G_{\text{tr}}}[\mu], \quad \mu \in \mathcal{M}_X.$$

Consequently a fixed linear permanent coefficient does not change the optimum at fixed X and fixed feasible schedules.

Proof. The constant-kernel term is $G_\infty \iint \mu(ds)\mu(dt) = G_\infty X^2$. □

Changing quantity, coefficient, information, benchmark or the feasible set lies outside this invariance. It is not an unconditional statement that every execution improvement must leave all measured permanent effects unchanged.

6 The linear theory as potential theory

A kernel is positive semidefinite (PSD) on the admissible measures when its energy is nonnegative for all of them. Here *strictly positive definite* means that a nonzero admissible measure has strictly positive energy.

Theorem 6.1 (The zero-mass energy condition). *No negative-cost round trip on $[0, T]$ is equivalent to $\mathcal{E}_G[\nu] \geq 0$ for every admissible $\nu \in \mathcal{M}_0$. This is conditional PSD on the zero-mass subspace. Full PSD is sufficient. Strict positivity on that subspace gives at most one minimizer at fixed mass; it does not by itself prove existence.*

A bounded continuous, nonnegative, nonincreasing, convex function G on $[0, \infty)$ gives a PSD kernel $G(|t - s|)$. In particular the exponential kernel with positive decay rate is strictly positive definite.

Proof. The first assertion follows from (4.3). For distinct fixed-mass schedules, the quadratic convexity gap is a positive multiple of the energy of their zero-mass difference, proving uniqueness when strict positivity holds. For the convex-kernel criterion, the distributional second derivative is a nonnegative measure ν and

$$G(r) = G_\infty + \int_0^\infty (u - r)_+ \nu(du).$$

The triangular kernel $(u - |t - s|)_+$ is the overlap of two interval indicators: its energy equals an integral of a squared convolution. It is therefore PSD, and so are nonnegative mixtures and the constant term. This is the relevant convex-kernel construction ([Alfonsi et al., 2012](#)). For an exponential, the Fourier density $2\rho/(\rho^2 + \xi^2)$ is strictly positive, giving strict positivity of energy for nonzero finite measures. □

Convex decreasing kernels are mixtures of triangular kernels under these conditions. Mixtures of exponentials form the more restrictive completely monotone class. Under appropriate Fourier integrability, a nonnegative spectral density characterizes full PSD on the whole line. It must not be substituted without qualification for the fixed-horizon, zero-mass condition. Oscillation alone also does not imply negative energy: a shifted nonnegative spectral measure can generate an oscillatory positive-definite function.

Example 6.2 (Full PSD is stronger on a fixed grid). On three time points take the symmetric Toeplitz matrix with first row $(1, 0.8, 0.21)$. For $z = (a, b, -a-b)$ its quadratic form is $1.58a^2 + 1.58ab + 0.4b^2 \geq 0$: the associated two-dimensional matrix has positive determinant 0.0079. But the full matrix has determinant $(1 - 0.21)(1 + 0.21 - 2(0.8)^2) < 0$. Thus this fixed-grid model has nonnegative round-trip costs despite failure of full PSD. This example does not assert conditional PSD for a continuum extension of the same sampled kernel.

Theorem 6.3 (Stationarity and the one-sided constraint). *Suppose a symmetric energy form is strictly positive on zero-mass variations. A fixed-mass schedule μ^* is the signed minimizer if and only if*

$$\iint G(|t-s|)\mu^*(ds)\nu(dt) = 0 \quad \text{for every admissible zero-mass variation } \nu. \quad (6.1)$$

For bounded continuous kernels this says that $U^{\mu^}(t) = \int G(|t-s|)\mu^*(ds)$ is constant on the entire interval. For a buy-only problem, the condition is instead $\int U^{\mu^*} d(\eta - \mu^*) \geq 0$ for every feasible positive measure η of mass X . In the bounded continuous case this is equivalent to $U^{\mu^*} \geq c$ everywhere and $U^{\mu^*} = c$ on its support.*

Whenever the relevant constant-potential condition holds,

$$\mathcal{C}[\mu^*] = \frac{1}{2}cX = \frac{1}{2}X^2W_G(T), \quad W_G(T) = c/X, \quad (6.2)$$

for $X \neq 0$. Singular kernels use finite-energy variations; pointwise statements require the corresponding potential-theoretic qualifications.

Proof. Expanding around μ^* gives $\mathcal{E}_G[\mu^* + \nu] - \mathcal{E}_G[\mu^*] = 2 \int U^{\mu^*} d\nu + \mathcal{E}_G[\nu]$. For signed variations, both signs of a small multiple of ν are feasible, so the linear term must vanish. Nonnegative energy proves sufficiency. For positive measures only feasible directions are tested, giving the variational inequality. Bounded continuity allows point-mass competitors and identifies the support condition. Integrating the constant potential against the optimal measure gives cX . $\square$

Corollary 6.4 (Size scaling at fixed inputs). *With a fixed linear kernel, fixed horizon and a feasible family closed under quantity rescaling, the optimal energy cost scales as X^2 whenever it is finite. This also holds for the infimum when it is not attained.*

Proof. The map $\mu \mapsto X\mu$ takes unit-mass schedules to mass- X schedules and multiplies energy by X^2 ; apply the inverse rescaling as well. $\square$

An empirical subquadratic size law rejects this complete conditional specification only if horizon, kernel and constraint changes have been controlled. Endogenous duration or liquidity can change an unconditional size-cost relationship even when conditional impact is linear.

7 The solved linear cells

The following benchmarks use distinct response and admissibility conventions. [Gatheral and Schied \(2013\)](#) reviews their regularity, manipulation and existence issues. Risk-averse liquidation with general utility has richer state dependence than the quadratic benchmark; see [Schied and Schöneborn \(2009\)](#).

7.1 Constant kernel

For $G \equiv G_\infty$, every fixed-mass schedule has cost $G_\infty X^2/2$. There is no unique schedule. Additional temporary cost, risk, constraints or signals determine timing.

7.2 Temporary cost and inventory risk

Let $\gamma_p \geq 0$ be the linear permanent-impact coefficient, $\eta > 0$ the temporary-cost coefficient, $\lambda \geq 0$ the risk weight, and $\sigma \geq 0$ the unaffected-price volatility, all constant. With zero predictive drift, minimize over absolutely continuous paths with square-integrable derivative and $x_0 = X, x_T = 0$:

$$J[x] = \frac{1}{2}\gamma_p X^2 + \int_0^T \{\eta \dot{x}_t^2 + \lambda \sigma^2 x_t^2\} dt.$$

Proposition 7.1 (Almgren–Chriss trajectory). *For $\kappa = \sqrt{\lambda\sigma^2/\eta} > 0$ the unique optimum is*

$$x_t^* = X \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \quad J[x^*] = \frac{1}{2}\gamma_p X^2 + \eta \kappa X^2 \coth(\kappa T).$$

At $\kappa = 0$ the solution is $X(1-t/T)$ with cost $\gamma_p X^2/2 + \eta X^2/T$.

Proof. The Euler–Lagrange equation is $\ddot{x} = \kappa^2 x$; the boundary data give the stated path. Strict convexity in $\dot{x}$ with fixed endpoints proves uniqueness. Since $(x\dot{x})' = \dot{x}^2 + \kappa^2 x^2$, the integral is $\eta[x\dot{x}]_0^T = \eta \kappa X^2 \coth(\kappa T)$. The zero-risk formula is the continuous limit. $\square$

Uniform execution in calendar time is TWAP; it is VWAP only for a constant volume curve or after a specified volume-clock transformation (Almgren and Chriss, 2001).

7.3 Exponential resilience

Theorem 7.2 (Exponential kernel). *For $G(t) = e^{-\rho t}$, $\rho > 0$, the unique fixed-mass minimizer is*

$$\mu^* = a(\delta_0 + \delta_T) + b \text{Leb}|_{[0,T]}, \quad a = \frac{X}{2 + \rho T}, \quad b = \rho a,$$

with

$$\mathcal{C}[\mu^*] = \frac{X^2}{2 + \rho T}, \quad W_G(T) = \frac{2}{2 + \rho T}. \quad (7.1)$$

For $X > 0$ it is also feasible and optimal in the buy-only problem.

Proof. For $0 \leq t \leq T$ the potential of the proposed measure is

$$a(e^{-\rho t} + e^{-\rho(T-t)}) + \frac{b}{\rho}(2 - e^{-\rho t} - e^{-\rho(T-t)}) = 2a.$$

Its mass is $2a + bT = X$. The signed stationarity condition and strict positive definiteness prove optimality and uniqueness; its cost is aX . $\square$

This is the linear resilient-book benchmark (Obizhaeva and Wang, 2013). As $\rho \rightarrow 0$ its cost tends to $X^2/2$, while its selected schedule tends to equal endpoint blocks, one among many optimal constant-kernel schedules. As $\rho \rightarrow \infty$ at fixed amplitude, the endpoint fraction vanishes and cost tends to zero; obtaining a nonzero temporary-cost limit would also require rescaling the kernel amplitude.

7.4 Power-law equilibrium

Let $G(t) = t^{-\beta}$, $0 < \beta < 1$, and exclude atoms of infinite energy.

Theorem 7.3 (Riesz kernel on an interval). *The unique finite-energy fixed-mass minimizer has density*

$$v^*(t) = \frac{X}{T^\beta B(a, a)} [t(T-t)]^{a-1}, \quad a = \frac{1+\beta}{2}, \quad 0 < t < T, \quad (7.2)$$

and cost

$$\mathcal{C}[\mu^*] = \frac{X^2}{2} \left(\frac{T}{2}\right)^{-\beta} \frac{\Gamma(1+\beta/2)\Gamma((1-\beta)/2)}{\sqrt{\pi}}. \quad (7.3)$$

Proof. Scale first to $T = X = 1$ and set $b = (1-\beta)/2$, so $a = 1-b$. The density $[s(1-s)]^{-b}/B(a, a)$ integrates to one. For $0 < x < 1$, the left part of its unnormalized potential satisfies

$$\int_0^x (x-s)^{-\beta} [s(1-s)]^{-b} ds = bB(a, 2b)B_x(b, b),$$

where $B_x(b, b) = \int_0^x t^{b-1}(1-t)^{b-1} dt$. To verify the identity, expand $(1-s)^{-b}$ by its binomial series, integrate each term with the beta integral, and use $(b)_j/(b+1)_j = b/(b+j)$; the nonnegative terms justify termwise integration. The right part is the same expression with x replaced by $1-x$. By symmetry, $B_x(b, b) + B_{1-x}(b, b) = B(b, b)$, so the potential is constant:

$$U(x) = \frac{bB(a, 2b)B(b, b)}{B(a, a)} = \frac{\Gamma(a)\Gamma(b)}{B(a, a)}.$$

The endpoint integrals have the same finite value. The duplication formula for Γ gives $U = 2^\beta \Gamma(1+\beta/2)\Gamma(b)/\sqrt{\pi}$. The representation $|r|^{-\beta} = \Gamma(\beta)^{-1} \int_0^\infty u^{\beta-1} e^{-u|r|} du$ and the positive exponential energies establish strict positivity for finite-absolute-energy signed measures. The stationarity condition proves optimality. Scaling time by T multiplies energy by $T^{-\beta}$ and scaling mass by X multiplies it by X^2 , proving both formulas. This is consistent with the classical potential-theoretic treatment (Landkof, 1972; Gatheral et al., 2012). $\square$

Corollary 7.4 (Limits of the power-law family). *Fix $X \neq 0$. As $\beta \uparrow 1$, the optimal densities converge to X/T , while the unscaled cost diverges like $X^2/[(1-\beta)T]$. As $\beta \downarrow 0$, they converge to $X/[\pi\sqrt{t(T-t)}]$, and cost tends to $X^2/2$.*

Proof. Apply the beta-density normalization and continuity of the beta and gamma functions; at the upper endpoint use $\Gamma(\varepsilon) \sim 1/\varepsilon$. Density convergence is also in L^1 after division by X , by nonnegativity, pointwise convergence and equality of total masses. $\square$

The arcsine limit has no atoms. It does not become an exponential-kernel block-constant-rate-block schedule. Both families can approach the degenerate constant kernel while selecting different optimizers. Only the normalized uniform profile is shared with the zero-risk temporary-cost problem.

8 Nonlinear impact

8.1 A local temporary-cost model

Proposition 8.1 (Uniform execution under convex local cost). *Let $\eta > 0$, $\delta > 0$, $T > 0$. Among rates of total quantity $X > 0$,*

$$\mathcal{C}[v] = \eta \int_0^T |v_t|^{1+\delta} dt \quad \text{has unique minimizer} \quad v_t = X/T$$

up to null sets, with cost $\eta X^{1+\delta} T^{-\delta}$.

Proof. Jensen's inequality for the strictly convex function $|u|^{1+\delta}$ gives $T^{-1} \int |v|^{1+\delta} \geq |X/T|^{1+\delta}$, with equality only for an almost-everywhere constant rate. $\square$

At $\delta = 1/2$ the average premium scales as $\sqrt{X/T}$, whereas total cost scales as $X^{3/2}/\sqrt{T}$. This is a cost-model result; it is not a proof of the empirical square-root law for peak metaorder price response. Rate caps, predictive drift or time-varying liquidity can change the optimum.

8.2 A direct short-horizon manipulation construction

Theorem 8.2 (Concave rate response and a bounded initial kernel). *Suppose f is odd and strictly concave on the positive half-line, $f(0) = 0$, and $G(t) \rightarrow G_0 \in (0, \infty)$ as $t \downarrow 0$. If arbitrarily short round trips at two fixed positive rates are admissible and there are no additional charges, the separable rate propagator admits a negative-cost round trip.*

Proof. Choose $0 < a < b$, so $f(a)/a > f(b)/b$. Buy ε shares at rate a for time ε/a , then sell them at rate b for time ε/b . Replacing G by G_0 in the three triangular/rectangular cost terms gives

$$\mathcal{C}_\varepsilon = \frac{G_0 \varepsilon^2}{2} \left(\frac{f(b)}{b} - \frac{f(a)}{a} \right) + o(\varepsilon^2) < 0$$

for small enough ε . Uniform convergence of G to G_0 over this shrinking interval bounds the replacement error by $o(\varepsilon^2)$. $\square$

No derivative of G is required. A zero kernel, a positive spread charge, minimum trade sizes, a sell prohibition or a nonlinear state-dependent book changes the assumptions. The result restricts this propagator class (Gatheral, 2010; Curato et al., 2017); it does not exclude every nonlinear resilient order book.

8.3 A necessary power–power restriction

Theorem 8.3 (A round-trip test for power kernels). *For $G(t) = t^{-\beta}$, $0 < \beta < 1$, and $f(v) = \text{sgn}(v)|v|^\delta$, $0 < \delta \leq 1$, nonnegative cost for all admissible signed round trips requires*

$$\beta + \delta \geq 1. \tag{8.1}$$

The inequality alone is not a sufficiency theorem.

Proof. Buy at rate one on $[0, 1]$ and sell at rate $1/\varepsilon$ on $[1, 1 + \varepsilon]$. With $D = (1 - \beta)(2 - \beta) > 0$, direct integration gives

$$\mathcal{C}_\varepsilon = \frac{1}{D} \left[1 + \varepsilon^{1-\beta-\delta} - \frac{(1 + \varepsilon)^{2-\beta} - 1 - \varepsilon^{2-\beta}}{\varepsilon} \right]. \tag{8.2}$$

If $\beta + \delta < 1$, this converges to $-1/(2 - \beta) < 0$. Rescaling both durations fits the same construction inside any positive horizon and multiplies cost by a positive factor. Thus a model allowing these strategies cannot be manipulation-free. This gives the necessary restriction associated with Gatheral (2010). $\square$

At $\beta + \delta = 1$, the displayed strategy instead has the positive limit $\beta/[(1 - \beta)(2 - \beta)]$. Equality in the exponent restriction does not make this round trip costless. Other strategies and restrictions still matter; numerical work documents problems not excluded by the simple bound (Curato et al., 2017). Within $0 < \delta \leq 1$, only $\delta = 1$ satisfies this necessary inequality for every $\beta \in (0, 1)$; this does not make a linear response compatible with arbitrary non-PSD kernels.

9 Nonlinear stationarity, scaling and concentration

For smooth strictly positive interior rates, differentiation of the cost gives

$$\int_0^t G(t-s)f(v_s)ds + f'(v_t) \int_t^T G(u-t)v_u du = c. \quad (9.1)$$

This requires differentiation under the integral and feasible two-sided local variations. At zero rates or binding bounds one needs the corresponding variational inequalities. Nonlinear response can destroy convexity, so the stationarity equation need not identify a global minimum. There is no general closed-form solution supplied here, and no theorem ruling out closed forms for all special nonlinear models.

Proposition 9.1 (Exact homogeneity and time dilation). *For power response and a fixed kernel, $\mathcal{C}_G[av] = a^{1+\delta}\mathcal{C}_G[v]$ for $a > 0$. If feasible schedules rescale with quantity, optimal normalized shape is therefore independent of X , up to possible nonuniqueness. For $X > 0$, $G(t) = t^{-\beta}$ and $v(t) = (X/T)u(t/T)$, $\int_0^1 u = 1$,*

$$\mathcal{C}_G[v] = X^{1+\delta}T^{1-\delta-\beta} \int_0^1 \int_0^t (t-s)^{-\beta} \text{sgn}(u_s)|u_s|^\delta u_t ds dt. \quad (9.2)$$

Proof. The two rate factors contribute $a^{1+\delta}$ under size scaling. For time dilation the rates contribute $(X/T)^{1+\delta}$, the two integration measures contribute T^2 , and the kernel contributes $T^{-\beta}$. Applying the rescaling and its inverse to feasible competitors proves the claim about minimizers or infima. $\square$

The decay exponent changes horizon scaling, not merely a prefactor. A rate cap fixed in shares per second, nonhomogeneous response, or an additional cost with another size exponent can break shape invariance.

Proposition 9.2 (Concentration under a bounded kernel). *Suppose $f(v) = \text{sgn}(v)|v|^\delta$, $0 < \delta < 1$ and $0 < G(T) \leq G(t) \leq G_0 < \infty$ on $[0, T]$. In the buy-only rate problem with fixed $X > 0$ and no rate bound, the infimum cost is zero and is not attained. For $v_\varepsilon = (X/\varepsilon)\mathbf{1}_{[0, \varepsilon]}$,*

$$0 \leq \mathcal{C}[v_\varepsilon] \leq \frac{1}{2}G_0X^{1+\delta}\varepsilon^{1-\delta} \longrightarrow 0.$$

Proof. The displayed bound integrates a constant rate over a triangle of area $\varepsilon^2/2$. Every nonzero nonnegative rate of positive integral has positive cost because its support has positive measure and $G(T) > 0$. Hence zero is the infimum but no admissible rate attains it. $\square$

A constant memory kernel is not a temporary-cost functional: it retains the second time integral. Applying local Jensen bounds to it would give an incorrect positive lower bound. This example establishes nonattainment for this particular unbounded-rate class, not for all nonlinear transient models.

10 Extensions

10.1 Causal cross-impact

Let $A(\tau)$ be a real matrix mapping earlier signed asset trades into later price changes. On a finite ordered time grid, define a symmetric block matrix K by $K_{ij} = A(t_i - t_j)$ for $i > j$, $K_{ji} = K_{ij}^\top$, and $K_{ii} = \text{Sym } A(0)$. With average-price diagonal charging,

$$\mathcal{C}(x) = \frac{1}{2}x^\top Kx = \sum_{i>j} x_i^\top A(t_i - t_j)x_j + \frac{1}{2} \sum_i x_i^\top \text{Sym } A(0)x_i.$$

Specification	Optimum or condition	Qualification
Constant linear kernel	Cost $G_\infty X^2/2$	Every fixed-mass schedule is optimal
Temporary quadratic cost and risk	Hyperbolic-sine inventory path	Constant coefficients, zero drift
Exponential linear kernel	Endpoint blocks and constant interior rate	Average-price block convention
Power-law linear kernel	Beta-type density	Finite energy; atoms excluded
Local power temporary cost	Uniform rate	Fixed horizon and constant liquidity
Concave response, bounded initial kernel	Negative round trip exists	Unrestricted signed small trades, no spread
Power response and power kernel	$\beta + \delta \geq 1$ necessary	Not a sufficiency certificate

Table 3: Benchmarks and the assumptions that make them valid.

Proposition 10.1 (Cross-impact admissibility on a grid). *Absence of negative-cost round trips is equivalent to nonnegativity of $x^\top Kx$ on the subspace $\sum_i x_i = 0$ asset by asset. Full PSD of K is sufficient. For an instantaneous cost $q^\top Aq/2$, only $\text{Sym } A$ contributes; this last statement does not extend to an arbitrary causal transient matrix kernel.*

Proof. The first statements follow from the cost and round-trip definitions. For the last, $q^\top (A - A^\top)q = 0$. In contrast, take the constant causal matrix $A = \begin{pmatrix} 0 & a \\ -a & 0 \end{pmatrix}$, $a > 0$, and trades $e_1, e_2, -e_1, -e_2$. They sum to zero, but their causal cross-cost is $-2a$. Antisymmetric cross-time effects therefore need not vanish. $\square$

The distinction agrees with the dynamic cross-impact restrictions studied by [Schneider and Lillo \(2016\)](#). A negative one-shot eigenvector is a negative-cost trade, not by itself a zero-inventory round trip. Matrix estimation must retain temporal causality and test the actual round-trip subspace.

10.2 Volterra kernels and existence

Nonconvolution kernels permit more general temporal liquidity patterns. Positive definiteness provides convexity and uniqueness when a minimizer exists; it does not guarantee attainment in an infinite-dimensional space. A useful sufficient construction on L^2 adds a temporary term $\eta \|v\|_2^2$, $\eta > 0$, to a bounded PSD quadratic form. The result is coercive and strictly convex, so a continuous linear term and a nonempty closed convex feasible set admit a unique minimizer. Singular kernels require their own domain and operator conditions.

[Abi Jaber and Neuman \(2025\)](#) treat general propagators with signals through an infinite-dimensional stochastic-control formulation; [Neuman and Voß \(2022\)](#) combine temporary and transient effects. These results use additional hypotheses and are not consequences of kernel positivity alone.

10.3 Dynamic portfolios and hedging

Classical result 10.2 (A linear-quadratic policy form). *In the stationary quadratic-cost, quadratic-risk and mean-reverting-predictor model of [Gârleanu and Pedersen \(2013\)](#), the optimal position moves toward an aim portfolio according to $x_t = x_{t-1} + M(\text{aim}_t - x_{t-1})$, with the adjustment matrix and aim weights determined by the model’s Riccati system.*

This is a literature result under its stationary model, not a new proof here. Quadratic temporary-impact hedging tracks an average of expected future target positions (Bank et al., 2017); it is not generally a proportional-cost no-trade-band model. Stochastic execution and order placement are developed in Cartea et al. (2015); Guéant (2016); Gatheral and Schied (2011); Lehalle and Mounjid (2017).

11 Identification

Permanent versus slow. Over a short observation window a slowly decaying component can approximate a permanent one arbitrarily closely. Without restrictions on later decay, the unobserved tail is unidentified. Under a fully specified parametric family, however, sufficiently precise data inside the window can distinguish nonconstant curves; short-horizon difficulty is not universal exact observational equivalence.

Response and flow dependence. A response function mixes direct impact with autocorrelated trading and common information. Estimation must model or control these sources, and distinguish event time from calendar time (Bouchaud et al., 2004, 2018).

Actions and selection. Order size and urgency are chosen using information about prices and liquidity. Observed size-cost regressions therefore need a counterfactual interpretation. Randomized variation, credible instruments, natural experiments, structural restrictions or sufficiently rich conditional identification assumptions can supply different routes. No one route is guaranteed valid, and instruments are not the only possible identifying design. Execution and forecasting comparisons must state their conditioning sets and remaining uncertainty.

Part III

Analytic cost functionals and the geometry of impact

The preceding part solved a list of cells: linear kernels with explicit schedules, nonlinear rate responses with partial obstructions, cross-impact on a grid. Each result carries its own hypotheses, and the hypotheses do not nest. This part asks what the cells have in common. We treat execution cost as a functional on trading schedules and ask which structures are forced by invariance rather than assumed by parametrisation.

The common structure is conditional. Analyticity yields a graded expansion on rate space, not a regular measure kernel without additional assumptions. Permanent cross-asset round trips admit a geometric representation; transient costs retain their symmetric-energy criterion. Kernel violations can be measured by a homogeneous defect. A power-law metaorder tail and an asymptotic fair-pricing condition link the large-size exponent and permanent fraction. Finally, a positive convexification gap can be realised by an ordinary two-rate schedule; only a failure of finite contact permits non-attainment. Each statement below keeps its domain and boundary conditions explicit.

Domain of this part. Assumption 12.1 is a genuine restriction, and it partitions the cells of Part II. It holds for the linear propagator models of Sections 6–7, for local quadratic temporary cost, and for

analytic transformations of a bounded energy such as the crossover of Proposition 12.10. It fails for the nonlinear rate propagators of Section 8 and Proposition 9.2: with $f(v) = \text{sgn}(v)|v|^\delta$, $0 < \delta < 1$, the cost has no bounded second derivative at zero, as Corollary 12.9 records. Within this part, only Section 12 uses Assumption 12.1. Sections 13–16 hold under their own hypotheses—a linear permanent or transient operator, a bounded kernel, a metaorder-size law, or a lower semicontinuous local cost—and therefore also apply to models that Section 12 excludes.

12 The graded hierarchy of cost functionals

12.1 Axioms

Fix $T > 0$ and work on the rate space $V = L^2(0, T)$, real valued and signed. A *cost functional* is a map $\mathcal{C} : V \rightarrow \mathbb{R}$ with $\mathcal{C}[0] = 0$, interpreted as the expected execution cost of the schedule v relative to the arrival price p_0 (the arrival benchmark, whose full shortfall accounting is Proposition 23.1), computed under a fixed market model with a zero-mean unpredictable component.

Assumption 12.1 (Regularity at zero). $\mathcal{C}$ is real analytic on a ball $\{\|v\|_2 < r\}$ about zero. Each derivative $D^n \mathcal{C}[0]$ is a bounded symmetric n -linear form on V , and its Taylor series converges absolutely and uniformly on every smaller ball.

Assumption 12.2 (Truncation consistency). There is a family $\mathcal{C}_t$, $t \in [0, T]$, with $\mathcal{C}_T = \mathcal{C}$ and $\mathcal{C}_t[v] = \mathcal{C}[v\mathbf{1}_{[0,t]}]$ for every admissible v : each partial-horizon functional is the restriction of the same functional, and so depends on v only through $v|_{[0,t]}$. Moreover $t \mapsto \mathcal{C}_t[v]$ is nondecreasing for v of one sign. Truncation consistency is stronger than causality. The family $\mathcal{C}_t[v] = (t/T) \int_0^t v^2$, with $\mathcal{C}[v] = \int_0^T v^2$, depends only on the past and agrees with $\mathcal{C}$ at $t = T$, but it is not truncation consistent: $\mathcal{C}[v\mathbf{1}_{[0,t]}] = \int_0^t v^2 \neq \mathcal{C}_t[v]$ whenever $t < T$ and v is not zero almost everywhere on $[0, t]$.

Assumption 12.3 (Stationarity). Extending schedules by zero to $\mathbb{R}$, $\mathcal{C}[\tau_h v] = \mathcal{C}[v]$ for every shift $\tau_h v(\cdot) = v(\cdot - h)$ with support in $[0, T]$.

Assumption 12.4 (Side symmetry). $\mathcal{C}[-v] = \mathcal{C}[v]$: buying X and selling X along mirrored schedules cost the same.

Assumption 12.1 is the substantive one and is not innocuous; the point of Theorem 12.5 and Corollary 12.9 is exactly to isolate what it costs. Assumption 12.4 is a modelling convention appropriate to a symmetric book and no short-sale asymmetry; it is stated so that its consequences can be separated from the rest.

Theorem 12.5 (Graded representation). *Under Assumption 12.1 there exist symmetric distributional kernels K_n on $(0, T)^n$, $n \geq 1$, such that for $\|v\|_2 < r$*

$$\mathcal{C}[v] = \sum_{n \geq 1} \frac{1}{n!} \int_{(0, T)^n} K_n(t_1, \dots, t_n) v(t_1) \cdots v(t_n) dt_1 \cdots dt_n, \quad (12.1)$$

the series converging absolutely. Under Assumption 12.3 each K_n is locally invariant under simultaneous translations that remain in $(0, T)^n$ and depends only on increments, in the distributional sense specified in Remark 12.6; K_1 is a constant. Under Assumption 12.4 $K_n = 0$ for every odd n . Under truncation consistency (Assumption 12.2), $\mathcal{C}_t$ has the same expansion with the domain of integration restricted to $(0, t)^n$.

For nonsmooth rates, the distributional integrals denote the continuous multilinear extension from smooth test schedules. They do not assign pointwise values to a distribution or define its pairing with arbitrary trading measures.

Proof. Taylor's theorem in Banach space gives $\mathcal{C}[v] = \sum_n D^n \mathcal{C}[0][v^{\otimes n}]/n!$ with convergence on the stated ball. Each $D^n \mathcal{C}[0]$ is a bounded symmetric n -linear form on $L^2(0, T)$, hence continuous on $\mathcal{D}((0, T))^n$ with the test-function topology; by the Schwartz kernel theorem it is represented by a unique distribution $K_n \in \mathcal{D}'((0, T)^n)$, symmetric because the form is. Polarization transfers stationarity to the multilinear form whenever the common shift keeps its test schedules inside the horizon. Thus the derivative of K_n along the common time coordinate vanishes locally; it is a distribution in the $n - 1$ increments. For $n = 1$ it is constant. Replacing v by $-v$ multiplies the n th term by $(-1)^n$, so [Assumption 12.4](#) forces the odd terms to vanish identically as forms, hence as kernels. Truncation consistency restricts the support statement to $(0, t)^n$ by the same argument applied to $\mathcal{C}_t$. $\square$

Remark 12.6 (The translation action is partial). The shifts in [Assumption 12.3](#) act only while supports remain inside the horizon. For $n \geq 2$, use $u = t_1$ and $r_j = t_j - t_1$, $j = 2, \dots, n$. The increment domain is

$$D_{n,T} = \left\{ r \in \mathbb{R}^{n-1} : \max(0, r_2, \dots, r_n) - \min(0, r_2, \dots, r_n) < T \right\}.$$

For each r the allowed u form a connected open interval. Local translation invariance gives $\partial_u K_n = 0$, so K_n is the pullback of a distribution $k_n \in \mathcal{D}'(D_{n,T})$: $K_n(t_1, \dots, t_n) = k_n(t_2 - t_1, \dots, t_n - t_1)$ in that sense. On a strictly ordered chamber one may equivalently use successive positive gaps. The full domain also retains diagonal-supported terms such as $\delta_{t=s}$; restricting to the strict chamber alone would lose them. No extension beyond the finite-horizon increment domain is asserted or needed in [Corollary 12.7](#).

12.2 The second layer and its domain

Side symmetry gives $K_1 = 0$. The quadratic layer is

$$\mathcal{C}_2[v] = \frac{1}{2} \langle K_2, v \otimes v \rangle. \quad (12.2)$$

Stationarity and symmetry give an even convolution distribution. When this distribution is represented by a function $G(|t - s|)$ for which the energy is finite, [\(12.2\)](#) is exactly $\mathcal{E}_G[v]/2$. The function representation is an additional assumption, not a consequence of analyticity. For example, $\mathcal{C}[v] = \|v\|_2^2$ has $K_2 = 2\delta_{t=s}$, a local temporary charge. Its approximation to a unit block by $h^{-1}\mathbf{1}_{[0,h]}$ costs h^{-1} and does not define a finite atomic-block cost.

Corollary 12.7 (Placement of the solved cells). *Suppose the second kernel admits the propagator representation above. The results of [Sections 6](#) and [7](#) apply to that layer under their own kernel and admissibility assumptions. On rate space, the second-order round-trip condition is $\mathcal{E}_G[\nu] \geq 0$ for $\nu \in V_0 := \{v \in L^2(0, T) : \int_0^T v = 0\}$. An extension to signed measures requires a separately defined finite-energy pairing. This condition alone does not constrain higher layers.*

Corollary 12.8 (The first possible correction is quartic). *Under [Assumptions 12.1](#), [12.3](#) and [12.4](#), the first permitted correction to the quadratic layer is $\mathcal{C}_4[v] = D^4 \mathcal{C}[0][v, v, v, v]/24$; it may vanish. If $\mathcal{C}[\epsilon v] \geq 0$ for every sufficiently small real ϵ and every fixed $\nu \in V_0$, then $D^2 \mathcal{C}[0][\nu, \nu] \geq 0$, and on its null set $D^4 \mathcal{C}[0][\nu, \nu, \nu, \nu] \geq 0$.*

Proof. Odd terms vanish. Divide the expansion on the ray $\epsilon\nu$ first by ϵ^2 and let $\epsilon \rightarrow 0$. If the resulting quadratic term is zero, divide instead by ϵ^4 . The two necessary conditions follow. They are pointwise conditions on directions, not a uniform sufficiency theorem over all shrinking perturbations. $\square$

Side symmetry excludes a cubic *cost* term. It does not fix the sign of the quartic layer away from quadratic null directions. A negative quartic coefficient can reduce curvature over a size range, and a quartic cost gives a cubic correction to marginal impact. Neither the first nonzero correction nor global convexity is fixed by symmetry alone.

12.3 What analyticity forbids

Corollary 12.9 (Pure subquadratic size laws at zero). *Suppose the cost of a schedule of total size X over a fixed horizon behaves as $\mathcal{C}[Xu] = c(u)X^{1+\delta}$ for a fixed unit-mass shape u with $c(u) > 0$ and $\delta \in (0, 1)$. Then $\mathcal{C}$ violates [Assumption 12.1](#): the second Gateaux derivative at 0 in the direction u is infinite, so no bounded second-order form exists and no expansion (12.1) is available at any order.*

Proof. $\lambda^{-2}\mathcal{C}[\lambda Xu] = c(u)X^{1+\delta}\lambda^{\delta-1} \rightarrow \infty$ as $\lambda \downarrow 0$ because $\delta < 1$. $\square$

Empirical concavity usually describes impact per share or peak impact as a function of metaorder size, not concavity of total dollar cost ([Tóth et al., 2011](#); [Zarinelli et al., 2015](#); [Bucci et al., 2018](#); [Bershova and Rakhlin, 2013](#)). A cost proportional to $X^{3/2}$ is convex but subquadratic. [Corollary 12.9](#) concerns that law holding all the way to zero for a fixed schedule shape: an even analytic functional is $O(X^2)$ there. It places no obstruction on a subquadratic large-size asymptotic separated from zero by a transition.

Proposition 12.10 (A non-degenerate analytic functional with both regimes). *Let G be a kernel whose energy $\mathcal{E}_G$ in (4.3) is a bounded nonnegative quadratic form on $L^2(0, T)$ and is not constant across all schedules of the same total mass. Fix $e_\star > 0$ and set*

$$\mathcal{C}[v] = \Phi(\mathcal{E}_G[v]), \quad \Phi(e) = e_\star[(1 + e/e_\star)^{3/4} - 1]. \quad (12.3)$$

Then $\mathcal{C}$ is side symmetric, stationary whenever the energy is, and real analytic at zero in the sense of [Assumption 12.1](#). For a fixed unit-mass shape u with $e_u := \mathcal{E}_G[u] > 0$,

$$\begin{aligned} \mathcal{C}[Xu] &= \frac{3}{4}e_u X^2 - \frac{3}{32}\frac{e_u^2}{e_\star}X^4 + O(X^6) & (X \rightarrow 0), \\ \mathcal{C}[Xu] &\sim e_\star^{1/4}e_u^{3/4}X^{3/2} & (X \rightarrow +\infty). \end{aligned} \quad (12.4)$$

For every common admissible fixed-mass class $\mathcal{V}_X$ on which the energy is finite and nonnegative,

$$\arg \min_{v \in \mathcal{V}_X} \mathcal{C}[v] = \arg \min_{v \in \mathcal{V}_X} \mathcal{E}_G[v]. \quad (12.5)$$

This equality of minimizer sets does not assert their nonemptiness. If $e_u = 0$, the cost is identically zero along that ray instead.

Proof. Choose $M > 0$ with $\mathcal{E}_G[v] \leq M \|v\|_2^2$. The scalar binomial series for Φ converges absolutely for $|e| < e_\star$, so its composition with the bounded quadratic form gives a Taylor series of continuous homogeneous polynomials, uniformly and absolutely on every smaller ball inside $\|v\|_2 < \sqrt{e_\star/M}$.

This proves the asserted analytic regularity. Evenness and stationarity are inherited from the energy. Substituting $\mathcal{E}_G[Xu] = X^2 e_u$ gives both expansions. Finally,

$$\Phi'(e) = \frac{3}{4}(1 + e/e_\star)^{-1/4} > 0 \quad (e \geq 0),$$

so Φ preserves all comparisons of energy values on the same admissible class. This proves (12.5) and preserves any schedule dependence of the energy. $\square$

Remark 12.11 (What the crossover changes). [Proposition 12.10](#) changes the *level* of impact cost as a function of size while preserving the minimizers of the underlying energy. For this transformed impact objective, the optimal shapes are therefore those of [Section 7](#) whenever their domain and existence hypotheses hold. A separately defined nonnegative extension to finite-energy measures permits the same argument for block schedules. In particular, the exponential optimizer in [Theorem 7.2](#) has endpoint atoms; it belongs to that measure domain, rather than to $L^2(0, T)$. The analytic statement above concerns the rate space and does not itself supply a measure extension.

An observed subquadratic cost curve is therefore not by itself evidence of a size-dependent optimal schedule. The mechanisms in item 2 of [Section 41](#)—rate caps, endogenous horizons, changing liquidity or nonhomogeneous response—are candidates for such a dependence; nonhomogeneity alone is not sufficient, as (12.3) shows.

For comparison, the minimal size-only example

$$\mathcal{C}[v] = F\left(\int_0^T v\right), \quad F(X) = (1 + (X/V_{\text{lat}})^2)^{3/4} - 1, \quad V_{\text{lat}} > 0, \quad (12.6)$$

is analytic at zero and has a three-halves large-size asymptotic, but is constant across schedules of equal size. It illustrates compatibility of the two regimes and supplies no schedule information.

Two pure powers cannot be joined smoothly at a positive crossover: matching $\kappa X^2/2$ with $cX^{3/2}$ in value gives $X_{\text{value}} = (2c/\kappa)^2$, while matching slopes gives $X_{\text{slope}} = (3c/(2\kappa))^2$. Their ratio is 16/9 for $c, \kappa > 0$. A transition function such as (12.3) specifies the intermediate-size behavior; fitting its form and scale is a separate empirical task.

13 Manipulation is a curvature

[Proposition 10.1](#) shows that antisymmetric cross-impact creates profitable round trips while anti-symmetric one-shot impact does not. We now give the reason, which is geometric and exact.

Let $q_t \in \mathbb{R}^d$ denote cumulative signed inventory in d assets along an absolutely continuous path $\gamma : [0, T] \rightarrow \mathbb{R}^d$, and let prices respond to inventory through a linear permanent operator $M \in \mathbb{R}^{d \times d}$: $p_t = p_0 + Mq_t$. Execution cost along γ is the line integral $\mathcal{C}(\gamma) = \int_\gamma p \cdot dq$. Write $M = S + A$ with $S = \frac{1}{2}(M + M^\top)$ and $A = \frac{1}{2}(M - M^\top)$. For a closed loop, define the signed area (the Lévy area of the path in the (i, j) plane)

$$\mathcal{A}_{ij}(\gamma) = \frac{1}{2} \oint_\gamma (q^i dq^j - q^j dq^i). \quad (13.1)$$

Theorem 13.1 (Loop cost is an area pairing). *For every closed inventory loop γ (a round trip, $q_T = q_0$),*

$$\mathcal{C}(\gamma) = \oint_\gamma p \cdot dq = -2 \sum_{i < j} A_{ij} \mathcal{A}_{ij}(\gamma). \quad (13.2)$$

Consequently: (i) the symmetric part S and the initial price p_0 contribute nothing to any round trip; (ii) no closed loop is profitable, for every loop, if and only if $A = 0$, i.e. M is self-adjoint; (iii) if

$A \neq 0$ the profit is unbounded above over loops of unbounded area, and over loops confined to the plane of a pair (i, j) the cost is $-2A_{ij}\mathcal{A}_{ij}$ exactly. As a check, $M = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and the counterclockwise unit square give $A_{12} = 1$, $\mathcal{A}_{12} = 1$ and cost -2 , which is what direct integration returns.

Proof. $\oint p_0 \cdot dq = 0$ and $\oint (Sq) \cdot dq = \oint d(\frac{1}{2}q^\top Sq) = 0$ because the integrand is exact. For the antisymmetric part, $(Aq) \cdot dq = \sum_{i,j} A_{ij}q^j dq^i$, so $\oint (Aq) \cdot dq = \sum_{i<j} A_{ij} \oint (q^j dq^i - q^i dq^j) = -2 \sum_{i<j} A_{ij}\mathcal{A}_{ij}$, using $A_{ji} = -A_{ij}$ and (13.1). Statement (ii) is immediate in one direction; conversely, if $A_{ij} \neq 0$ a small rectangular loop in the (i, j) plane has nonzero area and both orientations are admissible, so one of them has negative cost. Scaling a loop by λ scales its area by λ^2 , giving (iii). $\square$

Corollary 13.2 (Zero loop cost in one asset). *For $d = 1$ every loop has zero area, so $\mathcal{C}(\gamma) = 0$: with linear permanent impact, round trips in a single asset are costless and never profitable. This is the geometric content of Proposition 5.3 and complements the linearity forced by splitting invariance in Theorem 5.2: linearity removes the size channel, closedness of the loop removes the schedule channel, and in one dimension nothing remains. The statement concerns the permanent channel in the stated frictionless model only. Transient and temporary channels are governed by Proposition 13.6 and Corollary 13.8, and a scalar market with a non-admissible transient kernel still admits profitable round trips.*

Corollary 13.3 (Admissible cross-impact is self-adjoint). *The cone of permanent cross-impact matrices admitting no profitable round trip is exactly the set of symmetric matrices. After subtracting the arrival-price benchmark, a symmetric matrix charges $X^\top MX/2$ for net vector X . Requiring this to be nonnegative for every $X \in \mathbb{R}^d$ gives the symmetric positive semidefinite cone. If only componentwise nonnegative X are allowed, the additional condition is instead copositivity. Estimated cross-impact matrices that are asymmetric are therefore not merely inefficiently estimated: within this frictionless model the asymmetry is a measurable arbitrage, of size given by (13.2). The scope distinction of Schneider and Lillo (2016) should be preserved when reading this: model-level manipulability is not profitability once spreads, fees and the finite depth of the book are charged, and (13.2) bounds only the former.*

Remark 13.4 (The invariant is a level-two signature). The right side of (13.2) is a linear functional of the second level of the path signature of q over $[0, T]$: $\mathcal{A}_{ij}$ is the antisymmetric part of $\int \int_{s<t} dq_s^i dq_t^j$. The cost of a loop is thus determined by the depth-two truncated signature and nothing finer. Section 17 builds the objective on the same object; the constraint and the objective of the execution problem therefore live on the same graded algebra.

Definition 13.5 (Kernel-weighted area). Let $g \in L^1(0, T)$ be real valued and $v \in L^2((0, T); \mathbb{R}^d)$ a deterministic trading rate. For $i < j$ define

$$\mathcal{A}_{ij}^g[v] = \frac{1}{2} \int_{0 < s < t < T} g(t-s)(v_s^i v_t^j - v_s^j v_t^i) ds dt. \quad (13.3)$$

Write $E_g[a, b] = \int_{(0, T)^2} g(|t-s|) a_s b_t ds dt$. Young's inequality gives absolute convergence of these integrals. For $g \equiv 1$ and a closed inventory path $q_t = q_0 + \int_0^t v_s ds$, (13.3) equals $\mathcal{A}_{ij}(\gamma)$ in (13.1).

Proposition 13.6 (Transient cost is energy minus weighted area). *Suppose $p_t = p_0 + M \int_0^t g(t-s)$*

$s)v_s ds$, where $M = S + A$ as above. The impact cost relative to p_0 is

$$\begin{aligned}\mathcal{C}[v] &= \int_0^T v_t^\top (p_t - p_0) dt \\ &= \underbrace{\frac{1}{2} \sum_{i,j} S_{ij} E_g[v^i, v^j]}_{\mathcal{E}_{g,S}[v]} - 2 \underbrace{\sum_{i<j} A_{ij} \mathcal{A}_{ij}^g[v]}_{\mathcal{B}_{g,A}[v]}.\end{aligned}\tag{13.4}$$

Under orientation reversal $\bar{v}_t = -v_{T-t}$, the energy is unchanged and the weighted area changes sign. Consequently, on a deterministic rate class $\mathcal{V}$ closed under this reversal,

$$\mathcal{C}[v] \geq 0 \quad \text{for every } v \in \mathcal{V} \quad \Longleftrightarrow \quad |\mathcal{B}_{g,A}[v]| \leq \mathcal{E}_{g,S}[v] \quad \text{for every } v \in \mathcal{V}.\tag{13.5}$$

Taking $\mathcal{V}$ to be the zero-net-trade class gives a round-trip criterion. Reversal invariance is a separate assumption for a restricted class; it does not follow from stochastic adaptedness.

Proof. Expand the cost as $\int_{s<t} g(t-s) v_t^\top M v_s ds dt$. Since $S = S^\top$, exchanging s, t turns its triangular integral into one half of the full symmetric energy. The rate formulation has no atomic trades, and the diagonal has zero Lebesgue measure. The A part is

$$\sum_{i<j} A_{ij} \int_{s<t} g(t-s) (v_t^i v_s^j - v_t^j v_s^i) ds dt = -2 \sum_{i<j} A_{ij} \mathcal{A}_{ij}^g[v].$$

Under reversal, both rate factors change sign and the lag is preserved. The substitution $(s, t) \mapsto (T-t, T-s)$ leaves the symmetric term fixed and reverses the bracket in (13.3). Thus the two orientations cost $\mathcal{E}_{g,S}[v] \mp \mathcal{B}_{g,A}[v]$. Both costs are nonnegative exactly under (13.5). $\square$

Corollary 13.7 (Scalar and permanent limits). *For $d = 1$ and $M = m > 0$, the criterion is $E_g[v, v] \geq 0$ on the prescribed zero-integral rate space, the rate version of Theorem 6.1. For $g \equiv 1$, $\mathcal{E}_{1,S}[v] = X^\top S X / 2$, where $X = \int_0^T v_t dt$. On closed loops this is zero, and (13.4) becomes Theorem 13.1; if all rectangular loops in every asset pair are admissible, nonnegative cost forces $A = 0$. Under (13.5), zero symmetric energy forces only $\mathcal{B}_{g,A}[v] = 0$ on that class. Inferring $A = 0$ additionally requires weighted areas that detect every antisymmetric matrix.*

Proof. In one dimension there is no area term and the positive scalar $m/2$ does not change the energy sign. For $g = 1$, the double integral factors through the net vector; rectangular loops isolate the entries of A . The final assertion follows directly from (13.5). For example, if $g(r) = r$ and $S = 0$, direct symmetrization gives $\mathcal{C}[v] = (\int_0^T t v_t dt)^\top A X$, which vanishes on every round trip for any A . Thus nondegeneracy of the weighted areas cannot be omitted. $\square$

Corollary 13.8 (Short loops detect instantaneous asymmetry). *Suppose g extends continuously to 0 with $g(0) > 0$, and every arbitrarily short rectangular round trip is admissible. For the impact cost in (13.4), absence of profitable round trips forces $A = 0$.*

Proof. Choose a unit-horizon rectangular rate w with $-2 \sum_{i<j} A_{ij} \mathcal{A}_{ij}^1[w] < 0$ if $A \neq 0$. Compress it to $v_h(t) = h^{-1} w(t/h)$ on $[0, h]$, and set it to zero afterward. Changing variables shows

$$\mathcal{C}[v_h] \longrightarrow g(0) \int_{0<s<t<1} w_t^\top M w_s ds dt = -2g(0) \sum_{i<j} A_{ij} \mathcal{A}_{ij}^1[w] < 0.$$

Continuity of g justifies the limit; the symmetric permanent energy vanishes because $\int w = 0$. Hence sufficiently short loops are profitable. This conclusion concerns the stated impact-only cost; additional temporary charges change the comparison. $\square$

Corollary 13.9 (The exponential lift, including its endpoint). *For $g(r) = \sum_{m=1}^{m_0} w_m e^{-\rho_m r}$ with $w_m, \rho_m > 0$, let $I_t^m = \int_0^t e^{-\rho_m(t-s)} v_s ds$. Define the area of this possibly open impact path by $\mathcal{A}_{ij}(I^m) = \frac{1}{2} \int_0^T (I_t^{m,i} dI_t^{m,j} - I_t^{m,j} dI_t^{m,i})$. Then*

$$\begin{aligned} \mathcal{C}[v] = & \sum_m w_m \left[\frac{1}{2} (I_T^m)^\top S I_T^m + \rho_m \int_0^T (I_t^m)^\top S I_t^m dt \right] \\ & - 2 \sum_m w_m \sum_{i < j} A_{ij} \mathcal{A}_{ij}(I^m). \end{aligned} \quad (13.6)$$

In particular, I_T^m need not vanish even when the inventory round trip is closed. If $A = 0$ and $S \succeq 0$, the endpoint and dissipation terms are both nonnegative.

Proof. The state identity $dI_t^m = v_t dt - \rho_m I_t^m dt$ and $I_0^m = 0$ give $\int v_t^\top S I_t^m dt = \frac{1}{2} (I_T^m)^\top S I_T^m + \rho_m \int (I_t^m)^\top S I_t^m dt$. For the antisymmetric part, $(I_t^m)^\top A I_t^m = 0$, so $\int v_t^\top A I_t^m dt = \int (dI_t^m)^\top A I_t^m = -2 \sum_{i < j} A_{ij} \mathcal{A}_{ij}(I^m)$. Summing proves (13.6). These are the impact paths generated by v ; arbitrary loops in the enlarged state space have not been admitted. $\square$

Remark 13.10 (Lag-dependent matrix kernels). The decomposition does not require a common scalar kernel. Let $G \in L^1((0, T); \mathbb{R}^{d \times d})$ and write $S(r) = (G(r) + G(r)^\top)/2$, $A(r) = (G(r) - G(r)^\top)/2$. Set

$$\begin{aligned} \mathcal{E}_S[v] &= \frac{1}{2} \int_{(0, T)^2} v_t^\top S(|t - s|) v_s ds dt, \\ \mathcal{B}_A[v] &= \sum_{i < j} \int_{s < t} A_{ij}(t - s) (v_s^i v_t^j - v_s^j v_t^i) ds dt. \end{aligned}$$

The same symmetrization proves $\mathcal{C}[v] = \mathcal{E}_S[v] - \mathcal{B}_A[v]$. The lag is preserved by time reversal, so $\mathcal{E}_S$ is invariant and $\mathcal{B}_A$ changes sign, even when $A(r)$ varies with r . Thus (13.5) holds with these two forms. This is consistent with the standard matrix-kernel energy representation of [Alfonsi et al. \(2016\)](#); it does not assert that a matrix which is positive semidefinite at each lag defines a positive energy kernel. A kernel depending on both calendar times, rather than their difference, needs a separate check of reversal invariance.

14 The manipulation defect

[Theorem 6.1](#) is a dichotomy: a kernel is admissible or it is not. For model selection and for estimation error one wants a magnitude.

Definition 14.1 (Defect). For a bounded continuous kernel G and horizon T , define

$$\delta_T(G) = \sup\{-\mathcal{E}_G[\nu] : \nu \in \mathcal{M}_0, \|\nu\|_{TV} \leq 1\} \geq 0. \quad (14.1)$$

The zero measure makes the supremum nonnegative. A fixed-grid version uses only measures supported on that grid. Singular kernels require a specified finite-energy domain and are not included in the normed kernel space below.

Proposition 14.2 (Properties and stability of the defect). *On $C([0, T])$, the defect is convex, positively homogeneous and 1-Lipschitz in the uniform norm:*

$$|\delta_T(G) - \delta_T(H)| \leq \|G - H\|_\infty. \quad (14.2)$$

It vanishes exactly on the closed convex cone $\mathcal{G}_T$ of kernels with no negative-cost round trip under the energy convention. It is nondecreasing with the horizon for a fixed kernel. The supremum of achievable profit with gross traded volume at most L is $L^2\delta_T(G)/2$: execution cost is one half of the energy by (4.3). Moreover,

$$\delta_T(G) \leq \inf_{H \in \mathcal{G}_T} \|G - H\|_\infty. \quad (14.3)$$

The same statements hold on a fixed finite grid, with its own admissible cone. Grid monotonicity requires nested grids evaluated under the same kernel.

Proof. For fixed ν , the negative energy is linear in G . Taking its supremum proves convexity and positive homogeneity, while $|\mathcal{E}_G[\nu] - \mathcal{E}_H[\nu]| \leq \|G - H\|_\infty \|\nu\|_{TV}^2$ proves the Lipschitz bound. Homogeneity of energy in ν proves the zero-set and profit statements; allowing the zero trade is essential when the defect vanishes. Embedding zero-mass measures on a shorter interval into a longer one proves monotonicity. Applying the Lipschitz inequality to any H of zero defect gives (14.3). For continuous kernels, the total-variation ball with fixed mass is weak-* compact and the energy is continuous on it, so the supremum is attained. On a finite grid this is ordinary compactness. $\square$

Remark 14.3 (Singular kernels and the defect). The norm and compactness statements in Proposition 14.2 use bounded continuous kernels. For $G(r) = r^{-\beta}$, $0 < \beta < 1$, restrict (14.1) to zero-mass measures of finite absolute energy. Dilation preserves that domain and gives $\delta_{\lambda T} = \lambda^{-\beta}\delta_T$; positive definiteness gives zero defect. This separate observation asserts neither uniform-norm continuity nor a general attainment theorem for unbounded kernels.

Example 14.4 (A decreasing kernel with positive defect). On the three-point grid $\{0, 1, 2\}$ take $G(0) = 1$, $G(1) = 0.95$, $G(2) = 0.2$: nonnegative and nonincreasing, but not convex. The round trip $\nu = (+1, -2, +1)$ has zero mass and energy $\mathcal{E}_G[\nu] = 6 + 2(0.95)(-2 - 2) + 2(0.2)(1) = -1.2$, i.e. execution cost -0.6 : a profit. The defect is available in closed form. Writing a zero-mass variation as $\nu = (u, -u - v, v)$ and setting $p = u + v$, $d = u - v$ gives

$$\mathcal{E}_G[\nu] = \frac{2}{5}d^2 - \frac{3}{10}p^2. \quad (14.4)$$

Since $\|\nu\|_{TV} = |u| + |v| + |u + v| \leq 1$ forces $|p| \leq \frac{1}{2}$, $\mathcal{E}_G[\nu] \geq -\frac{3}{40}$, with equality at $d = 0$, $p = \frac{1}{2}$, that is, $u = v = \frac{1}{4}$. Hence $\delta_T(G) = \frac{3}{40} = 0.075$ exactly, attained at $\nu = (\frac{1}{4}, -\frac{1}{2}, \frac{1}{4})$ and its negative — the normalised version of the round trip above. Its unnormalized volume is 4, so the profit also equals $4^2\delta_T(G)/2 = 0.6$, confirming the cost convention. On this grid, making the sampled kernel convex, for instance $0.2 \leq G(1) \leq 0.6$, sets the defect to zero. The defect is a usable magnitude to report when a kernel is estimated rather than assumed. Equation (14.3) gives a lower bound on uniform-norm distance to the admissible cone; equality is not asserted. The value computed here is for the stated three-point grid.

Proposition 14.5 (The three-point defect in closed form). *On the grid $\{0, 1, 2\}$ write $G_k = G(k)$ and assume $G_0 \geq G_2$. Let δ_T denote the fixed-grid defect of Definition 14.1. Then*

$$\delta_T(G) = \frac{(4G_1 - 3G_0 - G_2)^+}{8}, \quad a^+ = \max\{a, 0\}. \quad (14.5)$$

When positive, it is attained at $\nu = \pm(\frac{1}{4}, -\frac{1}{2}, \frac{1}{4})$. Consequently, the sampled kernel is round-trip admissible if and only if $4G_1 \leq 3G_0 + G_2$. Discrete convexity $G_0 - 2G_1 + G_2 \geq 0$ implies this inequality, with strict inequality whenever $G_0 > G_2$. These statements concern this grid, not a continuum extension.

Proof. For $\nu = (u, -u - v, v)$ set $p = u + v$ and $d = u - v$. Expanding the quadratic form and substituting $u^2 + v^2 = (p^2 + d^2)/2$ and $uv = (p^2 - d^2)/4$ gives

$$\mathcal{E}_G[\nu] = \frac{1}{2}(G_0 - G_2)d^2 + \frac{1}{2}(3G_0 - 4G_1 + G_2)p^2. \quad (14.6)$$

The total-variation constraint implies $|p| \leq \frac{1}{2}$. For each feasible p , replacing d by zero remains feasible, since the resulting total variation is $2|p|$, and cannot increase the energy because $G_0 \geq G_2$. If $3G_0 - 4G_1 + G_2 \geq 0$, every zero-mass variation has nonnegative energy and the zero trade attains defect zero. Otherwise $d = 0$ and $|p| = \frac{1}{2}$ minimize the energy, giving (14.5). Finally, discrete convexity gives

$$4G_1 \leq 2G_0 + 2G_2 \leq 3G_0 + G_2,$$

and the second inequality is strict if $G_0 > G_2$. $\square$

Remark 14.6 (Convexity, admissibility and sharp stability). The admissibility threshold is weaker than discrete convexity: $G = (1, 0.6, 0.2)$ is convex and has zero defect, whereas $G = (1, 0.7, 0.2)$ is not convex but still has $4G_1 = 2.8 < 3.2$ and zero defect. Nonnegativity and monotonicity alone imply neither property, as Example 14.4 shows. Under $G_0 \geq G_2$, the defect is the positive part of a single linear contrast, hence piecewise linear rather than linear.

Round-trip admissibility does not require a signed mandate optimum to trade only in the mandate's direction. For $G = (1, 0.7, 0.2)$, the signed unit-buy optimum is $(3/4, -1/2, 3/4)$ with cost $11/40$, whereas the buy-only optimum is $(1/2, 0, 1/2)$ with cost $3/10$. To check this, symmetry and strict positivity on zero-mass variations reduce the fixed-mass problem to $x = (a, 1 - 2a, a)$, whose cost is $1/2 - 3a/5 + 2a^2/5$. The unconstrained minimum is at $a = 3/4$; the buy-only constraint gives $0 \leq a \leq 1/2$. Thus zero defect alone does not exclude cost reductions from an intermediate trade in the opposite direction.

On this grid the Lipschitz constant 1 in (14.2) is sharp, even among nonnegative, nonincreasing kernels. Indeed, take $G = (1, 0.95, 0.2)$ and $H = G + h(-1, 1, -1)$ with $0 < h < 1/40$. Then

$$\|H - G\|_\infty = h, \quad \delta_T(H) - \delta_T(G) = h.$$

Both equalities follow directly from (14.5). On a general finer grid, the defect is still the finite-dimensional optimization in Definition 14.1; this three-point formula does not determine it.

15 The large-size exponent and its boundary conditions

The following is the fair-pricing mechanism of Farmer et al. (2013), stated with a finite minimum order size. Its implication is asymptotic; the finite-size boundary term cannot be discarded in an exact identity.

Let $Q \geq q_0 > 0$ have Pareto survival probability $\mathbb{P}(Q > q) = (q_0/q)^\alpha$ for $q \geq q_0$, with $\alpha > 1$. Let $I(u)$ be the price displacement after u shares of an unfinished metaorder and let $\pi(Q)$ be its eventual permanent displacement.

Assumption 15.1 (Martingale pricing). $I(u) = \mathbb{E}[\pi(Q) \mid Q \geq u]$ for $u \geq 0$, with finite expectations.

Assumption 15.2 (Asymptotic fair pricing). The average displacement and permanent displacement agree at leading order: $\pi(Q)/[Q^{-1} \int_0^Q I(u) du] \rightarrow 1$ as $Q \rightarrow \infty$.

Theorem 15.3 (Tail exponent and finite-cutoff correction). Suppose $I(u) = Yu^\delta$ for $u \geq q_0$, $Y > 0$, $\delta > 0$, and $\pi(Q) = cYQ^\delta$ with $0 < c < 1$. Under Assumptions 15.1 and 15.2,

$$\delta = \alpha - 1, \quad c = \frac{1}{1 + \delta} = \frac{1}{\alpha}. \quad (15.1)$$

Martingale pricing fixes $I(u) = Yq_0^\delta$ below the cutoff, and

$$\frac{1}{Q} \int_0^Q I(u) du = cYQ^\delta + \frac{Y\delta q_0^{1+\delta}}{(1+\delta)Q}. \quad (15.2)$$

Thus exact fair pricing at every finite size is incompatible with this nonconstant pure-power model. For $1 < \alpha < 2$, impact is concave in size; $\alpha = 3/2$ gives $\delta = 1/2$ and permanent fraction $2/3$.

Proof. For $u \geq q_0$, conditional integration gives $\mathbb{E}[\pi(Q) \mid Q \geq u] = cY\alpha u^\delta/(\alpha - \delta)$; finiteness requires $\delta < \alpha$. Hence $c = (\alpha - \delta)/\alpha$. Below q_0 the event $Q \geq u$ has probability one, so this same expression at q_0 gives $I(u) = Yq_0^\delta$. Direct integration of the resulting piecewise path gives

$$\frac{1}{Q} \int_0^Q I(u) du = \frac{YQ^\delta}{1+\delta} + \frac{Y\delta q_0^{1+\delta}}{(1+\delta)Q}.$$

The second term is negligible relative to Q^δ ; asymptotic fair pricing therefore gives $c = 1/(1 + \delta)$. Equating the two expressions for c gives $\delta(\alpha - 1 - \delta) = 0$, and $\delta > 0$ excludes the constant solution. Substitution proves (15.2). $\square$

Corollary 15.4 (A conditional joint test). *In the large-size regime of Theorem 15.3, the impact exponent, permanent fraction and tail index satisfy two joint restrictions. A rejection rejects their combined modeling assumptions, including the tail and power specifications; it does not identify which assumption failed. Finite-size comparisons must retain (15.2) or specify another boundary model.*

Remark 15.5 (Local and large-size descriptions). The local hierarchy constrains small perturbations of a fixed rate shape. The tail calculation conditions on a metaorder-size model. They concern different limits and neither determines the other. A single analytic functional can have a quadratic small-size regime and subquadratic large-size cost, as in Remark 12.11; only a pure subquadratic law down to zero violates the local analytic assumption. Empirical applications must also separate changes in horizon and liquidity from changes in size (Lillo and Farmer, 2004; Bouchaud et al., 2018).

16 Order splitting as a convexification gap

Proposition 8.1 gives a uniform optimum for convex local costs and Proposition 9.2 exhibits concentration when the response is concave. The general statement is a classical relaxation result, and it identifies both the value and the structure of the limit.

Let $h : [0, \infty) \rightarrow [0, \infty)$ be Borel with $h(0) = 0$, and consider

$$\mathcal{J}[v] = \int_0^T h(v_t) dt, \quad v \geq 0, \quad \int_0^T v_t dt = X. \quad (16.1)$$

Write h^{**} for the lower convex envelope of h .

Classical result 16.1 (Value, attainment and concentration). *Let h be finite, nonnegative and lower semicontinuous, with $h(0) = 0$, and put $x = X/T > 0$. Over nonnegative integrable rates, the infimum of (16.1) is $Th^{**}(x)$. It is attained if and only if there are finite $v_1 \leq x \leq v_2$ and $\theta \in [0, 1]$ such that*

$$\theta v_1 + (1 - \theta)v_2 = x, \quad \theta h(v_1) + (1 - \theta)h(v_2) = h^{**}(x).$$

In that case a two-rate schedule with those time proportions is optimal. The uniform schedule is optimal exactly when $h(x) = h^{**}(x)$, and its excess cost is

$$\Delta(X, T) = T[h(x) - h^{**}(x)] \geq 0. \quad (16.2)$$

If no finite contact pair exists, no minimizing sequence of rate laws is uniformly integrable in its first moment. With a finite rate cap $\bar{V} \geq x$, the infimum for the restricted integrand is always attained by at most two rates in $[0, \bar{V}]$.

Proof. The cost and constraint depend only on the rate law under normalized time. Jensen gives the lower bound $Th^{**}(x)$. In one dimension the convex envelope is the infimum of two-point mixtures with mean x : any finite mixture can be reduced to at most two support points without increasing its cost by minimizing a linear objective with the two constraints of total probability and mean. The resulting finite convex envelope is continuous in the interior; nonnegativity and $h(0) = 0$ give continuity at zero. It is therefore the lower convex envelope. Approximate two-point mixtures prove the upper bound, and every such mixture is realised by a measurable two-rate schedule.

For necessity of finite contact, choose a supporting affine function L of h^{**} at x . If a rate law attains the bound, its expectation of $h - L \geq 0$ is zero; thus it is carried by the contact set $\{h = L\}$. Either x itself is a contact or contacts lie on both sides of x , giving the required pair. The converse is the stated construction.

All feasible rate laws are tight by Markov's inequality. If a minimizing sequence were uniformly integrable, a weakly convergent subsequence would retain mean x . Lower semicontinuity and the Portmanteau theorem would make its limit an attaining rate law, a contradiction in the non-attained case. With a rate cap, weak compactness preserves the mean automatically, so attainment follows and the same contact argument gives two rates. $\square$

Example 16.2 (Tight laws with lost first moment). For $T = X = 1$ and $h(v) = \sqrt{v}$, the rates $v_n = n\mathbf{1}_{[0, 1/n]}$ cost $n^{-1/2} \rightarrow 0$, while no feasible rate has zero cost. Their laws $(1 - 1/n)\delta_0 + (1/n)\delta_n$ are tight and converge weakly to δ_0 , whose mean is zero. The obstruction is lack of uniform integrability, not lack of tightness. An ordinary Young measure records that limit but loses the concentrating quantity; a generalized Young measure with a concentration component is needed to retain it.

Corollary 16.3 (Concave response can have convex cost). For $h(v) = \eta v^{1+\delta}$ with $\eta > 0$ and $\delta > 0$, the local cost is strictly convex and uniform execution is the unique optimum. The response ηv^δ is concave only for $0 < \delta < 1$. Bursting is governed by nonconvexity of the total local cost, or by a separate transient interaction; the local convex-envelope theorem does not automatically solve a nonlocal kernel problem.

The convexification result is classical (Ekeland and Temam, 1999). Its useful execution distinction is between a positive splitting gain that is attained by ordinary rates and a non-attained value requiring concentration. Neither case justifies replacing the executed rate by its average without recording the cost of the actual schedule.

Part IV

A general theory of execution

The impact layer is a functional on schedules. The execution layer is a functional on *policies*: measurable maps from observed histories to instructions. This part gives that layer the same

treatment. We choose a representation for policies in which the optimisation is exact and finite (Section 17), price the information that a policy consumes (Section 18), extend the problem from one trader to many and locate the crowding externality in the equilibrium operator (Section 19), and state the discretisation that connects the continuous theory to the discrete order-placement problem of Sections 24 and 27 (Section 20).

17 Signature policies, exact completion and error certificates

Signature execution policies are developed by Kalsi et al. (2020), including approximation of exponential transient impact. The results here identify an exact finite algebra after deterministic kernel augmentation, an adapted completion transformation, and a common feasible set on which kernel, moment and optimization errors yield an explicit regret certificate. These are conditional finite-feature results, not a claim that arbitrary endogenous order-placement dynamics reduce to a quadratic programme.

17.1 History representation and the initial state

Fix a deterministic initial state z_0 and a horizon T . A stopped history $H = (t, z)$ is a continuous finite-variation path on $[0, t]$ with $z(0) = z_0$ and time coordinate $z^1(s) = s$. Extend every coordinate constantly from t to T to define the 1-variation topology on histories. Write $\mathbb{S}(H) = \mathbb{S}_{0,t}(z)$ for the ordinary signature (Lyons, 1998), including the depth-zero constant. Policies of depth N have the form

$$v_t = \langle \ell, \mathbb{S}_{0,t}^{\leq N}(Z) \rangle, \quad \ell \in \mathbb{R}^{D(N)}, \quad D(N) = \sum_{n=0}^N d^n. \quad (17.1)$$

Theorem 17.1 (Universality on a compact history space). *Let $\mathcal{K}$ be a compact set of the stopped histories just defined, and let $\Phi : \mathcal{K} \rightarrow \mathbb{R}$ be continuous. For every $\epsilon > 0$ there are N and ℓ with $\sup_{H \in \mathcal{K}} |\Phi(H) - \langle \ell, \mathbb{S}^{\leq N}(H) \rangle| < \epsilon$. If an unconstrained policy cost is defined on all these continuous history maps and is continuous in the uniform norm, its infimum over finite-depth signature policies equals its infimum over all continuous history maps. This assertion does not by itself preserve execution constraints.*

Proof. The shuffle identity makes signature-linear functions an algebra; the depth-zero term supplies constants. The terminal time is read from the first time coordinate. At fixed initial point, the signature determines the path up to tree-like equivalence, and the strictly increasing time coordinate on the active interval removes that ambiguity. Thus the algebra separates stopped histories. Stone–Weierstrass gives density (Hambly and Lyons, 2010; Levin et al., 2013; Kalsi et al., 2020). Apply this density to each candidate continuous policy and then use continuity of its cost for the infimum claim. $\square$

Remark 17.2 (Variable initial states). A constant appended coordinate does not encode an initial state in the signature: the two paths (t, a, a) , $a = 0, 1$, have identical differentials and identical signatures. For varying initial states, use the algebra generated jointly by the explicit coordinates of z_0 and signature coordinates, for example $p(z_0)\langle \ell, \mathbb{S} \rangle$ with polynomial p . It separates a compact set of based stopped histories, so the same argument applies. The depth statements below condition on a fixed initial state and refer only to the ordinary iterated-integral degree.

17.2 An exact kernel-aware quadratic objective

Let $Y = (t, Y^2, \dots, Y^d)$ be a continuous finite-variation exogenous path: its law does not depend on the execution coefficients. For $G(r) = \sum_{m=1}^M w_m e^{-\rho_m r}$, $w_m > 0$, $\rho_m > 0$, augment Y by $E_m^\pm(t) = e^{\pm \rho_m t}$, and call the path $\hat{Y}$. For a coefficient functional a , let $A_j a$ append coordinate letter j ; thus $\langle A_j a, \mathbb{S}_{0,t} \rangle = \int_0^t \langle a, \mathbb{S}_{0,s} \rangle d\hat{Y}_s^j$.

Theorem 17.3 (Exact quadratic reduction and depth). *For*

$$v_t = \langle \ell, \mathbb{S}_{0,t}^{\leq N}(\hat{Y}) \rangle, \quad (17.2)$$

assume absolute integrability of all signature coordinates used below through depth $2N + 2$, and let $\eta \geq 0$. Then

$$\mathbb{E} \left[\int_0^T v_t \int_0^t G(t-s) v_s ds dt + \eta \int_0^T v_t^2 dt \right] = \ell^\top Q \ell, \quad (17.3)$$

where $Q \succeq 0$ and its entries are linear functionals of $\mathbb{E} \mathbb{S}_{0,T}^{\leq 2N+2}(\hat{Y})$. The exact transient functional for exponential m is

$$-\rho_m^{-2} A_{m,-} [(A_{m,+} \ell) \sqcup \ell]. \quad (17.4)$$

Temporary cost uses $\eta A_1(\ell \sqcup \ell)$ of depth $2N + 1$; expected quantity uses $A_1 \ell$ of depth $N + 1$. Consequently the objective and an expected-quantity equality define a convex quadratic programme. Adding a specified exogenous charge $\mathbb{E} \int_0^T a_t v_t dt$ adds a linear coefficient $b^\top \ell$; its own required moments must be supplied. If a_t is a signature functional of depth L , depth $N + L + 1$ suffices for that term.

Proof. Put $J_t^m = \int_0^t v_s dE_m^+(s)$, a depth- $(N + 1)$ signature functional. The impact state is $I_t^m = \rho_m^{-1} E_m^-(t) J_t^m$. Since $dE_m^- = -\rho_m E_m^- dt$,

$$\int_0^T I_t^m v_t dt = -\rho_m^{-2} \int_0^T J_t^m v_t dE_m^-(t).$$

The shuffle identity proves (17.4); the product has depth at most $2N + 1$ and its final integration adds one. The temporary and quantity formulas follow by the same append-letter rule. Taking expectations is justified by the stated moment assumption. Positive exponential mixtures have nonnegative symmetric energy on all signed rates, and $\eta \geq 0$ adds a nonnegative local form. Its restriction through the linear map $\ell \mapsto v$ is therefore positive semidefinite. The exogenous linear charge has the stated product degree. $\square$

No invertibility of Q is asserted: redundant or identically zero features can give singular coefficient matrices. For the impact-only objective, write the quantity constraint as $a^\top \ell = X$. Since every null direction of Q is a zero rate in the exponential energy, a annihilates $\ker Q$. For $X \neq 0$, a minimizing representative is $\ell^* = X Q^\dagger a / (a^\top Q^\dagger a)$, with null coefficients free; the constant policy ensures feasibility. Additional linear charges or inequalities require their own boundedness or compactness conditions.

Proposition 17.4 (Why ordinary finite moments are insufficient). *For the unaugmented path $Y_t^U = (t, (t - U)_+)$ on $[0, 1]$, the depth-one policy $v_t = (t - U)_+$ and kernel $G(r) = e^{-r}$, no fixed finite truncation of the expected ordinary signature determines the expected transient cost over all laws of $U \in [0, 1]$.*

Proof. The path has two straight increments $(U, 0)$ and $(1 - U, 1 - U)$. Every signature coordinate through depth D is a polynomial in U of degree at most D , by Chen's concatenation identity. With $L = 1 - U$ its transient cost is $f(U) = L^3/3 - L^2/2 + 1 - (L + 1)e^{-L}$, which is not a polynomial of any finite degree. If expectations of f were determined by the first D moments, every finite signed measure annihilating polynomials through degree D would annihilate f . Choose $D + 1$ interpolation points and a further point where the interpolation polynomial differs from f ; their interpolation weights give a signed measure violating that assertion. Its positive and negative parts have equal nonzero total mass, so normalization yields two probability laws with identical moments and different costs. Since every signature coordinate through depth D is a polynomial of degree at most D , those laws have identical entire expected ordinary signatures through that depth. Thus for each finite D there are two feature laws with the same depth- D expected signature and different transient costs. The augmentation in [Theorem 17.3](#) supplies the additional information needed for its exact finite-depth reduction.

Explicitly, the laws on $(0, 2/5, 4/5)$ with weights $(1, 10, 5)/16$ and on $(1/5, 3/5, 1)$ with weights $(5, 10, 1)/16$ match moments through degree four. They therefore have identical expected ordinary signatures through depth four. Both have expected executed quantity $3/20$ and temporary cost $1/15$ when $\eta = 1$, but transient expected costs 0.0148460680 and 0.0148890841 . $\square$

Remark 17.5 (Exogeneity and the optimization scope). If the feature path contains inventory, own fills or own impact, its law generally changes with ℓ and the quadratic reduction no longer follows. Even the feedback law $v_t = \ell x_t$, $\dot{x}_t = -v_t$, gives $v_t = \ell X e^{-\ell t}$, which is nonlinear in ℓ . Such policies can be useful, but require a different optimization argument. Likewise, $\mathbb{E} \int v = X$ allows pathwise under- and over-execution. The next construction enforces completion on every path without including endogenous state in the feature signature.

Corollary 17.6 (Required moments). *On a fixed exogenous feature class, the impact objective requires only the augmented moments specified in [Theorem 17.3](#). This is a statement about evaluating a given finite class, not identification of the market's full law. Estimation of the constraint itself can introduce feasibility error; the pathwise construction below avoids estimating a completion equality.*

17.3 Adapted features with pathwise completion

Let $X > 0$ and let $\phi_1, \dots, \phi_J$ be continuous adapted features of the exogenous history. Define

$$B_j(t) = \int_0^t \phi_j(s) ds, \quad \psi_j(t) = (T - t)\phi_j(t) - B_j(t), \quad v_t^\ell = X/T + \sum_j \ell_j \psi_j(t). \quad (17.5)$$

Theorem 17.7 (Pathwise completion and finite rate constraints). *Every rate in (17.5) is adapted and satisfies $\int_0^T v_t^\ell dt = X$ on every path. Suppose $|\psi_j(t)| \leq M_j$ uniformly over the prescribed path class and times, where $0 < M_j < \infty$, and let $\bar{V} > X/T$. The compact coefficient set*

$$\mathcal{L} = \left\{ \ell : \sum_j M_j |\ell_j| \leq r \right\}, \quad r = \min\{X/T, \bar{V} - X/T\}, \quad (17.6)$$

guarantees $0 \leq v_t^\ell \leq \bar{V}$ and exact completion on every path. It is a polytope, expressible by finitely many linear inequalities using auxiliary absolute-value variables. If the ϕ_j have signature depth at most N , then the rates have depth at most $N + 1$, and their impact objective uses augmented expected signatures through depth $2N + 4$. Optimization over $\mathcal{L}$ is a finite convex quadratic programme.

Proof. ψ_j is the derivative of $(T - t)B_j(t)$, whose values at both endpoints are zero. This proves the pathwise integral identity using only past features. The weighted coefficient bound gives $|v_t^\ell - X/T| \leq \sum_j M_j |\ell_j| \leq r$, proving both rate bounds. For signature features, multiplication by time and time integration each add at most one level; thus [Theorem 17.3](#) with rate depth $N + 1$ gives the moment bound. The affine baseline creates a constant and linear term but preserves convexity. Compactness guarantees attainment, including when the quadratic matrix is singular. $\square$

The coefficient bound is sufficient and can be conservative; it is not a characterisation of all feasible adapted policies. Uniform feature bounds are available on compact finite-variation path classes. Zero features can be removed before choosing positive M_j . The case $\bar{V} = X/T$ admits the uniform rate and is handled separately. For example, on $T = 1$ with $Y_t^U = (t, (t - U)_+)$, $U \in [0, 1]$, the features $1, t, (t - U)_+$ yield completion features with bounds $1, 1/2, 1/2$, respectively.

17.4 A common certificate for kernel and moment errors

For a fixed feasible set $\mathcal{L} \subset \{\|\ell\|_2 \leq R\}$, let $J_G(\ell)$ be the true expected cost, including any common temporary and exogenous linear terms. Let an exponential approximation H give $J_H(\ell) = c + \ell^\top Q \ell + b^\top \ell$, and let $\hat{J}(\ell) = \hat{c} + \ell^\top \hat{Q} \ell + \hat{b}^\top \ell$ be its estimated objective. The same feasible set is used by all three objectives.

Theorem 17.8 (Execution-regret certificate). *Suppose $\sup_{\ell \in \mathcal{L}} |J_G(\ell) - J_H(\ell)| \leq \epsilon_K$, $\|Q - \hat{Q}\|_{\text{op}} \leq \epsilon_Q$ and $\|b - \hat{b}\|_2 \leq \epsilon_b$. If $\hat{J}(\hat{\ell}) \leq \inf_{\ell \in \mathcal{L}} \hat{J}(\ell) + \tau$, then*

$$J_G(\hat{\ell}) - \inf_{\ell \in \mathcal{L}} J_G(\ell) \leq 2\epsilon_K + 2R^2\epsilon_Q + 2R\epsilon_b + \tau. \quad (17.7)$$

No error in the constant coefficient is needed, since it cancels in policy comparison. For nonnegative pathwise-completing rates, $\epsilon_K \leq X^2 \|G - H\|_\infty / 2$ when this norm is finite. For rates also bounded by $\bar{V}$,

$$\epsilon_K \leq \bar{V} X \|G - H\|_{L^1(0,T)}. \quad (17.8)$$

These are certificates within the prescribed policy class. Relative to a larger class, its best-achievable approximation gap must be added.

Proof. Subtract the constant coefficient from each estimated and exact quadratic. Their difference is bounded by $R^2\epsilon_Q + R\epsilon_b$ on $\mathcal{L}$. Compare $\hat{\ell}$ with an arbitrary ϵ -minimizer of J_G , insert J_H and $\hat{J}$ on both sides, apply the optimization tolerance and let $\epsilon \downarrow 0$. This gives (17.7). For the uniform kernel bound, symmetrize the cost difference and use $\int v = X$ and $v \geq 0$. For the integrable bound, extend v by zero; Young's convolution inequality and Cauchy-Schwarz give $|\langle v, (G - H) * v \rangle| \leq \|G - H\|_{L^1(0,T)} \|v\|_2^2$ for the causal convolution. Finally $\|v\|_2^2 \leq \bar{V} \int v = \bar{V} X$ pathwise. $\square$

Estimated moments need not produce a positive semidefinite matrix. One may fit the Gram form directly from sampled paths, or project the symmetric estimate onto the positive semidefinite cone and measure ϵ_Q after that operation. Neither procedure supplies a statistical error bound by itself. Given deterministic or high-probability bounds on the coefficient errors, (17.7) inherits exactly that interpretation. When $\hat{Q} \succeq 0$, a computable optimization certificate is the convex duality gap $\tau = \nabla \hat{J}(\hat{\ell})^\top \hat{\ell} - \min_{\ell \in \mathcal{L}} \nabla \hat{J}(\ell)^\top \ell$.

Corollary 17.9 (An explicit finite-sample certificate). *In the setting of Theorems 17.7 and 17.8, suppose n independent exogenous paths have the true feature law and $|a_t| \leq A$ for the common exogenous linear charge. Fix the features and positive exponential kernel H before drawing the sample. Estimate Q, b by averaging their per-path quadratic coefficients. Put*

$$D = \int_0^T (T-s)H(s) ds + \eta T, \quad B = 2(X/T)D + AT, \quad z_{n,\delta} = \sqrt{\frac{2 \log(J(J+3)/\delta)}{n}}.$$

The logarithm counts $J(J+1)/2$ distinct entries of the symmetric matrix and J vector entries, each with a two-sided bound. For $0 < \delta < 1$, with probability at least $1 - \delta$, simultaneously

$$\epsilon_Q \leq D \|M\|_2^2 z_{n,\delta}, \quad \epsilon_b \leq B \|M\|_2 z_{n,\delta}. \quad (17.9)$$

Substitution into (17.7) gives an explicit high-probability class-regret bound. The sample quadratic matrix is positive semidefinite.

Proof. The per-path coefficient bounds are $|Q_{ij}| \leq DM_i M_j$ and $|b_j| \leq BM_j$, by the two ordered impact integrals, the local quadratic term and the bounded linear charge. For an average of n independent variables in $[-L, L]$, Hoeffding's inequality (Hoeffding, 1963) bounds the probability of a centered error exceeding Lz by $2e^{-nz^2/2}$. Apply it to the $J(J+1)/2$ distinct matrix entries and the J vector entries and take a union bound. Symmetry supplies the duplicate matrix entries; no independence between entries is needed. If $|E_{ij}| \leq DM_i M_j z$, then $|u^\top E v| \leq Dz(\sum_i M_i |u_i|)(\sum_j M_j |v_j|) \leq Dz \|M\|_2^2 \|u\|_2 \|v\|_2$; the vector estimate is analogous. Each per-path impact-plus-temporary form is positive semidefinite, and averaging preserves that property. $\square$

The independence and bounded-feature assumptions are substantive. Dependent market paths, estimated feature bounds or a changed response law require a different concentration argument. The deterministic certificate remains valid whenever its stated errors are bounded, irrespective of how those bounds were obtained.

Proposition 17.10 (Positive exponential approximation of a singular kernel). *For $0 < \beta < 1$, $G(t) = t^{-\beta}$ on $(0, T]$ admits positive finite exponential mixtures converging to it in $L^1(0, T)$. Consequently the kernel term in (17.8) tends to zero on any common pathwise-completing rate-bounded class.*

Proof. Use $t^{-\beta} = \Gamma(\beta)^{-1} \int_0^\infty e^{-\rho t} \rho^{\beta-1} d\rho$. The integrated contribution of $0 < \rho < a$ is at most $Ta^\beta/[\beta\Gamma(\beta)]$; that of $\rho > b$ is at most $b^{\beta-1}/[(1-\beta)\Gamma(\beta)]$. On $[a, b]$, partition the interval and replace each exponential by its value at an interior representative, using the exact positive mass of $\rho^{\beta-1} d\rho/\Gamma(\beta)$ as weight. Uniform continuity on $[a, b] \times [0, T]$ makes the L^1 quadrature error tend to zero as the mesh shrinks. First choose a small and b large, then the mesh fine. Apply Theorem 17.8. This approximation is a statement about bounded rates; it does not assign finite Riesz energy to atomic blocks. $\square$

The geometric inventory signature of Remark 13.4 and the feature signature used here describe different paths. A manipulation condition on a market model is not thereby a linear constraint on policy coefficients. The rate and completion constraints in Theorem 17.7, by contrast, are derived explicitly on the coefficients.

18 The information ledger: nats and basis points

Liquidity data can change an execution decision, but that observation alone does not say how much the change can be worth. This section supplies the exchange rate between information and cost in both directions—an upper ceiling and an attaining family—which converts a qualitative statement about beliefs into a budget. The policies of [Section 17](#) and the hidden-liquidity model developed in [Section 28](#) provide applications. [Section 31](#) later applies the same accounting to a misspecified liquidity belief, where the relevant divergence is between the trader’s belief and the truth rather than between two information sets.

At a decision epoch, condition on the current history and let θ be the relevant uncertain state. A common feasible action set $\mathcal{A}$ contains instructions or fully specified continuation plans. Let $0 \leq C(a, \theta) \leq \bar{C}$ be their loss. The supplementary data change the posterior, not the loss function or action set. For nested information sets $\mathbb{G} \subseteq \mathbb{F}$, write $P_{\mathbb{H}}$ for the posterior, $m(P) = \inf_{a \in \mathcal{A}} \int C(a, \theta) P(d\theta)$ and $V(\mathbb{H}) = \mathbb{E}m(P_{\mathbb{H}})$. Assume measurable minimizers or measurable ϵ -minimizers exist. Total variation is $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$.

Theorem 18.1 (Sharp bounded-loss information ceiling). *With $I = I(\theta; \mathbb{F} \mid \mathbb{G})$ in nats,*

$$0 \leq V(\mathbb{G}) - V(\mathbb{F}) \leq \bar{C} \mathbb{E} \|P_{\mathbb{F}} - P_{\mathbb{G}}\|_{\text{TV}} \leq \bar{C} \min\{1, \sqrt{I/2}\}. \quad (18.1)$$

The coefficient of $\sqrt{I}$ is sharp uniformly over bounded-loss decision problems as $I \downarrow 0$.

Proof. For a function in $[0, \bar{C}]$, its expectation changes by at most $\bar{C} \|P - Q\|_{\text{TV}}$. Taking infima over the same action set gives $|m(P) - m(Q)| \leq \bar{C} \|P - Q\|_{\text{TV}}$. The Bayes envelope m is concave, and $P_{\mathbb{G}} = \mathbb{E}[P_{\mathbb{F}} \mid \mathbb{G}]$, which proves nonnegative information value. The Lipschitz bound, conditional Pinsker inequality and Jensen give the other inequalities; total variation is at most one. Sharpness follows from [Theorem 18.2](#). $\square$

Theorem 18.2 (Binary identity and an attaining family). *For a uniform binary state and wrong-instruction loss $\Delta > 0$, let P_0, P_1 be the conditional signal laws. The exact value of the signal is*

$$V(\emptyset) - V(\mathbb{G}) = \frac{\Delta}{2} \|P_0 - P_1\|_{\text{TV}}. \quad (18.2)$$

If a binary signal reports the correct state with probability $1/2 + u$, $0 < u < 1/2$, its saving is Δu and $I = \log 2 - h_2(1/2 + u) = 2u^2 + O(u^4)$, where h_2 is binary entropy. Thus the ratio of saving to $\Delta\sqrt{I/2}$ tends to one. For conditional Gaussian laws $\mathcal{N}(0, 1), \mathcal{N}(\epsilon, 1)$, the saving is $\Delta[\Phi(|\epsilon|/2) - 1/2]$ and $I = \epsilon^2/8 + o(\epsilon^2)$.

Proof. The optimal binary test has error $[1 - \|P_0 - P_1\|_{\text{TV}}]/2$, whereas the uninformed error is $1/2$. This proves (18.2). In the binary channel the posterior success probability is $1/2 + u$, and direct evaluation of binary entropy gives the expansion. For the Gaussian channel the likelihood-ratio threshold is $\epsilon/2$ when $\epsilon \neq 0$, with the decision inequality reversed for $\epsilon < 0$. The error is $\Phi(-|\epsilon|/2)$; at $\epsilon = 0$ the signal is uninformative. This gives the displayed saving. Its mutual information is the Jensen–Shannon divergence; a second-order expansion of the two Gaussian densities about their common limit gives $\epsilon^2/8 + o(\epsilon^2)$. The exponential moments of a Gaussian dominate the derivatives in a neighborhood of zero and justify integration of that expansion. $\square$

Remark 18.3 (An upper ceiling is not an individual value law). A signal revealing a uniform binary state with probability ϵ and otherwise nothing has $I = \epsilon \log 2$ and saving $\Delta\epsilon/2$, linear in information. Information about a component irrelevant to every action can have zero value. The attaining family above proves a worst-case sharp rate; actual value still depends on the loss and feasible decisions.

Remark 18.4 (Mutual information and residual error). Mutual information also gives valid error lower bounds. For a uniform binary state, conditional entropy and Jensen imply $h_2(p_e) \geq \log 2 - I$, so $p_e \geq h_2^{-1}(\log 2 - I)$, with the inverse on $[0, 1/2]$. This bound vanishes for perfect information. Pairwise KL can instead be used through Pinsker in (18.2), but it is not interchangeable with mutual information: the rare-revelation channel above has arbitrarily small I and infinite pairwise KL. Smooth local families such as the Gaussian channel permit a local comparison; arbitrary low-information families do not (Tsybakov, 2009).

Corollary 18.5 (An execution information budget). *Under the common-action and bounded-loss conditions, data with conditional information I cannot improve expected decision cost by more than $\bar{C} \min\{1, \sqrt{I/2}\}$. Expressing $\bar{C}$ in basis points makes this a comparable ceiling for the value of that decision's data. Applying the statement to repeated decisions requires the corresponding conditional decision model at each epoch; it does not automatically price an arbitrary change to a controlled observation process or to the feasible action set.*

This complements Proposition 28.2: liquidity data can change execution decisions without improving the midpoint forecast, but information quantities alone do not identify their actual dollar value.

19 Many traders and the equilibrium operator

The mandate of Section 18 is one trader's. Impact is shared. Transient-impact execution games were analysed by Schied and Zhang (2019), who give existence, uniqueness and explicit equilibria for two players under related conditions; Theorem 19.1 is the variational-inequality statement for N players and Theorem 19.2 the operator decomposition, and neither should be read as prior to that literature. Let $i = 1, \dots, N$ traders execute mandates X_i over $[0, T]$ with a common kernel. Write Γ for the causal convolution $(\Gamma w)(t) = \int_0^t G(t-s)w_s ds$, so that the price is $p = p_0 + \Gamma w$ with $w = \sum_j v_j$ the aggregate rate, and let trader i face

$$C_i(v_i, v_{-i}) = \int_0^T (\Gamma w)_t v_{i,t} dt + \eta \|v_i\|_2^2, \quad v_i \in \mathcal{V}_i = \left\{v \in L^2 : \int_0^T v = X_i\right\}. \quad (19.1)$$

Theorem 19.1 (Existence and uniqueness of equilibrium). *Assume the causal convolution Γ is bounded on $L^2(0, T)$. If $\eta > 0$ and G is positive semidefinite in the sense that $\langle \Gamma x, x \rangle \geq 0$ for all $x \in L^2$, then the game (19.1) has a unique Nash equilibrium in $\prod_i \mathcal{V}_i$.*

Proof. Each C_i is convex and continuous in v_i and the strategy sets are closed, convex and nonempty. The pseudo-gradient is $F(v)_i = \Gamma w + \Gamma^* v_i + 2\eta v_i$. For $d = v - u$ with $D = \sum_i d_i$,

$$\langle F(v) - F(u), d \rangle = \langle \Gamma D, D \rangle + \sum_i \langle \Gamma^* d_i, d_i \rangle + 2\eta \sum_i \|d_i\|_2^2 \geq 2\eta \sum_i \|d_i\|_2^2,$$

using $\langle \Gamma^* d_i, d_i \rangle = \langle \Gamma d_i, d_i \rangle \geq 0$. So F is strongly monotone on the affine feasible set, and the associated variational inequality has a unique solution (Facchinei and Pang, 2003), which is the unique equilibrium. $\square$

Theorem 19.2 (The crowding operator). *Under the hypotheses of Theorem 19.1, let the mandates be symmetric, $X_i = X/N$. The unique equilibrium is symmetric, $v_i = w/N$, and its aggregate w satisfies, for a Lagrange multiplier c ,*

$$\left(\Gamma + \frac{1}{N}\Gamma^* + \frac{2\eta}{N}\right)w = c, \quad (19.2)$$

whereas the cooperative optimum satisfies $(\Gamma + \Gamma^* + \frac{2\eta}{N})w = c$. Writing $\Sigma = \frac{1}{2}(\Gamma + \Gamma^*)$ for the symmetric energy operator and $\Lambda = \frac{1}{2}(\Gamma - \Gamma^*)$ for its antisymmetric part,

$$\Gamma + \frac{1}{N}\Gamma^* = \frac{N+1}{N}\Sigma + \frac{N-1}{N}\Lambda. \quad (19.3)$$

Equilibrium execution is therefore governed by a non-self-adjoint operator: the self-adjoint part has ratio $(N+1)/(2N)$ to 2Σ in the cooperative problem, and an antisymmetric component of relative weight $(N-1)/(N+1)$ appears, vanishing at $N=1$ and increasing to 1 as $N \rightarrow \infty$.

Proof. Permuting the players maps the game to itself and equilibria to equilibria. By uniqueness in Theorem 19.1, the equilibrium is fixed by every permutation and is therefore symmetric.

Differentiate (19.1) in v_i at a symmetric profile: the own contribution to the aggregate gives Γw from the price the trader pays and $\Gamma^* v_i = \Gamma^* w/N$ from the effect of the trader's own rate on prices paid later, plus $2\eta v_i$. Stationarity on the mass-constrained set gives (19.2). Summing the individual costs and differentiating the total gives the cooperative condition. The decomposition (19.3) is algebra. $\square$

Corollary 19.3 (Interpretation). *Relative to the cooperative operator $\Gamma + \Gamma^* = 2\Sigma$, the equilibrium operator differs by*

$$-\frac{N-1}{N}\Gamma^* = -\frac{N-1}{N}\Sigma + \frac{N-1}{N}\Lambda :$$

crowding weakens the symmetric energy the trader internalises and introduces a temporally non-self-adjoint component, both with weight $(N-1)/N$. As $N \rightarrow \infty$ the operator tends to $\Sigma + \Lambda = \Gamma$, the purely causal operator: each trader then internalises only the price it pays, not the price its own rate imposes on later fills. This is an operator limit; convergence of equilibrium schedules requires additional estimates, since the coercive temporary coefficient also tends to zero.

Remark 19.4 (Temporal skew and potential games). The skew part here is temporal: scalar exponential convolution is not self-adjoint on $L^2(0, T)$, yet its symmetric energy is positive. It is different from the permanent cross-asset skew matrix of Theorem 13.1. The pseudo-gradient has off-diagonal derivative blocks Γ between distinct players. A differentiable scalar potential with this gradient on the affine strategy space requires these blocks to be adjoints after restriction to zero-integral directions. Thus a nonzero compressed temporal skew prevents that exact potential representation. This does not say that the equilibrium cannot minimize some other function, or that every nonzero skew on the ambient space survives the constraint projection.

Remark 19.5 (Interpretation for estimation). Under the symmetric game specification, the relative temporal-skew coefficient is $(N-1)/(N+1)$. Testing that restriction requires distinguishing the equilibrium first-order operator from the market response kernel and retaining the temporary-cost scaling. Price-flow regressions alone need not identify these objects: participation breadth, correlated flow and latent liquidity can co-vary (Bouchaud et al., 2018). The operator identity therefore supplies a conditional model restriction, not an unconditional prediction that a measured market-impact matrix becomes more asymmetric as trader counts rise.

20 From the continuum to the order book

The impact theory is stated on measures and rates; the execution layer sends discrete messages to discrete price levels. The link is a convergence statement.

Proposition 20.1 (Grid consistency and the diagonal charge). *Let G be bounded and continuous on $[0, T]$, fix $K \geq |X|$, and let $\mathcal{M}_X^{h,K}$ be the schedules of mass X supported on a grid of mesh h with $\|\mu\|_{\text{TV}} \leq K$. Then $\mathcal{E}_G$ restricted to $\mathcal{M}_X^{h,K}$ Γ -converges as $h \downarrow 0$, for the weak-* topology on measures, to $\mathcal{E}_G$ on $\mathcal{M}_X^K = \{\mu \in \mathcal{M}_X : \|\mu\|_{\text{TV}} \leq K\}$; minimum values converge and every sequence of discrete minimisers has weak-* limit points, each a minimiser on $\mathcal{M}_X^K$. The total-variation bound cannot be dropped: fixed mass alone does not give equicoercivity for signed measures, as $\mu_n = (X + n)\delta_0 - n\delta_T$ shows for $G \equiv 1$, where every mass- X schedule has energy X^2 but $\|\mu_n\|_{\text{TV}} \rightarrow \infty$ and no subsequence converges. For nonnegative schedules the bound is automatic with $K = X$.*

For a Riesz kernel $G(t) = t^{-\beta}$, $0 < \beta < 1$, use a different discretization: a uniform partition into cells C_i of width $h = T/n$ and piecewise constant rates reconstructed from cell quantities x_i . Their energy is exactly $\sum_{i,j} K_{ij}^h x_i x_j$, where

$$K_{ij}^h = h^{-2} \int_{C_i} \int_{C_j} |t - s|^{-\beta} ds dt. \quad (20.1)$$

In particular,

$$K_{ii}^h = \frac{2h^{-\beta}}{(1-\beta)(2-\beta)}, \quad K_{i,i+k}^h = \frac{h^{-\beta}[(k+1)^{2-\beta} - 2k^{2-\beta} + (k-1)^{2-\beta}]}{(1-\beta)(2-\beta)}, \quad k \geq 1.$$

This matrix is positive semidefinite and gives the exact continuum energy of the reconstructed rate, including the uniform rate. Replacing only the diagonal by d_h changes the uniform schedule's energy by

$$\frac{X^2 h}{T} (K_{ii}^h - d_h). \quad (20.2)$$

For bounded d_h , the resulting underestimate is of order $h^{1-\beta}$; it tends to zero but is not a higher-order correction. These statements do not assert a signed-measure Γ -limit for the singular kernel.

Proof. The bounded-kernel feasible set $\mathcal{M}_X^K$ is weak-* compact. On a uniformly bounded total-variation set, weak-* convergence implies convergence of product integrals against continuous kernels: approximate the kernel uniformly by finite sums of products of one-variable continuous functions. The lower-limit inequality follows, and pushing each cell's mass to a grid representative preserves total mass, does not increase total variation, and gives a recovery sequence. Compactness then yields convergence of minima and cluster points of minimizers.

For the singular calculation, integrate $|t - s|^{-\beta}$ twice over each cell pair to obtain (20.1). Positive definiteness of the Riesz energy on piecewise constant rates gives positive semidefiniteness of the matrix. The uniform rate has $x_i = Xh/T$, so altering $n = T/h$ diagonal entries changes its energy by (20.2). The leading term is $2X^2 h^{1-\beta} / [T(1-\beta)(2-\beta)]$ when d_h is bounded. $\square$

Discretization must specify what a cell quantity represents. A point mass has infinite self-energy for a Riesz kernel, whereas a finite-width constant rate has the finite charge above. Cell-averaged costs therefore preserve the rate interpretation and provide a direct comparison across bar widths.

21 What the two general theories buy

The pieces above are not independent.

1. *Local structure has a domain.* Analyticity gives distributional kernels on rate space. Ordinary propagator and measure-schedule results enter only when their additional kernel assumptions hold. Side symmetry excludes odd cost layers; a pure subquadratic law down to zero is non-analytic.

2. *Cross-impact compares energy and area.* Closed permanent loops have the signed-area cost in [Theorem 13.1](#); transient cost combines symmetric energy and weighted area in [Proposition 13.6](#). The exponential lift retains an explicit endpoint term. The defect in [Proposition 14.2](#) quantifies violations within its specified kernel space. [Proposition 14.5](#) gives the exact three-point threshold, and [Corollary 30.2](#) translates a fitted kernel’s defect into the largest phantom round-trip gain under a gross-volume cap.
3. *Size asymptotics retain boundary terms.* The Pareto and asymptotic fair-pricing model yields [Theorem 15.3](#); its finite-cutoff correction is explicit. It is not an exact finite-size universality law.
4. *Splitting and non-attainment are distinct.* Finite contact with the convex envelope yields an ordinary two-rate optimizer. Non-attainment requires loss of uniform integrability, not absence of a weak rate-law limit.
5. *Kernel-aware signatures give an exact finite objective.* [Theorem 17.3](#) requires augmented depth $2N + 2$ for a depth- N rate. [Theorem 17.7](#) supplies pathwise completion and sufficient finite rate constraints, with no estimated quantity equality.
6. *Approximation errors have a common execution consequence.* [Theorem 17.8](#) separates kernel error, coefficient error and optimization tolerance on a common feasible class. [Proposition 17.10](#) handles an integrable singular kernel by positive exponential approximation on bounded rates.
7. *Information gives a sharp ceiling under a specified decision model.* [Theorem 18.1](#) is sharp in its leading constant, while actual signal value can be linear in information or zero.
8. *Crowding changes both parts of the operator.* The equilibrium differs from cooperation by $-(N - 1)\Gamma^*/N$. Temporal skew is not identified with permanent cross-asset manipulation.

The limits of the construction are substantive. Exogenous features do not encode unrestricted feedback through own queue position or own impact. The finite coefficient polytope is sufficient for feasibility and may exclude better policies. The certificate bounds regret within that class. [Corollary 17.9](#) supplies moment bounds for independent paths and bounded features; it does not establish those conditions for market data. The crossover and fair-pricing examples specify conditional mechanisms rather than an estimated order-book model.

Part V

The execution mandate

22 Institutional scope

The institutional focus is cash-equity execution on the New York Stock Exchange (NYSE, XNYS) and The Nasdaq Stock Market (Nasdaq, XNAS). Order types below refer to those execution destinations. The security’s listing market, the destination of a child order, and the broker’s access arrangement are separate state variables. For NYSE, the venue designation means the NYSE itself; the rules of NYSE Arca or another affiliated exchange cannot be substituted. The same principle applies to other markets operated by Nasdaq. The mathematical control framework is general, but its institutional implementation here is organized around these two exchanges.

Official order-entry, allocation and auction sources were consulted on 11 September 2026. An instruction is admissible only for the chosen instrument, session, protocol and participant access. Historical applications require the rules and reference data effective on the traded date; a published future feature is not evidence that it was active in a historical session.

23 The parent mandate and the execution ledger

23.1 Quantity, benchmarks, and units

Consider a buy order for $Q > 0$ shares over $[0, T]$. Sell orders follow by reversing the signed price convention. Let $F_i \geq 0$ denote the shares in fill i , P_i its price in dollars per share, and f_i its net fee in dollars per share. A rebate is a negative fee. If the order completes, arrival-price implementation shortfall is

$$\text{IS} = \sum_i F_i(P_i - S_0 + f_i), \quad \sum_i F_i = Q. \quad (23.1)$$

The dollar denominator matters. Reporting $10^4 \text{IS} / (QS_0)$ gives basis points of initial notional; dividing by Q instead gives dollars per share. Confusing these quantities can make a small per-share improvement appear to be a large portfolio return.

If $q_T = Q - \sum_i F_i > 0$, a completed-order score is not defined without a treatment of the residual. One may mark it to S_T , apply an explicit completion price, or retain it and report non-completion separately. A useful marked ledger is

$$\text{IS}^{\text{mark}} = \sum_i F_i(P_i - S_0 + f_i) + q_T(S_T - S_0) + \Phi(q_T), \quad (23.2)$$

where Φ represents mandate-specific costs beyond marking to the terminal reference. Taking $\Phi(q) = +\infty$ for $q > 0$ imposes hard completion, but only when the modeled action set can actually deliver it. A halt or a failed final auction can invalidate that feasibility assumption.

The benchmark is part of the mandate. An arrival-price strategy manages the opportunity cost of waiting after the decision. A VWAP-relative strategy tracks a market-volume-weighted reference. A closing-price strategy may value participation in the closing auction. These objectives can prefer different actions even under the same market forecast.

23.2 An exact cost decomposition

At discrete decision times $k = 0, \dots, N-1$, aggregate the fills occurring before the next reference-price observation into F_k . Let $q_{k+1} = q_k - F_k$, $q_0 = Q$, and suppose $q_N = 0$. The algebra below applies to any reference path $S_0, \dots, S_N$. When several prices occur in a bucket, P_k is their volume-weighted execution price; buckets with no fill contribute zero.

Proposition 23.1 (Shortfall accounting). *For every completed execution path,*

$$\text{IS} = \sum_{k=0}^{N-1} F_k(P_k - S_k + f_k) + \sum_{k=0}^{N-1} q_{k+1}(S_{k+1} - S_k). \quad (23.3)$$

The first term records local execution premiums and fees; the second records reference-price changes while shares remain unexecuted.

Proof. Since $F_k = q_k - q_{k+1}$, summation by parts gives

$$\sum_{k=0}^{N-1} F_k S_k = q_0 S_0 - q_N S_N + \sum_{k=0}^{N-1} q_{k+1}(S_{k+1} - S_k).$$

Use $q_N = 0$, subtract QS_0 , and add the execution premiums and fees. □

Table 4: Core state and its economic meaning.

Symbol	Meaning	Units or domain
q	Unexecuted parent quantity	Shares
S, M	Structural reference and observed midpoint	Dollars/share
B^v	Visible book at venue v	Prices, sizes, counts
ℓ^v	Own live orders, including pending cancellations	Order records
z^v	Own queue/allocation state	Venue dependent
H, R	Hidden resting shares and iceberg reserve	Latent shares
b	Conditional distribution of latent state	Probability measure
I	Impact/resilience state	Model dependent
r	Session, halt and auction state	Discrete state
a	New, cancel, replace or route instruction	Admissible action

The identity does not assume that fills and subsequent returns are independent. That distinction is essential for passive orders: being filled can be informative about the next price movement. The identity is also not an impact decomposition. If S_k is the observed midpoint, its changes may already include the trader’s impact. Adding a separate permanent-impact charge to the same observed-price ledger may count the same effect twice. A structural model must state whether its reference is an unaffected price or an observed price.

23.3 Risk and adaptation

Suppose that after observing a bucket’s fills, the next unaffected reference increment has conditional mean $\alpha_k \Delta t_k$ and variance $\sigma_k^2 \Delta t_k$. The remaining quantity q_{k+1} is then measurable before the next centered price innovation. Under square integrability, its martingale exposure has variance

$$\mathbb{E} \sum_k q_{k+1}^2 \sigma_k^2 \Delta t_k. \quad (23.4)$$

This motivates an additive inventory-risk penalty. It is not generally the variance of total shortfall: random impact, fees, fill selection, and predictable drift can also fluctuate and covary. Moreover, a static mean–variance objective is generally time inconsistent. In the dynamic sections, an additive exposure penalty is specified as the objective rather than silently substituted for the variance of all trading costs.

24 NYSE and Nasdaq order instructions

An order instruction changes the set of possible executions. It should therefore be represented by an admissibility condition and a transition mechanism, not only by a categorical feature. In a general representation,

$$a = (v, n, p, d, \tau, m, \rho, \chi), \quad (24.1)$$

where v is destination, n submitted quantity, p the price or pegging rule, d display size, τ lifetime, m minimum execution size, ρ routing instructions, and χ other eligibility flags. The tuple is subject to venue-, protocol-, instrument-, and session-specific compatibility rules. Not every combination is valid.

The market/limit distinction concerns price instructions; IOC and FOK concern execution conditions. They are different axes. Likewise, a passive limit order and a post-only instruction are

Table 5: Order instructions as economic controls. Availability and exact handling must be checked for the chosen venue and protocol.

Instruction	Decision being made	Main modeling consequence
Market	Seek immediate liquidity without an explicit limit price	Depth and price protection determine achievable fills
Marketable limit	Cross available prices up to a limit	Residual quantity depends on prices reached and lifetime
Resting limit	Exchange immediacy for a price bound	Queue position, adverse selection and deadline risk
Post-only / ALO	Request passive treatment	A marketable arrival may reprice, cancel or receive special handling
Midpoint or other peg	Delegate repricing to a reference rule	Reference availability, price moves and eligibility enter the state
Non-displayed limit	Withhold displayed interest	Different eligibility/priority and uncertain matching
Reserve/iceberg	Choose visible clip and reserve	Replenishment transfers size and can change displayed priority
Immediate-or-cancel	Permit immediate partial execution only	Unfilled residual is canceled
Fill-or-kill	Require immediate complete execution or cancellation	An all-or-nothing immediate fill event
Minimum quantity	Restrict acceptable executions	Removes some matches; not equivalent to fill-or-kill
Auction order	Commit to a scheduled or event-driven crossing process	Clearing, allocation, cutoff and residual rules

not equivalent: the former may become marketable before it reaches the matching engine. A stop instruction introduces a trigger followed by a resulting order; it is not itself a guaranteed execution price. A broker-facing label must be mapped to an accepted exchange instruction. The generic definition of FOK above does not establish native availability: the consulted NYSE Pillar FIX matrix marks FOK unsupported for its NYSE equity rows ([New York Stock Exchange, 2026i](#)). A controller must not silently turn an unsupported instruction into several independent orders.

24.1 Mapping the control to native instructions

Table 6 connects economic choices with exchange terminology. It is a compact mapping, not a claim that every displayed combination of attributes is valid. NYSE’s gateway matrix explicitly distinguishes the NYSE rows from other group exchanges and distinguishes Tape A from Tape B/C symbols ([New York Stock Exchange, 2026i](#)). A symbol’s auction eligibility is checked separately.

A passive label is not a sufficient execution rule. Nasdaq Post-Only can execute on entry under the rule’s price-improvement conditions; otherwise its handling can involve repricing or cancellation. The optimizer must use the resulting acceptance and fill law rather than impose zero immediate fills ([Nasdaq, 2026d](#)).

NYSE D Orders are Floor Broker instructions with price discretion. Access to that service is an action constraint, not a flag every electronic participant can assume. NYSE describes their use in continuous trading and auctions ([New York Stock Exchange, 2019](#)). Their value in the model comes from the distribution of permitted prices and allocations, net of access costs, rather than a presumed unconditional fill advantage. A resting MPL order with minimum-size conditions can lose parity eligibility, so increasing the minimum fill changes allocation as well as the set of acceptable

Table 6: Selected native instructions for NYSE and Nasdaq. Exact protocol and attribute compatibility remains part of the action constraint. Sources: [New York Stock Exchange \(2026i\)](#); [Nasdaq \(2026d,b\)](#); [New York Stock Exchange \(2019\)](#).

Economic choice	NYSE (XNYS)	Nasdaq (XNAS)
Immediate / priced	Market Day; Limit Day; Limit IOC	Market Order; limit-priced instructions; IOC attribute
Passive entry	Add Liquidity Only (ALO) on supported limit instructions	Post-Only Order
Hidden limit	Limit Non-Displayed	Non-Displayed Order
Visible clip / reserve	Limit Reserve	Reserve attribute on an eligible order
Reference pricing	Midpoint Liquidity (MPL); Primary Peg; Dark Primary Peg; Last Sale Peg	Pegging attributes, including midpoint; Midpoint Peg Post-Only is a distinct type
Opening / closing	MOO, LOO, MOC and LOC, subject to auction eligibility	MOO, LOO, MOC and LOC for the corresponding cross
Specialized access	Floor Broker D Orders; Closing IO interest	Opening and closing Imbalance-Only instructions

counterparties ([New York Stock Exchange, 2026g](#)).

For a parent order, these distinctions enter three linked choices. Price and urgency determine whether immediate execution is worth its premium. Display and reference-price instructions determine what liquidity the child can meet. Lifetime and auction commitments determine how much future flexibility remains. The controller compares these choices through conditional fill costs and the continuation value of remaining shares, developed below.

24.2 Conditional fill costs and adverse selection

For a buy, a fill below the current midpoint looks favorable locally. Its value depends on the price path conditional on being filled. Define the signed post-fill markout

$$A_\delta = -\mathbb{E}[M_{t+\delta} - M_t \mid \text{buy fill at } t]. \quad (24.2)$$

A positive value indicates an adverse subsequent midpoint movement for the buyer. It is a diagnostic, not a fee. If future-price costs are already included in a joint continuation model or in (23.3), adding the markout again would double count it. For a sell, the sign convention reverses.

An empirically useful forecast is therefore the joint law of fill size, fill time, price, and subsequent price change. Multiplying a marginal fill probability by an unconditional expected return discards the very dependence that makes passive execution difficult. Similarly, a high rebate can be outweighed by adverse selection, delay, or failure to complete the order.

24.3 The queue has economic value

For a simplified FIFO queue, let A be units ahead of an order of size ℓ . Assume the only queue depletion is N unit services during the waiting interval, and that no priority-changing event occurs. Then

$$F = \min\{\ell, (N - A)_+\}, \quad \mathbb{E}F = \sum_{j=1}^{\ell} \mathbb{P}(N \geq A + j). \quad (24.3)$$

The second equality is the tail-sum formula for a nonnegative integer random variable. Under Poisson service of rate ν over duration Δ , the first-fill probability is $\mathbb{P}(N \geq A + 1)$ for $N \sim \text{Poisson}(\nu\Delta)$.

Table 7: Exchange-specific state required by the execution model.

Destination / phase	Allocation distinction	State retained by the controller
Nasdaq continuous	Price, display and time priority	Ranked/displayed prices, order class, queue history and reserve refreshes
NYSE continuous	Setter priority and parity allocation	Eligible participant, setter status, allocation class and within-book ordering
Nasdaq cross	Cross-specific clearing and allocation	Cross eligibility, NOII information, accepted orders and phase restrictions
NYSE auction	Auction eligibility and DMM-led opening process	Listing/auction eligibility, imbalance information, broker access and commitments

The service count here is a deliberately narrow model. Real queue dynamics include cancellations ahead and behind, partial executions, changes in price, and possible priority classes. A cancellation behind the order does not bring it closer to service. Aggregate depth alone cannot distinguish these events. The queue-reactive approach of [Huang et al. \(2015\)](#) motivates estimating event intensities conditional on state instead of treating each price level as an independent reservoir with constant turnover.

Canceling and replacing an order can improve price while surrendering an existing place in the queue. If V^{keep} and V^{replace} include the resulting queue states, replace only when

$$V^{\text{replace}} + c^{\text{message}} < V^{\text{keep}}. \quad (24.4)$$

An optimization that ignores the queue state effectively assigns its option value to zero.

25 NYSE and Nasdaq as allocation mechanisms

Let $\mathcal{K}_v$ denote a venue’s joint kernel for acceptance, queue/allocation changes, fills, execution prices, and reports. Different destinations can expose identical best quotes and charge identical fees while having different $\mathcal{K}_v$. The venue index belongs in the dynamics and the feasible action set.

Nasdaq’s continuous algorithm ranks better prices first and then distinguishes displayed and non-displayed interest at an equal price. Reserve replenishment can create a new displayed order when the residual drops below a normal trading unit; the old residual and newly displayed clip need not have the same priority ([Nasdaq, 2026d](#)). These mechanics require more detail than a rule that empties one aggregate bucket and refills it once per five-second observation.

NYSE’s published explanation describes setter priority followed by parity allocation. Its FAQ states that eligible setter priority may execute the full displayed quantity before parity allocation. This is not a size-weighted pro-rata rule, nor is it the same as a single FIFO queue across all participants ([New York Stock Exchange, 2026j,e](#)). The electronic book, individual Floor Brokers and DMM participation enter the allocation structure; an upstairs electronic order is not automatically a separate Floor Broker parity participant. NYSE also states that its public feeds do not carry a parity identifier ([New York Stock Exchange, 2026e](#)). Consequently, public depth alone is insufficient to assign an exact allocation state to a hypothetical child order.

The model therefore uses different execution kernels,

$$\mathcal{K}_{\text{XNAS}}(\cdot \mid B, z, b, a, r) \quad \text{and} \quad \mathcal{K}_{\text{XNYS}}(\cdot \mid B, z, b, a, r).$$

The same abstract cost function can call either kernel, but an estimated FIFO fill probability cannot be relabeled as a NYSE parity probability. Where participant state is unavailable, allocation should be represented by a conditional distribution or bounds. The controller’s own orders and execution reports provide information not contained in an aggregate public book.

Sharing the Pillar platform does not make all NYSE-affiliated markets identical ([New York Stock Exchange, 2026k](#)). Restricting the modeled destinations to XNYS and XNAS also does not remove other venues from the national reference market. Outside quotations can still affect eligibility, pegging and routing constraints even when the strategy sends children only to the two modeled destinations.

25.1 Reference prices, fees, and information boundaries

A venue best bid and offer need not equal the national best bid and offer. NBBO-dependent pegging or routing requires a suitable consolidated reference and an explicit synchronization policy. A single Nasdaq MBP-10 feed cannot reconstruct national price protection or the complete opportunity set across venues. Tick increments, round lots, session calendars and symbol definitions must be effective-dated. Nasdaq announced new round-lot definitions beginning in November 2025 and periodic reference updates ([Nasdaq, 2025](#)); a universal 100-share assumption is consequently inappropriate for historical replay.

Fees belong in net execution costs but must be distinguished from liquidity. Membership, volume tiers, flags, routing and clearing can affect net charges. The synthetic fees below are parameters, not a reproduction of a current fee schedule. A broker’s best-execution obligation is broader than selecting the largest rebate: FINRA Rule 5310 requires diligence in seeking favorable terms under the circumstances ([FINRA, 2026](#)). The optimization objective should represent the customer mandate and execution quality, with applicable legal requirements imposed on admissible decisions.

26 Schedules as an outer allocation problem

26.1 Time, volume, urgency and price limits

For a deterministic trading rate $v_t \geq 0$, let $\dot{q}_t = -v_t$, $q_0 = Q$, $q_T = 0$. A time-weighted schedule uses $v_t = Q/T$. A forecast-volume schedule uses

$$v_t = Q \frac{\hat{V}_t}{\int_0^T \hat{V}_s ds}, \quad (26.1)$$

where $\hat{V}_t$ is a volume-rate forecast available when the schedule is chosen. It must not be replaced in a backtest by the realized full-day volume curve. A participation strategy instead updates from observed market volume, for example targeting increments ρdV_t , subject to quantity and price constraints. Participation alone does not guarantee completion if volume is low.

An urgency objective can be written

$$\min_{v \geq 0} \int_0^T \left[\eta_t v_t^2 + \lambda \sigma_t^2 q_t^2 + \alpha_t q_t \right] dt, \quad \dot{q} = -v, \quad q_0 = Q, \quad q_T = 0. \quad (26.2)$$

Here η_t converts squared trading rate into dollars per time, $\lambda \sigma_t^2 q_t^2$ is the specified inventory-risk cost, and α_t is an unaffected expected price drift. A positive drift encourages earlier buying. Spread and linear fees that are constant in time add a fixed multiple of Q and do not change this particular optimum. State-dependent fees and spread can change it.

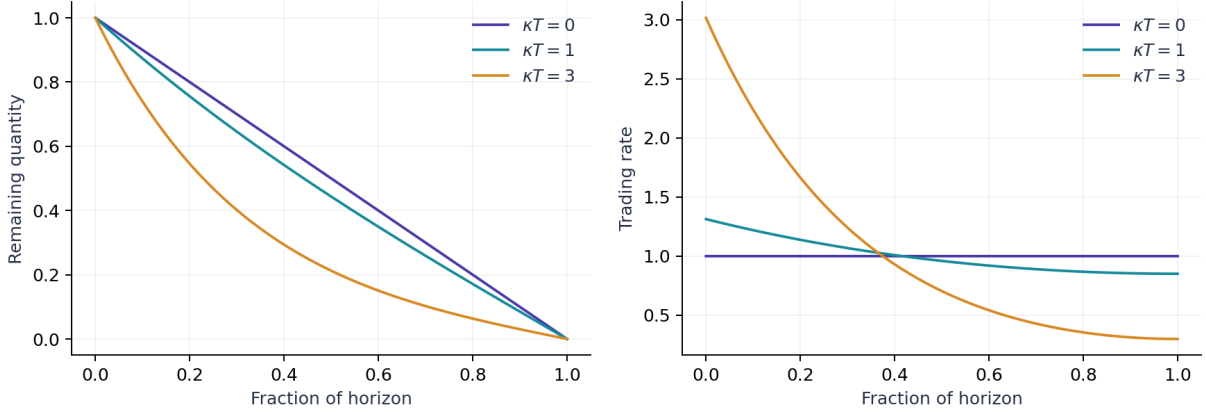


Figure 1: Analytic schedules with $Q = T = \eta = 1$ and three values of κT . Greater exposure aversion moves execution earlier. These are deterministic schedule calculations, not forecasts of available exchange liquidity.

This is the risk-augmented linear-impact problem solved as Proposition 7.1 in Part II; we do not repeat the derivation. For constant η, λ, σ and $\alpha = 0$, its minimiser is

$$q_t = Q \frac{\sinh(\kappa(T-t))}{\sinh(\kappa T)}, \quad v_t = Q \kappa \frac{\cosh(\kappa(T-t))}{\sinh(\kappa T)}, \quad \kappa = \sqrt{\lambda \sigma^2 / \eta}, \quad (26.3)$$

with the continuous limit $q_t = Q(1 - t/T)$ when $\kappa = 0$. Recall from Proposition 5.3 that the permanent component of impact enters this objective only as the additive constant $\frac{1}{2} G_\infty Q^2$ and therefore does not shape the trajectory at all: what (26.3) trades off is temporary cost against inventory risk.

This is the standard quadratic scheduling structure associated with Almgren and Chriss (2001). Its role here is to define an outer demand for execution. Equation (26.3) does not say that a limit order sent at time t will fill at rate v_t . The placement layer must supply a realistic cost of obtaining that rate. Figure 1 shows the increased initial urgency as inventory risk grows.

26.2 Liquidity-dependent scheduling and feasibility

For differentiable η_t and an interior solution, the variational equation becomes

$$(\eta_t \dot{q}_t)' = \lambda \sigma_t^2 q_t + \frac{1}{2} \alpha_t. \quad (26.4)$$

Rate caps and monotonicity require constrained optimization rather than this equation alone. On a discrete grid a convenient convex program is

$$\begin{aligned} \min_{x_0, \dots, x_{N-1}} \quad & \sum_k \left[C_k(x_k) + \lambda \sigma_k^2 q_{k+1}^2 \Delta t_k + \alpha_k q_{k+1} \Delta t_k \right], \\ \text{subject to} \quad & q_{k+1} = q_k - x_k, \quad 0 \leq x_k \leq \bar{x}_k, \quad q_0 = Q, \quad q_N = 0. \end{aligned} \quad (26.5)$$

For deterministic reliable capacities $\bar{x}_k$, feasibility is equivalent to $Q \leq \sum_k \bar{x}_k$. Necessity follows by summing the capacities; sufficiency follows by allocating greedily until Q is exhausted. Observed displayed depth is not a reliable future capacity: it can cancel before the order arrives. In a stochastic model, hard guarantees must be replaced by explicitly assumed capacity, a chance constraint, or a residual mandate.

26.3 Benchmarks are distinct objectives

TWAP, VWAP, participation and implementation-shortfall algorithms are related but not interchangeable. TWAP fixes a clock allocation. Forecast VWAP places more volume in anticipated liquid intervals. Participation follows realized activity and can lag a deadline. An arrival-price optimizer changes speed with impact, drift and risk. An auction-focused strategy may concentrate at a benchmark event. Benchmark tracking can be added through a term such as $\theta \sum_k (q_k - \bar{q}_k)^2$, but this creates a different optimization problem from minimizing arrival-price shortfall alone.

Part VI

Allocation, information and control

27 Joint allocation across venues

27.1 Immediate liquidity and marginal cost

Suppose a scheduler requests x shares immediately and venue v offers a convex total cost $C_v(x_v)$ for taking $x_v \in [0, c_v]$. Costs include price levels reached, specified impact and net fees, expressed relative to a common reference. Define the effective cost

$$C(x) = \min \left\{ \sum_{v=1}^V C_v(x_v) : \sum_v x_v = x, 0 \leq x_v \leq c_v \right\}. \quad (27.1)$$

The problem treats c_v as available at execution time. Price protection or directed-order restrictions must be imposed when applicable; the simple box formula below need not apply with additional routing constraints. Under latency, it is an approximation conditional on those capacities; random availability belongs in a stochastic extension.

Theorem 27.1 (Convex venue aggregation). *Let each C_v be continuous and convex on $[0, c_v]$, with $c_v > 0$. For $0 \leq x \leq \sum_v c_v$, the minimum in (27.1) is attained and C is convex. If the costs are strictly convex, the allocation is unique. For differentiable costs and an interior aggregate quantity, an optimum has a multiplier μ satisfying*

$$\begin{cases} C'_v(x_v) = \mu, & 0 < x_v < c_v, \\ C'_v(0) \geq \mu, & x_v = 0, \\ C'_v(c_v) \leq \mu, & x_v = c_v. \end{cases} \quad (27.2)$$

Such a multiplier is a subgradient of C at x ; where C is differentiable, $C'(x) = \mu$.

Proof. The feasible set is nonempty and compact, so continuity gives attainment. Take feasible optimizers for x and y . Their convex combination is feasible for $\theta x + (1 - \theta)y$, and convexity of each cost bounds its objective by $\theta C(x) + (1 - \theta)C(y)$. This proves convexity of C . Strict convexity rules out two distinct minimizing allocations. The multiplier conditions are the first-order conditions for minimizing $\sum_v C_v(x_v) - \mu(\sum_v x_v - x)$ over the box. Convexity makes them sufficient; standard finite-dimensional multiplier conditions apply because an aggregate quantity strictly between the endpoints admits a point interior to every box interval. Finally, their supporting-line inequalities, summed over venues, give $C(y) \geq C(x) + \mu(y - x)$. $\square$

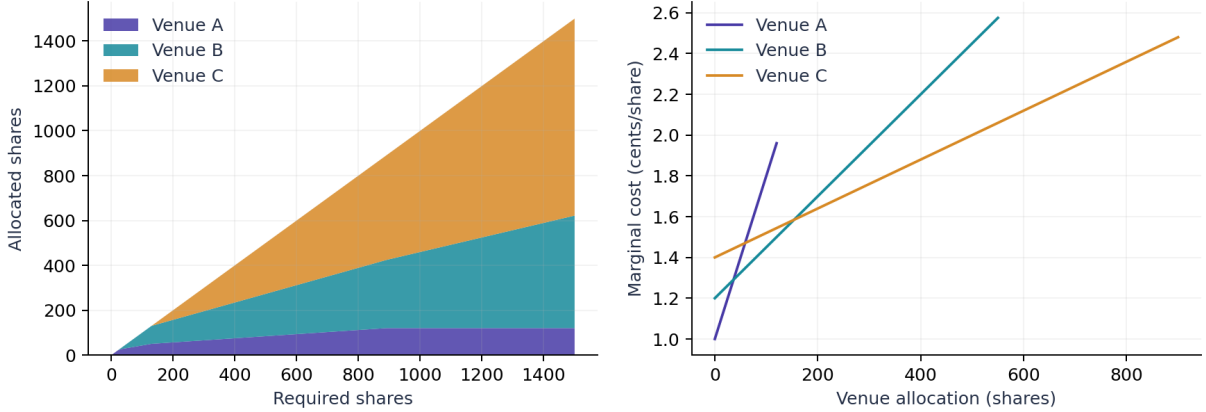


Figure 2: Synthetic immediate allocation among three venues. The left panel shows the allocation as required size increases. The right panel shows marginal costs up to each venue’s capacity. A venue with a low initial cost need not receive most of a large parent slice.

For $C_v(x) = a_v x + b_v x^2/2$, $b_v > 0$, the solution is

$$x_v(\mu) = \text{clip} \left(\frac{\mu - a_v}{b_v}, 0, c_v \right), \quad \sum_v x_v(\mu) = x. \quad (27.3)$$

This is a marginal-cost allocation, not a ranking by the smallest fee a_v . A cheap venue may have rapidly increasing marginal cost or a binding capacity. Piecewise-linear costs obtained by walking observed depth give the analogous discrete allocation, with lot and price-grid constraints retained.

27.2 When decomposition is exact

Substituting C_k from (27.1) into (26.5) preserves convexity. This gives an exact schedule-then-route decomposition under three specific assumptions: contemporaneous venue costs are separable; the chosen allocation has no effect on future states beyond total executed quantity; and all input states/capacities are specified independently of the allocation. The scheduler uses the actual optimized venue cost, not an arbitrary average.

These assumptions fail when a route changes a queue, reveals hidden capacity, creates venue-specific impact, or changes later fill dependence. Then the correct routing comparison contains a continuation term,

$$\min_{a \in \mathcal{A}(q, s, b)} \mathbb{E} [c(s, a, Y) + V_{k+1}(q - F, s', b') \mid s, b, a]. \quad (27.4)$$

Two child-order choices with the same expected fill can have different values because the variance and information in their fills differ.

Example 27.2 (A schedule cannot ignore its effective cost). There are two intervals, total quantity 1, no risk, and effective costs $C_1(x) = x^2$, $C_2(x) = 9x^2$. A flat schedule $(1/2, 1/2)$ costs 2.5. Minimizing $x^2 + 9(1 - x)^2$ gives $(x_1, x_2) = (0.9, 0.1)$ and cost 0.9. Routing optimally after committing to the flat schedule cannot recover the lost timing decision. The example does not identify any actual exchange; it shows the mathematical cost of imposing a schedule before modeling liquidity.

27.3 Correlation and reservations

Fill probabilities at different venues need not be independent. A common price move, institutional sweep, or cancellation burst can affect all of them. If two 80-share child orders are outstanding for a 100-share parent, each having a 0.6 chance of full execution, independence gives a 0.36 probability of a 60-share overfill and expected excess quantity 21.6. The same marginal probabilities permit joint full-fill probability anywhere between 0.2 and 0.6 by the Fréchet bounds. Marginals do not identify overfill risk.

One can penalize overfills in a stochastic allocation problem, as in the broader placement literature (Cont and Kukanov, 2017). A mandate that forbids them instead needs a reservation constraint. Fast cancellation messages do not create an atomic cancellation across independent venues.

Proposition 27.3 (Reservation invariant). *Let q be remaining parent quantity after authoritative reported fills. For each outstanding child, reserve an upper bound r_j on the additional quantity it can execute that is not already included in those reported fills. Reserve quantity before sending a new instruction. Maintain*

$$\sum_j r_j \leq q. \quad (27.5)$$

Assume executions never exceed their child bounds, reported cumulative fills are processed without duplication, and a cancellation releases a reservation only when its terminal cumulative fill is authoritative. Then actual total executions cannot exceed Q .

Proof. At each state, unreported and future executions on outstanding children sum to at most $\sum_j r_j$. Previously reported executions equal $Q - q$. Thus their combined upper bound is $Q - q + \sum_j r_j \leq Q$. A new submission preserves the inequality by construction. An authoritative partial fill of size d decreases both q and that child’s bound by d . A terminal cancellation removes only quantity that can no longer execute. Induction over processed events preserves the invariant. $\square$

Orders pending acceptance or cancellation still consume capacity if they may execute. A replace may require reserving both old and new exposure until the venue’s atomic semantics are known. A disconnected route cannot be assumed canceled. These facts belong in the state transition and in the optimizer’s action filter, not merely in a post-trade error report.

28 Hidden liquidity and execution information

28.1 What is observed and what is inferred

Let $H_t(p, v)$ be non-displayed resting liquidity and $R_t(p, v)$ iceberg reserve at a fixed price and venue. Neither variable is the inventory held by a financial institution. A dealer can hold a large position without offering it for execution; a reserve order is a trading instruction, not a complete balance sheet.

Displayed books, public prints, private fills, and order acknowledgments are different observations of these latent states. Public data may be unable to identify the hidden side or reserve identity. In Nasdaq’s native non-cross trade message, the order-reference field is zero and the buy/sell field is fixed to B, irrespective of the resting side. Separate messages report cross trades (Nasdaq, 2026a). Neither the fixed side field nor an auction print identifies continuous hidden buy liquidity. Message type and session state must be retained before assigning uncertain sides or updating hidden-stock estimates.

A simple execution-count model illustrates non-identification:

$$E_t \mid H_t, \kappa_t \sim \text{Poisson}(\kappa_t H_t \Delta t). \quad (28.1)$$

For $c > 0$, replacing (H_t, κ_t) by $(cH_t, \kappa_t/c)$ leaves this likelihood unchanged. Counts identify an activity product, not the two factors separately. Validated stock dynamics, multiple exposures, private information or stronger assumptions are needed to learn more.

The Poisson likelihood is a reduced activity model, not a stock-conserving execution law: it assigns positive probability to more executions than a fixed finite starting stock. With integer H_t , no arrivals, replenishment, cancellations or other competing exits, and independent per-share execution hazards, a compatible depletion law is instead

$$E_t \mid H_t, \kappa_t \sim \text{Binomial}(H_t, 1 - e^{-\kappa_t \Delta t}), \quad H_{t+1} = H_t - E_t.$$

Its small-interval mean is approximately $\kappa_t H_t \Delta t$, but its full likelihood does not retain the Poisson model's exact scaling invariance. This does not identify a hidden stock from one observed count; inference still depends on repeated observations, the hazard model, and assumptions about arrivals, cancellation and replenishment. Observed replenishment, cancellations and changing sides must enter the transition jointly; clipping an independently sampled execution after the fact changes the likelihood used for filtering.

The hidden-stock balance at a price should distinguish arrivals A^H , cancellations C^H , executions E^H and transfers. An iceberg balance uses

$$\begin{aligned} D_{t+1} &= D_t + A_t^D - C_t^D - E_t^D + U_t, \\ R_{t+1} &= R_t + A_t^R - C_t^R - E_t^R - U_t, \end{aligned} \quad (28.2)$$

where U_t is replenishment into display. It cancels in the combined share balance. Multiple such events can occur between two snapshots. The observed endpoint does not determine their sequence, queue priority or gross turnover.

Example 28.1 (A reserve transfer with residual priority). For an explicitly assumed 100-share normal unit, start with 200 displayed shares and 600 reserve shares. A 150-share execution leaves 50 displayed shares. Replenishing with a new 200-share displayed clip leaves 400 in reserve. The resulting 250 displayed shares comprise the old 50 and the new 200; their timestamps differ. The ledger is $250 + 400 + 150 = 800$ shares. This scenario illustrates the reserve rule; it does not infer the round lot or actual reserve of any particular stock ([Nasdaq, 2026d](#)).

28.2 Beliefs must condition on the trader's actions

Let s_t be observable state, h_t latent state, and $b_t(h)$ its posterior given the observation history and the trader's past instructions. With transition $P(h' \mid h, s, a)$ and observation likelihood $L(y \mid h', s, a)$, the next posterior is

$$b_{t+1}(h') = \frac{L(y \mid h', s, a) \sum_h P(h' \mid h, s, a) b_t(h)}{\sum_{\tilde{h}} L(y \mid \tilde{h}, s, a) \sum_h P(\tilde{h} \mid h, s, a) b_t(h)}. \quad (28.3)$$

Zero-probability observations require model repair or an explicit fallback, not division by zero. A fill and a non-fill can both update liquidity beliefs. Their informativeness depends on where, how much and how long the trader posted. Private fills should not be treated as passive observations generated independently of the policy.

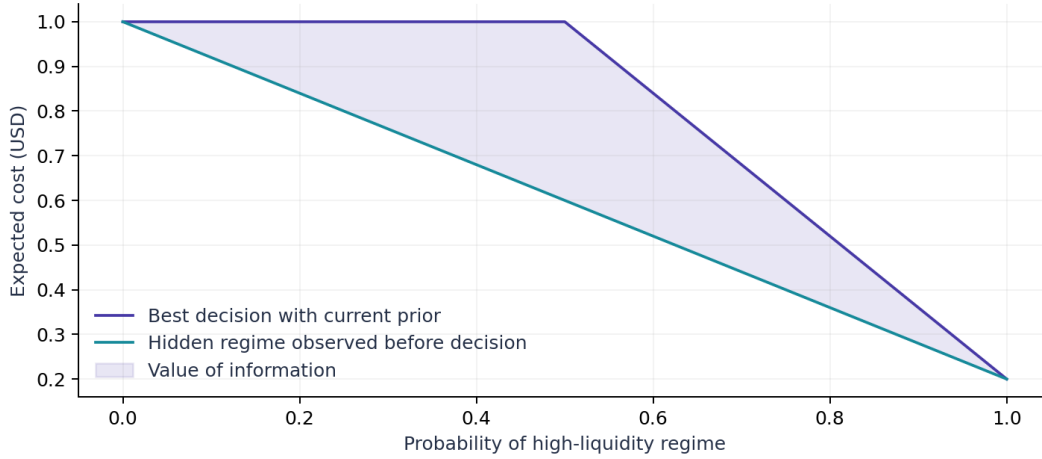


Figure 3: The exact information-value example of Proposition 28.2 over different priors. The shaded gap is positive even though the price process is independent of hidden regime. Information is assumed free and observed before choosing the order.

In an event simulator, executed shares are realized outcomes. A model trained on completed execution volume does not thereby learn unfilled aggressive demand. Interpreting its output as a request and clipping it against a guessed single-period reserve can manufacture enormous apparent non-fill rates. Such a statistic diagnoses the combined simulator, not an actual market rejection rate.

28.3 Execution value without price-prediction value

Proposition 28.2 (A separation example). *There exists an execution problem in which observing hidden liquidity strictly reduces optimal expected execution cost while leaving every price forecast unchanged.*

Proof. Let a hidden regime $H \in \{0, 1\}$ have equal prior probabilities. Let the entire reference-price process be independent of H , for example a constant price. One unit must be bought. Immediate execution costs 1 relative to the reference. Alternatively, submit a passive order that fills at cost zero with probability p_H , where $p_0 = 0.1$ and $p_1 = 0.9$. Failure leads to terminal completion at cost 2. Conditional passive costs are 1.8 and 0.2. Without regime information the passive expected cost is 1, so the optimum cost is 1. With H observed, execute immediately in regime 0 and passively in regime 1, at mean cost $(1 + 0.2)/2 = 0.6$. The information saves 0.4. Independence implies that conditioning any price forecast on H leaves its distribution unchanged. $\square$

This counterexample does not establish that hidden liquidity never predicts prices. It establishes that predictive and execution value are different properties. Conversely, a variable can affect real execution while its noisy estimate adds no useful information beyond observed depth and flow. The correct empirical question is whether a specified observation or estimate improves a specified decision at a specified horizon.

29 From order flow and clearing to execution kernels

A joint execution law can be constructed from primitive order arrivals, cancellations and a matching rule. [Cont et al. \(2025\)](#) separate stochastic order flow from deterministic clearing and factor the order-book generator through these two operations. Their construction supplies the starting point here. Execution control additionally needs order ownership, priority, cash charges and private reports: two books with the same displayed depth can give a particular trader different fills. This section retains those marks, derives the joint kernel used in [Section 30](#), obtains a propagator as a stationary linear response, and carries primitive rate error into a policy certificate. Two further subsections make the resulting certificates computable: [Section 29.7](#) encloses the partially observed optimum for binary hidden liquidity, and [Section 29.8](#) certifies a controller that discards reports against the richer-history optimum, in cost and in action. The bounded finite-state assumptions are explicit; the results do not assert a scaling limit or identify an empirical kernel.

29.1 Clearing with ownership and a conserved parent ledger

Fix a finite price grid and integer share units. At each price and side, store an ordered queue of orders, retaining remaining size and an ownership tag. Let S be a finite structural state space containing these queues, the parent residual q , live and pending own exposure, and any finite regime or session variables needed to determine event rates and reports. Finite queue capacities and a finite parent bound make this possible; continuous ages, impact coordinates or latent variables require an explicitly stated finite approximation if they are included in S . Accumulated cash and the full report history can be recorded as additive marks outside S , provided future rates and admissible actions do not require omitted information.

Consider first continuous matching under price–time priority. An arriving buy consumes the lowest compatible sell price, oldest order first at that price; a sell uses the highest compatible buy price. Each match removes the minimum of the two remaining sizes and executes at the resting order’s price. An accepted limit-order residual joins the back of its queue; an immediate-or-cancel residual expires. A cancellation names an existing order and removes no more than its remaining size. Invalid requests and capacity overflow have explicit rejection reports. Empty books remain possible: a request cannot execute against depth absent from the state. These conventions specify a deterministic map, including each trade and rejection, rather than only a post-clearing depth vector.

Let e be a primitive event mark and a the current control mode. Write

$$\mathfrak{T}^a(s, e) = (T^a(s, e), \omega^a(s, e)), \quad T^a(s, e) \in S. \quad (29.1)$$

The output ω contains the ordered trade list, own fills, execution prices, fees and observable reports. A deterministic observation rule may hide other participants’ tags. Random hidden allocation or reporting can be represented by enriching e with the corresponding random mark. Atomic decision-time messages use a separate map $\mathfrak{J}^a$ of the same kind. Non-atomic messages instead enter the pending-message state and take effect when their acknowledgment or execution events arrive.

Proposition 29.1 (Clearing and parent conservation). *Starting from an uncrossed book, the price–time construction terminates and returns an uncrossed book with nonnegative queue sizes. Write $B^\pm$ for total resting shares, $A^\pm$ for accepted arrivals, $C^\pm$ for cancellations, $D^\pm$ for expired incoming residuals, and V for matched volume during one update. Then*

$$B^{\pm'} = B^\pm + A^\pm - C^\pm - D^\pm - V. \quad (29.2)$$

Each own fill of F shares reduces the parent residual by F and increments cash by $\sum_j f_j p_j + \text{fees}$. The event and decision maps preserve the reservation invariant of Proposition 27.3 with $R = \sum_j r_j$; in particular the parent cannot overfill.

Proof. Every match exhausts the arriving residual or at least one finite resting order, so the matching loop terminates. It leaves no compatible resting counterparty for a residual that is posted, and a cancellation cannot create a crossed book. Each match removes the same volume from the two sides; arrivals, cancellations and expiry give (29.2). The parent statements are the event-by-event bookkeeping in the proof of Proposition 27.3, applied to the tagged own orders. $\square$

This is a continuous price–time matching specialization. The NYSE parity, reserve and auction mechanisms in Sections 25 and 33 require their own $\mathfrak{T}$, ownership categories and priority state. The operator construction below remains applicable once that map is specified; the FIFO example does not implement those additional rules.

29.2 The controlled generator and the joint law

Let E be a finite mark set and let $\nu^a(s, e) \geq 0$ be its primitive event intensities. Assume, uniformly over admissible states and actions,

$$\lambda^a(s) := \sum_{e \in E} \nu^a(s, e) \leq \Lambda, \quad 0 < \Lambda < \infty. \quad (29.3)$$

The intensities may depend on own displayed orders, inventory, regime and past activity retained in s . They are not required to be exogenous to execution. At deterministic times $0 = t_0 < \dots < t_N = T$, a policy chooses an admissible action from its observed history; apply $\mathfrak{J}^a$ and hold its control mode until the next decision. Finitely many interventions exclude an accumulation of own messages at one time.

Proposition 29.2 (A marked controlled book). *The common bounded rate (29.3), the event maps (29.1), and the decision maps define a nonexplosive controlled process and its joint execution/report law. Between decisions its state generator, acting on functions $f : S \rightarrow \mathbb{R}$, is*

$$(\mathcal{L}^a f)(s) = \sum_e \nu^a(s, e) \{f(T^a(s, e)) - f(s)\}. \quad (29.4)$$

For an additive event charge $c^a(s, e)$, the corresponding cost-plus- continuation term is

$$\sum_e \nu^a(s, e) \{c^a(s, e) + f(T^a(s, e)) - f(s)\}. \quad (29.5)$$

State-preserving events can still carry a charge or a report and therefore cannot in general be discarded from the marked model.

Proof. Use a Poisson clock of rate Λ . At a clock time in state s , draw a mark from $E \cup \{\partial\}$ with probabilities

$$\mu_s^a(e) = \frac{\nu^a(s, e)}{\Lambda}, \quad \mu_s^a(\partial) = 1 - \frac{\lambda^a(s)}{\Lambda}. \quad (29.6)$$

The dummy mark ∂ leaves state unchanged, has zero charge and gives no observable report. Apply $\mathfrak{T}^a$ at every other mark. There are almost surely finitely many clock times on $[0, T]$; the recursion therefore constructs the process and all its marks. The probabilities of one and two or more clock events in a short interval are $\Lambda dt + o(dt)$ and $o(dt)$, respectively. Conditioning on the first mark gives (29.4) and (29.5). $\square$

The state part is precisely the order-flow/clearing factorization. If $R^a(s, e)$ denotes the intermediate raw update, $\mathcal{C}$ clears it, $\tilde{\mathcal{C}}f = f \circ \mathcal{C}$, and Ξ restricts a function to valid states, then $T^a = \mathcal{C} \circ R^a$, $\mathcal{C}(s) = s$ on S , and

$$\mathcal{L}^a = \Xi \mathcal{L}_o^a \tilde{\mathcal{C}}. \quad (29.7)$$

On a valid state, the raw generator is $\mathcal{L}_o^a g(s) = \sum_e \nu^a(s, e) \{g(R^a(s, e)) - g(s)\}$; any generator extension to the other raw states gives the same restricted identity. The raw space retains the incoming-message data needed to determine priority and the resting execution price. Substituting $f \circ \mathcal{C}$ in the raw generator proves (29.7), following the decomposition of Cont et al. (2025). The additional marked pushforward, rather than this state identity alone, provides the execution ledger and observation likelihood.

For a fixed mode, let P^a be the state transition matrix of one uniformization clock. Its exact interval law is

$$e^{h\mathcal{L}^a} = e^{-\Lambda h} \sum_{n=0}^{\infty} \frac{(\Lambda h)^n}{n!} (P^a)^n. \quad (29.8)$$

To retain cash and reports, compose the *marked* updates along those n clock events before taking the desired marginal. Replacing the tail $n > M$ by any declared fallback trajectory gives a probability law whose total-variation distance from the exact interval law is at most $\mathbb{P}\{\text{Poisson}(\Lambda h) > M\}$. Clearing after every event is essential: combining several arrivals before clearing can change own priority, executions and reports.

Given the full state s , this construction produces a joint interval law $\mathcal{Q}_k^a(s', y \mid s)$, with y retaining the observed reports and own trade records. If only a history H_k is observed, let $b_k(s)$ be the conditional distribution of the full state. Then

$$\begin{aligned} \mathcal{K}_k(s', y \mid H_k, a) &= \sum_s b_k(s) \mathcal{Q}_k^a(s', y \mid s), \\ b_{k+1}(s') &= \frac{\sum_s b_k(s) \mathcal{Q}_k^a(s', y \mid s)}{\sum_{s, s''} b_k(s) \mathcal{Q}_k^a(s'', y \mid s)}. \end{aligned} \quad (29.9)$$

The posterior formula applies when the denominator is positive. Observable state components are included in y ; the corresponding inconsistent states receive zero posterior mass. Marginalizing $\mathcal{K}_k$ gives the fills, price records, next book, live orders and session variables in Section 30. Thus the same matching calculation generates the fill distribution, its associated prices, and the information used in the next decision.

A proposed reduced state $z = \rho(s)$ is sufficient if, whenever $\rho(s_1) = \rho(s_2)$, both states have the same admissible actions and the one-clock laws of $(\rho(s'), \omega)$ agree under each action. Induction through (29.8) then gives a Markov control reduction; the decision maps must obey the same condition. Aggregate displayed depth usually fails this test. With one own unit O and one external unit E at the best bid, queues (O, E) and (E, O) have equal depth. One incoming unit sell fills the parent only in the first queue. A reduced state must retain priority information or a posterior over it.

29.3 A propagator obtained from the book generator

The full controlled book generally has neither a linear price response nor a quadratic execution cost. A precise connection is available at a stationary small-intervention limit. In this subsection take a finite environmental book model, without a depleting parent inventory, with baseline generator L_0 and stationary row distribution π . Let $m : S \rightarrow \mathbb{R}$ be a bounded reference-price observable with an

explicit convention for empty sides. Consider the generator family

$$L_\epsilon(t) = L_0 + \epsilon u(t)B, \quad B\mathbf{1} = 0,$$

where u is a deterministic integrable signed input and this family consists of valid generators for the amplitudes under consideration. It can arise by differentiating primitive arrival intensities with respect to a trading intervention while retaining a fixed clearing map. This is an assumption about causal response, not a consequence of fitting an order-flow correlation.

Proposition 29.3 (Stationary book response). *Start with distribution π , set $m_0 = m - (\pi m)\mathbf{1}$, and let $b = \|B\|_{\infty \rightarrow \infty}$. Then*

$$\begin{aligned} \mathbb{E}_\epsilon[m(S_t)] - \pi m &= \epsilon \int_0^t G_{\text{book}}(t-s)u(s) \, ds + R_\epsilon(t), \\ G_{\text{book}}(r) &= \pi B e^{rL_0} m_0, \\ |R_\epsilon(t)| &\leq \frac{\epsilon^2 b^2 \|m_0\|_\infty}{2} \left(\int_0^t |u(s)| \, ds \right)^2. \end{aligned} \tag{29.10}$$

Proof. Let $U_\epsilon(0, t)$ be the inhomogeneous transition matrix. Variation of constants gives

$$U_\epsilon(0, t) = e^{tL_0} + \epsilon \int_0^t U_\epsilon(0, s)u(s)B e^{(t-s)L_0} \, ds.$$

Multiply by π and m_0 and replace $U_\epsilon(0, s)$ in the integral once by e^{sL_0} plus its variation-of-constants remainder. Stationarity gives the displayed first-order kernel. Every transition matrix is a contraction in the supremum norm. The second-order remainder is bounded by $\epsilon^2 b^2 \|m_0\|_\infty$ times $\int_0^t |u(s)| \int_0^s |u(r)| \, dr \, ds$, which is one half of the squared integral in (29.10). $\square$

Example 29.4 (An exact exponential response). Let the reference-price state take values $-\Delta$ and Δ . The upward and downward transition rates are $\kappa + \epsilon\beta u(t)$ and $\kappa - \epsilon\beta u(t)$, respectively, with $\kappa, \beta, \Delta > 0$ and $|\epsilon\beta u(t)| \leq \kappa$. Start from the symmetric stationary law. Its mean $M(t)$ solves

$$M'(t) = -2\kappa M(t) + 2\epsilon\beta\Delta u(t), \quad M(0) = 0.$$

Consequently $G_{\text{book}}(r) = 2\beta\Delta e^{-2\kappa r}$ and the convolution in (29.10) is exact. This is a stipulated two-state price model. It supplies an explicit exponential reduction, not a claim that an arbitrary tagged book has two states or exponential resilience.

For a general finite generator, $\pi B e^{rL_0} m_0$ need not be a positive mixture of exponentials: modal coefficients may have either sign, and nonreversible dynamics can have nonreal modes. Thus round-trip nonnegativity of the symmetric energy formed from this response remains a separate condition in Sections 6 and 14. Moreover, the proposition concerns a mean price under deterministic intervention. Turning it into the execution-cost energy requires additional assumptions about the own trade process, the block convention and the reference-price ledger. It does not justify replacing random action-dependent fills by their mean.

When ω already records realized execution prices from the book, mechanical depletion is already in the cash ledger. Adding the cost generated by the same fitted G_{book} would count that effect twice. A separate impact coordinate must either replace a declared component of the price law or represent an explicitly additional response, such as a shift of the external reference price. Its effects on future event rates and observations must then enter $\mathfrak{T}$ and ν , as well as the cost convention.

29.4 Primitive model error and execution regret

Consider two models on the same full state, action and primitive-mark spaces, with the same initial law, matching/report maps and decision-time maps. Let their intensities be ν and $\hat{\nu}$, both dominated by the common clock Λ . Use the total-variation convention $\text{TV}(P, Q) = \sup_A |P(A) - Q(A)|$. Define

$$\begin{aligned} \eta &= \sup_{s,a} \text{TV}(\mu_s^a, \hat{\mu}_s^a), & d &= \Lambda\eta, \\ d &= \frac{1}{2} \sup_{s,a} \left\{ \sum_e |\nu^a(s, e) - \hat{\nu}^a(s, e)| + |\lambda^a(s) - \hat{\lambda}^a(s)| \right\}. \end{aligned} \quad (29.11)$$

The last term retains the difference in total event rates. Agreement of the conditional event-type proportions alone is insufficient.

Theorem 29.5 (From primitive rates to joint execution paths). *For every common nonanticipating observed-history policy π , the two joint laws of full state paths, own trades and observed reports through time $t \leq T$ satisfy*

$$\text{TV}(\mathbb{P}_\pi^{[0,t]}, \hat{\mathbb{P}}_\pi^{[0,t]}) \leq \beta_t := 1 - e^{-dt} \leq dt. \quad (29.12)$$

In particular, a common deterministic matching/report map cannot increase the one-clock primitive error η .

Proof. Use the same Poisson clock, initial full state and policy randomization. While states and observed histories agree, the policy selects the same action. Couple the two primitive marks maximally at each clock time. Their conditional mismatch probability is at most η , uniformly over the common state and action. A matched primitive mark has identical state, execution and report outputs; identical interventions also preserve equality. Conditional on n clock times, the probability of no mismatch is at least $(1 - \eta)^n$. Averaging over the Poisson count gives at least $e^{-\Lambda\eta t}$. The coupling inequality proves (29.12). The one-clock statement is the same argument with a single mark, or the contraction of total variation under a pushforward. $\square$

The policy in this theorem is the same map from observable history to actions in both environments. In particular, a learned filter is held fixed when evaluating the learned policy in the true environment. The theorem does not require posterior distributions computed by the two different models to be uniformly close after every rare observation. Such a conditional claim would need extra assumptions. This direct full-path coupling is therefore useful even when a uniform history-by-history conditional bound of the form in [Corollary C.1](#) is unavailable. If the initial full-state laws instead have total-variation distance δ_0 , maximally coupling them first replaces β_t by $1 - (1 - \delta_0)e^{-dt}$.

If the matching or reporting maps differ, their discrepancy must also be charged. For finite marks a sufficient one-clock replacement is

$$\eta_{\text{eff}} = \min \left\{ 1, \sup_{s,a} \left[\text{TV}(\mu_s^a, \hat{\mu}_s^a) + \sum_e \hat{\mu}_s^a(e) \mathbf{1}_{\{\mathfrak{T}^a(s,e) \neq \hat{\mathfrak{T}}^a(s,e)\}} \right] \right\}. \quad (29.13)$$

Indeed, compare the pushforward of μ through $\mathfrak{T}$ first with that of $\hat{\mu}$ through the same map, then change the map. Theorem 29.5 applies with $d = \Lambda\eta_{\text{eff}}$ by coupling the joint outputs. Decision-time maps are still required to agree; differences at deterministic interventions need their own coupling error. Numerical truncation of (29.8) likewise contributes its Poisson-tail probability. A finite capacity rule is an assumption of the model, not an error-free approximation to unlimited depth.

Corollary 29.6 (A combined execution certificate). *Let $C(\omega)$ and $\widehat{C}(\omega)$ be the true and approximate total loss functionals on a common feasible path space, with*

$$\sup_{\omega} |C(\omega) - \widehat{C}(\omega)| \leq D_{\text{cost}}.$$

Suppose $\widehat{C}$ lies in one common interval of width L over all these paths and policies. Then every fixed policy satisfies

$$|J_{\pi} - \widehat{J}_{\pi}| \leq B_T, \quad B_T = D_{\text{cost}} + L(1 - e^{-dT}). \quad (29.14)$$

Let $\Pi_0 \subseteq \Pi$ be a nonempty common feasible policy class and let $\widehat{\pi} \in \Pi_0$ be τ -optimal for the approximate objective over Π_0 . Then

$$J_{\widehat{\pi}} - \inf_{\pi \in \Pi} J_{\pi} \leq 2D_{\text{cost}} + 2L(1 - e^{-dT}) + \tau + \Delta_{\Pi_0}, \quad \Delta_{\Pi_0} = \inf_{\pi \in \Pi_0} J_{\pi} - \inf_{\pi \in \Pi} J_{\pi}. \quad (29.15)$$

If the only same-path cost error is the single-asset quadratic impact term, the grid and block convention of Proposition 30.1, together with $\|x\|_1 \leq Q$ and entrywise kernel error ε_K , give $D_{\text{cost}} = Q^2 \varepsilon_K / 2$. The within-class bound becomes

$$J_{\widehat{\pi}} - \inf_{\pi \in \Pi_0} J_{\pi} \leq Q^2 \varepsilon_K + 2L(1 - e^{-dT}) + \tau. \quad (29.16)$$

Proof. First change the pathwise loss on a fixed true trajectory, at cost at most D_{cost} . Then change its law while using $\widehat{C}$ on both sides. The expectation of a function with range width L changes by at most L times total variation, proving (29.14). Apply this bound to $\widehat{\pi}$ and to an arbitrarily near-optimal member of Π_0 , and use approximate optimality between the two comparisons. Taking the infimum and adding Δ_{Π_0} proves (29.15). Finally, $|x^\top (K - \widehat{K})x|/2 \leq \varepsilon_K \|x\|_1^2/2$ gives (29.16). $\square$

The range L includes all approximate loss components, including variation in impact cost across different fill paths. This matters when changing a response model changes both prices and the distribution of fills. If its cost decomposes into adapted stage losses of range widths c_k , realized by t_{k+1} , the same proof sharpens the transition term to $\sum_k c_k(1 - e^{-dt_{k+1}})$. A bounded terminal charge can be included in the final stage. Unbounded penalties or uncapped cumulative message fees do not satisfy the stated bounded-loss hypothesis.

The impact norm and the transition metric are distinct hypotheses. A small numerical change in a deterministically reported price can give disjoint observation laws and total variation one. The same-path bound on ε_K therefore does not, by itself, supply a small d for a physically changed price or fill mechanism.

This construction also locates the scope of the signature certificate. Finite signature features may parameterize an admissible history-feedback class Π_0 in the controlled book, and (29.15) then separates model error, optimization tolerance and class approximation. However, the law of a book affected by those coefficients is endogenous. The fixed moment matrix and exact quadratic objective in Theorems 17.3 and 17.8 require their stated exogenous-path assumption. They do not extend to this endogenous case merely by using signature features. The book construction supplies the controlled law; a separate argument is required for an exact reduced optimization objective.

29.5 Feasible feedback policies and a vanishing class gap

The class term in (29.15) measures the restriction to a chosen family of feasible feedback policies. A complete observable history encoding and approximation in the laws induced by successive decisions give a vanishing class gap. This result concerns the richness of the policy class; the endogenous objective need not be quadratic or convex.

Assumption 29.7 (Finite controlled book and observable feasibility). Use the finite structural state and mark spaces of [Proposition 29.2](#), a common clock $0 < \Lambda < \infty$, and $K \geq 1$ deterministic decision times $0 = t_0 < \dots < t_{K-1} < t_K = T$. The action set A is finite. At stage k the nonempty admissible set $A_k(H) \subseteq A$ is a Borel function of the trader's full observable history. Its definition includes the ledger and terminal-feasibility requirements; an absorbing completed state has a no-operation action. The initial observation takes values in a finite alphabet. Each clock event produces either no report or one finite-alphabet report bundle, and each decision produces a finite-alphabet record of the chosen action and its observable intervention output. A bundle retains the ordering of its constituent reports and all observed ledger changes. The total loss C is measurable and lies in a common interval of width $L < \infty$ over feasible paths and policies. All policies are behavioral probability kernels given the observable history, including their own past actions.

Finite alphabets are explicit modeling restrictions on prices, sizes, reports and initial observations. Finite structural states alone would not imply a bounded total loss if, for example, uncapped message fees accumulated at every clock event. Nonempty observable masks and admissible terminal rules are also assumptions; a mask does not manufacture missing liquidity.

Replacing one decision at a time

Write Π for these feasible randomized policies and $J_\pi = \mathbb{E}_\pi C$.

Lemma 29.8 (Policy perturbation). *For $\pi, \pi' \in \Pi$, let $\pi^{(j)}$ use $\pi'_0, \dots, \pi'_{j-1}$ followed by $\pi_j, \dots, \pi_{K-1}$, and put*

$$\delta_j = \mathbb{E}_{\pi^{(j)}} [\text{TV}(\pi_j(\cdot | H_j), \pi'_j(\cdot | H_j))]. \quad (29.17)$$

Under [Assumption 29.7](#),

$$|J_\pi - J_{\pi'}| \leq L \sum_{j=0}^{K-1} \delta_j. \quad (29.18)$$

Proof. Consecutive hybrids have identical rules before t_j , so the law of H_j is the law in (29.17). Couple their initial states, clock events and primitive marks up to t_j , then couple their action draws maximally conditional on H_j . Their later decision rules are the same. Conditional on a matched action, the remaining paths, ledgers and reports can therefore coincide. The full-path total variation is at most δ_j , so the expected losses differ by at most $L\delta_j$. Sum from $\pi^{(0)} = \pi$ to $\pi^{(K)} = \pi'$. $\square$

The histories in this lemma have the law generated by the already replaced earlier decisions. Thus no likelihood-ratio comparison between different policies, or continuity of the controlled value function, is required.

A complete observable history on a compact set

Let M_T count the uniformization clock events on $[0, T]$. Its law is Poisson with mean ΛT , independently of which policy is used, and

$$\rho_n = \mathbb{P}\{M_T > n\} \downarrow 0. \quad (29.19)$$

The clock and its dummy marks are auxiliary objects, not observations used by the policy. Encode only the actual observable transcript: first the initial observation, then the timestamped report bundles and the trader's own decision records, in their known order. Use a disjoint finite alphabet $\mathcal{B}$ for these record types. Coincident timestamps retain their record order. Immediately before

decision k , the transcript has length $m \leq n + K + 1$ on $\{M_T \leq n\}$; the additional records account for the initial observation and own interventions, which do not consume clock events.

For records $(\tau_i, b_i)_{i=1}^m$, with $\tau_1 = 0$ for the initial observation, define the path Z_k on $u \in [0, 1]$ by linear interpolation from the common origin through the m nodes

$$Z_k(0) = 0, \quad Z_k(i/m) = \left(i/m, \tau_i, \sum_{j=1}^i e_{b_j} \right), \quad 1 \leq i \leq m. \quad (29.20)$$

Here e_b is the one-hot vector for record b . In particular $m \geq 1$, even when no clock report occurs. This encoding records a varying initial observation as an increment from a fixed origin; it does not append it as a constant coordinate. Policies and masks are Borel maps on the transcript space; their values on impossible transcripts can be assigned arbitrarily.

Lemma 29.9 (Compactness and separation of observable histories). *For fixed n, k , let $\mathcal{H}_{n,k}$ contain the lifts (29.20) with $1 \leq m \leq n + K + 1$, weakly ordered timestamps in $[0, t_k]$, and letters in $\mathcal{B}$. This is compact in 1-variation. The lift is observable and injective on transcripts, every truncated signature is continuous on $\mathcal{H}_{n,k}$, and signature-linear functions are uniformly dense in $C(\mathcal{H}_{n,k})$.*

Proof. For fixed m and record string, the ordered timestamp simplex is compact. Its interpolation map is continuous in 1-variation: segment endpoints vary continuously and the number and positions of interpolation segments are fixed. There are finitely many such strings and lengths, so their union is compact. All inputs are observed. The sum of the terminal record coordinates recovers m ; evaluating the path at i/m then recovers each timestamp and each one-hot record increment, proving injectivity. All paths start at zero, and the strictly increasing first coordinate u removes tree-like ambiguity. Signature uniqueness and the shuffle algebra therefore give separation and Stone–Weierstrass density, exactly as in Theorem 17.1; continuity follows in the 1-variation topology. $\square$

Finite observable summary coordinates can also be appended when their node values are determined by the discrete record string. This preserves compactness and separation. For a fully observed finite book, one example is the one-hot current state at each record node, starting at zero before the initial record. Its first-level signature is the current state indicator. Such a channel is unavailable when that state is hidden. In general the index parameterization is part of the specified encoding: signature invariance under reparameterization does not justify dropping record order, initial information, or the trader’s own decisions.

Approximation of scores with an exact admissibility mask

For a nonempty mask $B \subseteq A$, write $\sigma_B(g)$ for the softmax of finite scores $g \in \mathbb{R}^{|A|}$ restricted to B , with zero probability outside B . Along $g + t(h - g)$ its derivative is $p_a\{(h_a - g_a) - \sum_b p_b(h_b - g_b)\}$, and hence

$$\text{TV}(\sigma_B(g), \sigma_B(h)) \leq \min\{1, \|g - h\|_\infty\}. \quad (29.21)$$

This estimate compares scores at one fixed history and its common mask; it does not require the mask to vary continuously between histories.

Let Π^N consist of policies

$$\pi_k^\ell(a \mid H_k) = \frac{\mathbf{1}_{\{a \in A_k(H_k)\}} \exp\langle \ell_{k,a}, \mathbb{S}^{\leq N}(Z_k) \rangle}{\sum_{a' \in A_k(H_k)} \exp\langle \ell_{k,a'}, \mathbb{S}^{\leq N}(Z_k) \rangle}. \quad (29.22)$$

Nonempty masks make the denominator positive. Every such policy is observable and feasible, and $\Pi^N \subseteq \Pi^{N+1}$ by zero padding. Coefficients are finite but have no common norm cap.

Theorem 29.10 (Vanishing feasible signature class gap). *Under [Assumption 29.7](#), with the complete encoding [\(29.20\)](#),*

$$0 \leq \Delta_{\Pi^N} := \inf_{\pi \in \Pi^N} J_\pi - \inf_{\pi \in \Pi} J_\pi \downarrow 0. \quad (29.23)$$

More precisely, for every $\epsilon > 0$, integer $n \geq 0$, $\delta > 0$, $\alpha \in (0, 1)$ and $b > 0$, there are a finite depth N and a policy in Π^N such that

$$\Delta_{\Pi^N} \leq \epsilon + KL(\rho_n + \delta + \alpha + b). \quad (29.24)$$

The depth is existential; the stated hypotheses supply no uniform depth-versus-error or coefficient-norm bound.

Proof. Fix an ϵ -optimal feasible policy π . Construct its replacement forward, one stage at a time. At stage k , earlier replacement rules are already fixed. On $\mathcal{H}_{n,k}$ use the finite Borel subprobability measure

$$\zeta_k(D) = \mathbb{P}_{\pi^{(k)}}\{Z_k \in D, M_T \leq n\}.$$

Conditioning a measure on this auxiliary event introduces no future information into the implemented rule, which remains a function of Z_k .

First floor the target distribution only on its admissible actions: $p_k^\alpha = (1 - \alpha)\pi_k + \alpha \text{unif}(A_k(H))$. Define real-valued scores on every action by

$$g_{k,a}(H) = \begin{cases} \log p_k^\alpha(a | H), & a \in A_k(H), \\ 0, & a \notin A_k(H). \end{cases}$$

These are bounded Borel functions in $[\log(\alpha/|A|), 0]$, and $\sigma_{A_k(H)}(g_k(H)) = p_k^\alpha(\cdot | H)$ exactly. In particular no logarithm of zero is taken.

Apply Lusin's theorem to the finite vector g_k under ζ_k . There is a compact $F_k \subseteq \mathcal{H}_{n,k}$ with $\zeta_k(\mathcal{H}_{n,k} \setminus F_k) \leq \delta$ on which all its coordinates are continuous. By the bounded Tietze extension theorem each coordinate extends continuously to $\mathcal{H}_{n,k}$. The extensions are scores, so they need not satisfy a varying probability-simplex constraint. By [Lemma 29.9](#), approximate all extended scores by signature-linear functions with uniform error at most b , using some finite stage depth N_k . Masking their softmax gives a feasible rule π'_k on every history, including histories outside this compact set.

On F_k , flooring changes the target by at most α in total variation, and [\(29.21\)](#) adds at most b . On its complement total variation is at most one. Thus the mismatch in [\(29.17\)](#) obeys $\delta_k \leq \rho_n + \delta + \alpha + b$. Continue to the next stage under the law generated by the replacement rules already fixed. At the end take $N = \max_k N_k$ and zero-pad the coefficients, which changes none of these rules or laws. [Lemma 29.8](#) and ϵ -optimality give [\(29.24\)](#). Let $n \rightarrow \infty$ and $\epsilon, \delta, \alpha, b \downarrow 0$. Nestedness then proves [\(29.23\)](#) for the full sequence of depths. $\square$

The admissibility mask can be discontinuous. For example, if only a is allowed for a scalar history coordinate $h \leq 1/2$ and only b for $h > 1/2$, there is no globally continuous feasible probability vector. The proof above extends finite scores on a large compact set and retains the mask exactly, so it does not require such a vector. Initial information is equally essential: with an observed fair bit B and loss $\mathbf{1}_{\{a \neq B\}}$, an encoding that omits B has infimum $1/2$ while the unrestricted observed-history infimum is zero. The initial token in [\(29.20\)](#) excludes this failure.

Corollary 29.11 (Conditional convergence of the combined certificate). *Fix one true controlled book and loss. Let approximate models share its state and action spaces, initial law, decision and report*

maps, history encoding, and observable admissibility masks, as required in [Corollary 29.6](#). Suppose the true and approximate losses have uniformly bounded range width L , and use a common Λ, K . Let $\hat{\pi}_j \in \Pi^{N_j}$ have global approximate-objective gap at most τ_j within that class. If $d_j, D_{\text{cost},j}, \tau_j \rightarrow 0$ and $N_j \rightarrow \infty$, then

$$J_{\hat{\pi}_j} - \inf_{\pi \in \Pi} J_\pi \longrightarrow 0. \quad (29.25)$$

For an impact-only same-path error with entrywise kernel error $\varepsilon_{K,j}$ and gross volume at most Q , the error decomposition is

$$J_{\hat{\pi}_j} - \inf_{\pi \in \Pi} J_\pi \leq Q^2 \varepsilon_{K,j} + 2L(1 - e^{-d_j T}) + \tau_j + \Delta_{\Pi^{N_j}}. \quad (29.26)$$

Proof. The common encoding and masks give the same feasible class in both models. Apply (29.15) and then [Theorem 29.10](#) to the fixed true objective. Each term vanishes under the stated hypotheses. $\square$

This is a consistency statement conditional on global optimization tolerances, not an algorithm establishing them. Even a one-decision two-action softmax with costs 0 and 1 gives $J(\ell) = (1 + e^{-\ell})^{-1}$, whose second derivative changes sign. Endogenous signature-softmax optimization is therefore not generally convex. A local optimizer's stationarity residual cannot be substituted for τ_j . Nor does (29.24) make its Lusin defect and depth choice empirically estimable without further regularity and statistical assumptions. Fixed coefficient caps would define a different class and require a separate approximation argument.

Proposition 29.12 (An explicit finite-state softmax approximation). *In a fully observed finite-state Markov specialization with additive stage and terminal losses and a deterministic optimal policy $a_k^*(s)$, append the observable current-state indicator channel described after [Lemma 29.9](#). Set its depth-one logit for $a_k^*(s)$ to $r \geq 0$, and every other admissible logit to zero. For $m_A = |A|$, the resulting policy π^r satisfies*

$$0 \leq J_{\pi^r} - J^* \leq \min \left\{ L, KL \frac{m_A - 1}{e^r + m_A - 1} \right\}. \quad (29.27)$$

Proof. Finite backward induction supplies a_k^* . With $m \leq m_A$ admissible actions, its selection probability is $e^r / (e^r + m - 1)$; the total variation from the deterministic rule is at most $(m_A - 1) / (e^r + m_A - 1)$. Apply [Lemma 29.8](#); the common loss range supplies the cap L . The indicator channel's endpoint is a first-level signature coordinate, so stage- and state-specific logits have exactly the claimed depth. $\square$

This construction requires a finite-state optimization oracle and an observed state. Its explicit approximation parameter is the logit size r , not increasing depth. It gives a computable τ bound for the numerical specialization below; it does not provide a depth rate for the general partially observed transcript model.

29.6 Reference models and the price of an exogenous approximation

The reference model must be specified as a complete controlled law before it can be compared with the true book. Calling a fixed action a reference mode does not establish that its feature process is exogenous.

Proposition 29.13 (Reference-model defect and surrogate regret). *Let $\mathbb{P}_\pi$ and $\mathbb{P}_\pi^0$ have a common initial law, clock bound, full state and action spaces, and common feasible observed-history policies. Allow different marked event maps. Define*

$$\gamma = \min \left\{ 1, \sup_{s,a} \left[\text{TV}(\mu_s^a, \mu_s^{0,a}) + \sum_{e \in E \cup \{\partial\}} \mu_s^{0,a}(e) \mathbf{1}_{\{\mathfrak{T}^a(s,e) \neq \mathfrak{T}^{0,a}(s,e)\}} \right] \right\}. \quad (29.28)$$

At decision k , suppose the two joint intervention-output laws, given the same state and action, have total variation at most κ_k . Identical decision maps have $\kappa_k = 0$. Put

$$\beta_0 = 1 - e^{-\Lambda\gamma T} \prod_{k=0}^{K-1} (1 - \kappa_k).$$

For losses with $\sup_\omega |C(\omega) - C^0(\omega)| \leq D_0$ and C^0 in one interval of width L , every fixed policy satisfies

$$|J_\pi - J_\pi^0| \leq D_0 + L\beta_0. \quad (29.29)$$

For a common feasible coefficient family π^ℓ , $\ell \in \mathcal{L}$, let a computable surrogate F satisfy $\sup_{\ell \in \mathcal{L}} |F(\ell) - J_{\pi^\ell}^0| \leq D_{\text{red}}$. If $\hat{\ell}$ is globally τ -optimal for F over $\mathcal{L}$, then

$$J_{\pi^{\hat{\ell}}} - \inf_{\ell \in \mathcal{L}} J_{\pi^\ell} \leq 2D_0 + 2L\beta_0 + 2D_{\text{red}} + \tau. \quad (29.30)$$

Proof. The one-clock output comparison is (29.13). While the coupled paths agree, the probability of retaining agreement at an intervention is at least $1 - \kappa_k$. Conditional on n clock events, multiply these bounds with $(1 - \gamma)^n$ and average the Poisson count. The full-path mismatch probability is at most β_0 . Change first the pathwise cost and then its law to obtain (29.29). The uniform difference between J_{π^ℓ} and $F(\ell)$ is at most $D_0 + L\beta_0 + D_{\text{red}}$. Apply it to $\hat{\ell}$ and to any near-minimizer of the true objective, inserting the global τ -optimality inequality between them. $\square$

With identical interventions, the same loss, and an exact reduced reference objective, this becomes $2L(1 - e^{-\Lambda\gamma T}) + \tau$. Zero γ is a sufficient agreement condition for positive-probability clock outputs at a common state. It is not a characterization of policy-independent feature laws, nor is this primitive upper bound necessarily zero whenever two different primitive representations have the same output law. Intervention discrepancies and reduction error cannot be discarded.

Example 29.14 (Freezing a mode can leave an endogenous feature law). Take $S = \{0, 1\}$ and $A = \{0, 1\}$. The decision at time zero sets $s = a$. A state-preserving report event has rate $1 + s$ and reports the current state. At a fixed state these intensities and event maps do not depend on the action label, so replacing that label by $a_0 = 0$ gives $\gamma = 0$. Retaining the same decision map nevertheless gives

$$\mathbb{P}_a \{\text{at least one report by } T\} = 1 - e^{-(1+a)T}.$$

Thus the reference feature law still depends on the selected action. The coupling comparison correctly gives equality of true and reference costs for each fixed policy; it does not assert equality across policies. If the reference intervention also resets the state to zero for every action, its discrepancy for action one must instead be charged through $\kappa_0 = 1$. The report indicator is a bounded loss, so this example does not rely on an unbounded fee or queue.

A sufficient condition for an exogenous reference feature process is an autonomous observable projection. Specifically, suppose a projection $Y = \rho(s)$ has, at every reference clock event, a joint law of its next value and its external feature reports that depends only on Y , not on the own-order coordinates or chosen action. Suppose reference interventions preserve Y and emit no action-dependent feature reports, and the initial projected law is fixed. Induction along the common clock and decision times then proves that the projected feature history has a policy-independent law. Any prescribed nonanticipating continuous finite-variation transform of that history has the same independence. The feature process must be maintained on the fixed horizon even after parent completion; stopping it at a policy-dependent completion time would generally reintroduce dependence.

To set $F(\ell) = \ell^\top Q \ell + b^\top \ell + \text{const}$ and $D_{\text{red}} = 0$, one must additionally establish the rate-policy, cost, moment and common-feasibility assumptions of [Theorems 17.3](#) and [17.7](#). Autonomous external prices alone do not make queue-dependent passive fills a quadratic rate cost. If that identity is only approximate, its uniform objective discrepancy belongs in D_{red} . These conditions state exactly when the exogenous quadratic programme has a transferable guarantee; the controlled-book class-gap argument does not require them.

Remark 29.15 (Scope of the approximation result). The bounded clock supplies the uniform Poisson tail, finite records supply compact transcript sets, nonempty observable masks supply exact feasibility, and the bounded loss converts mismatch probability to objective error. Unbounded arrival rates, infinite record alphabets and unbounded penalties require separate tightness or integrability arguments. No extension to those cases, no general depth complexity, and no empirical identification guarantee is asserted by [Theorem 29.10](#) and [Corollary 29.11](#). The finite-state illustration supplies a computable special case under its declared oracle and observability assumptions.

29.7 Computable certificates under partial observation

The class gap in [Theorem 29.10](#) is existential. A finite-observation specialization admits a different, constructive route: compute a lower bound on the unrestricted partially observed optimum and an upper bound on the cost of an implementable controller. Their difference is a global certificate. Belief-state dynamic programming and its piecewise-linear value functions are classical ([Smallwood and Sondik, 1973](#)); computable bounds and finite-horizon point-based planning are developed by [Lovejoy \(1991\)](#) and [Walraven and Spaan \(2019\)](#). The construction below combines those foundations with the execution ledger, an observable finite-memory signature policy, and the model-error certificate of [Corollary 29.6](#).

Assumption 29.16 (Finite observations and a binary hidden state). There are K decision stages, a finite observable state space $\mathcal{X}$, a hidden state $h \in \{0, 1\}$, and finite action and observation alphabets. The nonempty feasible set $A_k(x)$ depends only on the observable state, which retains every ledger, reservation and session variable needed for feasibility. The joint controlled kernel is

$$M_{k,x,a,o}(h, h') \geq 0, \quad \sum_{o, h'} M_{k,x,a,o}(h, h') = 1.$$

An outcome o determines the next observable state $x' = F_k(x, a, o)$ and the stage charge $c_k(x, a, o)$. A terminal charge $c_K(x, h)$ is allowed. Actions may change both the hidden transition and the observation law. For each stage j , all remaining total losses lie in a common interval of width S_j over states and feasible continuation policies. All primitive arrays, charges and these bounds are supplied. The initial state x_0 and belief $b_0 = \mathbb{P}(h_0 = 1)$ are known.

The analytic statements allow real data. Algorithmic claims require rational primitives and grid nodes, or validated enclosures together with certified comparisons for the controller's memory update.

The observation alphabet is the information actually available to this controller. If a marked continuous-time model is reduced to these symbols, discarding timestamps or reports changes the information problem unless sufficiency is proved. No equivalence with a richer observation filtration is assumed. Section 29.8 supplies a separate certificate for that richer comparison. Binary hidden liquidity is the explicit computable case below; it does not exhaust the finite hidden-state setting.

For $b \in [0, 1]$, put $\mathbf{b} = (1 - b, b)$ and define

$$\begin{aligned} p_b^a(o) &= \sum_{h, h'} \mathbf{b}_h M_{k, x, a, o}(h, h'), & \Phi_b^a(o) &= \frac{\sum_h \mathbf{b}_h M_{k, x, a, o}(h, 1)}{p_b^a(o)}, \\ r_k(x, b, a) &= \sum_o p_b^a(o) c_k(x, a, o). \end{aligned} \quad (29.31)$$

Zero-probability outcomes contribute zero and may be assigned any posterior. The optimal value $V_k(x, b)$ minimizes expected remaining cost over all policies using the declared observation history, including their own actions.

Lemma 29.17 (Concavity and a computable belief modulus). *For fixed k, x , $V_k(x, \cdot)$ is concave, piecewise linear and S_k -Lipschitz on $[0, 1]$.*

Proof. A fixed deterministic continuation tree has conditional costs v_0, v_1 , so its expected cost is $(1 - b)v_0 + bv_1$. The common feasible action masks give the same finite family of trees for every b . Their minimum is concave and piecewise linear. The range assumption gives $|v_1 - v_0| \leq S_k$ for every tree; taking minima preserves that Lipschitz bound. Behavioral randomization cannot improve the minimum of these expected costs. $\square$

Two finite backward recursions

Let $G_m = \{0, 1/m, \dots, 1\} \cup \{b_0\}$, and let $Q_m b$ be a nearest node, with a fixed tie rule. Write $I_m f$ for linear interpolation of node values: if $b = (1 - \theta)g_i + \theta g_{i+1}$, then $(I_m f)(b) = (1 - \theta)f(g_i) + \theta f(g_{i+1})$. Adjacent nodes have distance at most $1/m$; inserting the initial prior therefore preserves the mesh estimates without restricting its value. The nearest-node planning problem is the finite Markov decision process

$$\begin{aligned} W_K(x, g) &= (1 - g)c_K(x, 0) + g c_K(x, 1), \\ W_k(x, g) &= \min_{a \in A_k(x)} \left\{ r_k(x, g, a) + \sum_o p_g^a(o) W_{k+1}(F_k(x, a, o), Q_m \Phi_g^a(o)) \right\}. \end{aligned} \quad (29.32)$$

Let $a_k^m(x, g)$ be a minimizing action. Independently compute

$$\begin{aligned} \underline{V}_K^m(x, g) &= (1 - g)c_K(x, 0) + g c_K(x, 1), \\ \underline{V}_k^m(x, g) &= \min_{a \in A_k(x)} \left\{ r_k(x, g, a) + \sum_o p_g^a(o) (I_m \underline{V}_{k+1}^m)(F_k(x, a, o), \Phi_g^a(o)) \right\}. \end{aligned} \quad (29.33)$$

Interpolation here gives a lower relaxation for cost minimization. It is not the deployed controller's state update.

The deployed controller starts at $\gamma_0 = b_0$, selects $a_k^m(x_k, \gamma_k)$, and updates its observable memory by

$$\gamma_{k+1} = Q_m \Phi_{\gamma_k}^{a_k}(o_{k+1}). \quad (29.34)$$

This recursively rounded belief need not equal the true posterior conditional on the entire observed history. The physical hidden state always evolves according to M ; it is never redrawn from γ_k in deployment.

Evaluation in the physical model

For any feasible finite-memory action probabilities $\sigma_k(a \mid x, \gamma)$ and the update (29.34), evaluate on the augmented state (x, γ, h) :

$$\begin{aligned} E_K(x, \gamma, h) &= c_K(x, h), \\ E_k(x, \gamma, h) &= \sum_{a \in A_k(x)} \sigma_k(a \mid x, \gamma) \sum_{o, h'} M_{k,x,a,o}(h, h') \{c_k(x, a, o) \\ &\quad + E_{k+1}(F_k(x, a, o), Q_m \Phi_{\gamma}^a(o), h')\}, \\ U_m(\sigma) &= (1 - b_0)E_0(x_0, b_0, 0) + b_0E_0(x_0, b_0, 1). \end{aligned} \quad (29.35)$$

The hidden coordinate is integrated out offline. Neither the online action nor its memory update observes h . Evaluation may also use a different physical kernel while retaining the planner's filter in the memory update.

Theorem 29.18 (A computable global execution certificate). *Under [Assumption 29.16](#), write $V^* = V_0(x_0, b_0)$. For every feasible controller evaluated by (29.35),*

$$\underline{V}_0^m(x_0, b_0) \leq V^* \leq U_m(\sigma), \quad 0 \leq J_\sigma - V^* \leq \mathcal{G}_m(\sigma) := U_m(\sigma) - \underline{V}_0^m(x_0, b_0). \quad (29.36)$$

All quantities on the right are obtained by finite backward induction. An outward lower enclosure of $\underline{V}_0^m$ and outward upper enclosure of U_m give a valid numerically enclosed certificate. The result holds for any candidate table; its validity does not require a local optimizer to have found a global minimum.

Proof. At K the grid terminal values are exact. If $\underline{V}_{k+1}^m(x, g_i) \leq V_{k+1}(x, g_i)$ at all nodes, concavity from [Lemma 29.17](#) gives $I_m \underline{V}_{k+1}^m(x, b) \leq V_{k+1}(x, b)$ for every belief. Nonnegative observation probabilities and minimization preserve the inequality in the backward step. This proves the lower bound. Conditioning on the actual hidden state and the finite controller memory gives (29.35), hence $U_m(\sigma) = J_\sigma$. Feasibility places this controller in the class defining V^* and proves the upper bound. Subtraction gives the certificate. Replacing either endpoint by an enclosure in the indicated direction can only enlarge it. $\square$

This certificate closes both an optimization gap and a representation gap relative to the full history-policy optimum of this model. Successive mesh values need not yield monotonically improving deployed policies. A stopping rule uses the computed gap of the retained best certified controller, not the difference between two grid values.

Proposition 29.19 (Explicit mesh and numerical optimization bounds). *Set*

$$\varepsilon_m = \frac{1}{2m} \sum_{j=1}^K S_j.$$

For the exact nearest-node minimizer σ^m ,

$$J_{\sigma^m} - V^* \leq 2\varepsilon_m, \quad \mathcal{G}_m(\sigma^m) \leq 3\varepsilon_m. \quad (29.37)$$

If a computed table has global gap at most τ_m in the finite problem (29.32), add τ_m to both bounds. One computable choice is $\tau_m = \sum_k \tilde{e}_k$, where $\tilde{e}_k$ bounds, at every grid state, the selected action's excess over the minimum of its true Bellman action values. Enclosures $\tilde{Q}_k^-(z, a) \leq \tilde{Q}_k(z, a) \leq \tilde{Q}_k^+(z, a)$ give

$$\tilde{e}_k = \max_z \left[\tilde{Q}_k^+(z, \hat{a}_k(z)) - \min_{a \in A_k(z)} \tilde{Q}_k^-(z, a) \right]_+. \quad (29.38)$$

Here the action values use the optimal continuation of the finite model, enclosed recursively by backward induction.

Proof. Nearest rounding changes a binary belief by at most $1/(2m)$ in total variation. For interpolation between neighboring nodes, the mean distance of its two nodes from b is $2\theta(1-\theta)(g_{i+1} - g_i) \leq 1/(2m)$. Using Lemma 29.17 in the two backward recursions gives, at grid nodes,

$$|W_k - V_k| \leq \frac{1}{2m} \sum_{j=k+1}^K S_j, \quad 0 \leq V_k - \underline{V}_k^m \leq \frac{1}{2m} \sum_{j=k+1}^K S_j.$$

The terminal interpolation is in fact exact; retaining S_K is a valid conservative bound.

To compare the deployed controller with the nearest-node planning model, introduce an artificial hidden-state resampling after each observation. Given the preceding artificial history, the pre-resampling posterior is $\Phi_\gamma^a(o)$; replace it by $Q_m \Phi_\gamma^a(o)$. This defines the finite Markov model (29.32). Starting from the physical model, introduce resamplings one at a time in forward order: the j th hybrid has the first j resamplings and no later ones. Before the next replacement, the common prefix therefore has all preceding resamplings, so the stated posterior is its correct conditional law. Any fixed continuation rule, with no later resampling in either compared hybrid, has conditional remaining-cost span at most S_j . Changing this distribution changes its expected cost by at most $S_j/(2m)$. Summing gives a uniform objective difference at most ε_m for every common history policy. This argument compares unconditional policy costs; it needs no uniform Lipschitz bound on Bayes' formula after a rare observation.

The artificial model has the observable sufficient state (x, γ) , so its optimum over all history policies is $W_0(x_0, b_0)$. Comparing the true and artificial optima and their selected policy gives $J_{\sigma^m} - V^* \leq 2\varepsilon_m$. Adding the lower-relaxation defect proves the factor three. A finite-model optimization gap contributes τ_m once. Finally, the finite-horizon performance-difference recursion bounds that gap by $\sum_k \tilde{e}_k$; (29.38) encloses each selected-action excess. $\square$

In the resampling comparison, the continuation uses the same observable memory update and action map in both models. A sampled hidden state is not revealed. The artificial model exists only for the comparison proof; (29.35) evaluates the physical model.

Corollary 29.20 (Finite-memory signature implementation and model transfer). *Append an observable summary channel which starts at zero and at decision k has value $e_{(x_k, \gamma_k)}$. Linear interpolation between its update nodes gives a finite-variation path whose first signature level equals this current one-hot vector. Give $a_k^m(x, \gamma)$ score $\log w$, $w \geq 1$, and every other admissible action score zero, retaining the exact mask. This is a depth-one signature-softmax controller $\sigma^{m,w}$ on the specified augmented observation path. With $m_A = |A|$,*

$$\mathcal{G}_m(\sigma^{m,w}) \leq 3\varepsilon_m + \tau_m + K S_0 \frac{m_A - 1}{w + m_A - 1}. \quad (29.39)$$

Its usually sharper certificate is the directly computed $U_m(\sigma^{m,w}) - \underline{V}_0^m$. If these quantities are computed in an approximate physical model and $|J_\pi - \hat{J}_\pi| \leq B_T$ uniformly over the common feasible policies, then the deployed policy satisfies

$$J_{\sigma^{m,w}} - \inf_{\pi \in \Pi} J_\pi \leq 2B_T + \hat{U}_m(\sigma^{m,w}) - \hat{V}_0^m(x_0, b_0). \quad (29.40)$$

For the hypotheses of [Corollary 29.6](#), one may take $B_T = D_{\text{cost}} + S_0(1 - e^{-dT})$ when S_0 also bounds the approximate loss range.

Proof. The first signature level is the channel's endpoint minus its zero origin. Stage-specific linear coefficients therefore reproduce the table scores. With $m_k(z) := |A_k(z)| \leq m_A$ feasible actions, where $z = (x, \gamma)$, the preferred-action probability is $w/(w + m_k(z) - 1)$. The bound [\(29.39\)](#) uses the monotone worst case $m_k(z) = m_A$. [Lemma 29.8](#) bounds its cost difference from the deterministic table by the last term in that bound. Apply [Proposition 29.19](#). For transfer, insert the approximate objective on both sides of a policy comparison and use [Theorem 29.18](#); the two model discrepancies contribute $2B_T$. The filter and table remain fixed when this policy is evaluated in the true environment. $\square$

The summary is computed from observations, actions and the supplied filter, not from the hidden regime. Its dimension is $|\mathcal{X}||G_m| \leq |\mathcal{X}|(m+2)$, and the policy has $K|\mathcal{X}||G_m|$ preferred-action entries. With at most B outcome branches, the recursions take $O(K|\mathcal{X}||G_m||A|B)$ arithmetic operations for a binary hidden state. The explicit accuracy parameter is the belief mesh and table size. This is not a dimension-free depth bound for the unaugmented transcript signature, nor does it make endogenous coefficient optimization convex. A higher-dimensional hidden-state simplex requires a mesh whose size can grow combinatorially. Unknown primitive-model error still requires evidence; the computed planning gap alone cannot certify an unvalidated market model.

29.8 Certifying the loss from an observable summary

A finite controller may retain fills and a rounded filter while discarding timestamps or finer reports. The lower bound in [Theorem 29.18](#) then concerns the declared finite observations; it need not be a lower bound for an agent observing the richer history. The following construction closes this comparison under explicit conditional-law bounds. It uses the robust dynamic-programming foundations of [Iyengar \(2005\)](#); [Nilim and El Ghaoui \(2005\)](#). The role here is to certify a feasible execution controller against a richer information class, without assuming that its summary is Markov or sufficient.

Assumption 29.21 (Conditional action-law envelopes for an observable summary). There are K decisions under a supplied controlled law on standard Borel history spaces. Let H_k be the full observed history and let $Z_k = \Psi_k(H_k) \in \mathcal{Z}_k$ be a finite summary, updated from the preceding summary, action and new observations. The initial history is fixed. Stage and terminal charges are bounded. The nonempty finite feasible action set satisfies

$$A_k(H_k) = A_k(Z_k).$$

In particular, the summary retains every unresolved reservation, instruction status and session variable needed for feasibility. For every feasible (z, a) , a nonempty compact set $\mathcal{U}_k(z, a) \subset \mathbb{R} \times \Delta(\mathcal{Z}_{k+1})$ is supplied, and contains

$$(r_k(H, a), p_k(H, a)) := (\mathbb{E}[c_k \mid H, a], \mathcal{L}(Z_{k+1} \mid H, a)) \quad \text{whenever } \Psi_k(H) = z. \quad (29.41)$$

Supplied terminal functions l_K, u_K satisfy $l_K(\Psi_K(H)) \leq \mathbb{E}[c_K \mid H] \leq u_K(\Psi_K(H))$. These inclusions hold for the conditional versions of the controlled model on all admissible histories, or almost surely under every feasible policy, including each feasible counterfactual action. Bounds established only under one logging policy do not meet this assumption.

Write $p \cdot v = \sum_{z'} p(z')v(z')$. Define the finite recursions

$$\begin{aligned} L_K &= l_K, & U_K &= u_K, \\ L_k(z) &= \min_{a \in A_k(z)} \inf_{(r,p) \in \mathcal{U}_k(z,a)} \{r + p \cdot L_{k+1}\}, \\ U_k(z) &= \min_{a \in A_k(z)} \sup_{(r,p) \in \mathcal{U}_k(z,a)} \{r + p \cdot U_{k+1}\}. \end{aligned} \tag{29.42}$$

Fix a minimizing action $\sigma_k(z)$ in the upper recursion, with a deterministic tie rule. Compactness and finite action sets ensure attainment. The recursions become numerical algorithms when their extrema can be evaluated or enclosed, as in the two explicit constructions below. Let $V_k^{\text{rich}}(H)$ be the infimum of remaining expected cost over all feasible policies using H and subsequent full observations.

Theorem 29.22 (A certificate against the richer history optimum). *Under [Assumption 29.21](#), for $z = \Psi_k(H)$,*

$$L_k(z) \leq V_k^{\text{rich}}(H) \leq J_k^\sigma(H) \leq U_k(z). \tag{29.43}$$

In particular, with $z_0 = \Psi_0(H_0)$,

$$0 \leq J_\sigma - V_0^{\text{rich}}(H_0) \leq U_0(z_0) - L_0(z_0). \tag{29.44}$$

For any other feasible summary controller ρ , including a randomized finite-score signature controller, replace the minimum in the upper recursion by the action average under $\rho_k(\cdot \mid z)$ and call the result U_k^ρ . Then $J_\rho - V_0^{\text{rich}}(H_0) \leq U_0^\rho(z_0) - L_0(z_0)$. Any independently certified upper evaluation $\bar{E}_\rho \geq J_\rho$ may replace $U_0^\rho(z_0)$.

Proof. At the terminal stage the asserted conditional bounds are the assumption. Suppose the lower bound holds at $k+1$ for the conditional cost of every feasible continuation. At history H , conditional on any chosen action a , the tower property bounds that cost below by $r_k(H, a) + p_k(H, a) \cdot L_{k+1}$. Membership in (29.41) bounds it below by the corresponding infimum, and then by $L_k(z)$. An average over randomized actions preserves the inequality. Taking the infimum over all full-history continuations proves $L_k(z) \leq V_k^{\text{rich}}(H)$, without restricting those continuations to summary policies.

For the selected summary policy, the induction hypothesis and the tower property instead give

$$J_k^\sigma(H) \leq r_k(H, \sigma_k(z)) + p_k(H, \sigma_k(z)) \cdot U_{k+1} \leq U_k(z).$$

The selected action is feasible at every history in the summary fiber, so this policy belongs to the class defining V_k^{rich} . This proves the middle inequality. For a fixed randomized controller, average the actionwise suprema instead. Subtraction and an upper evaluation give the remaining assertions. $\square$

The stagewise envelope choices in (29.42) constitute a rectangular bounding relaxation. A single physical model, or a fixed unknown parameter, need not attain all those choices along one path. The proof requires only conditional inclusion; it does not reset the hidden state or assume that a physical adversary reselects the model at each step. This distinction may make the certificate conservative.

Corollary 29.23 (Exact value sufficiency and model transfer). *If every envelope is the singleton $\{(\hat{r}_k(z, a), \hat{P}_k(\cdot | z, a))\}$ and $l_K = u_K$, then $L = U$ and the summary policy is optimal among all full-history policies. More generally, suppose the envelope certificate is computed in a supplied approximate model and $|J_\pi - \hat{J}_\pi| \leq B_T$ uniformly over the common feasible full-history policy class. Keep the summary update and the candidate ρ fixed in both models. Then*

$$J_\rho - \inf_{\pi \in \Pi_{\text{rich}}} J_\pi \leq 2B_T + \hat{E}_\rho - \hat{L}_0(z_0). \quad (29.45)$$

Proof. Singleton envelopes make both recursions the same Bellman recursion, so [Theorem 29.22](#) forces equality. For transfer, $J_\rho \leq \hat{J}_\rho + B_T$ and $\inf_\pi J_\pi \geq \inf_\pi \hat{J}_\pi - B_T$. Apply the approximate-model certificate and subtract. $\square$

A quantitative bound from conditional model discrepancies

The envelopes themselves can retain state and action dependence. A simpler uniform rate follows when they have the rectangular form

$$\begin{aligned} \mathcal{U}_k(z, a) &= [\hat{r}_k(z, a) - \alpha_k(z, a), \hat{r}_k(z, a) + \alpha_k(z, a)] \times \mathcal{P}_k(z, a), \\ \mathcal{P}_k(z, a) &= \{p \in \Delta(\mathcal{Z}_{k+1}) : \|p - \hat{P}_k(\cdot | z, a)\|_{\text{TV}} \leq \epsilon_k(z, a)\}, \end{aligned} \quad (29.46)$$

where $\alpha_k \geq 0$, $0 \leq \epsilon_k \leq 1$, and $l_K = \hat{g} - \delta_K$, $u_K = \hat{g} + \delta_K$, $\delta_K \geq 0$. Let $\hat{V}$ solve the nominal finite Bellman recursion with $(\hat{r}, \hat{P}, \hat{g})$, and write $\text{span}(v) = \max v - \min v$.

Proposition 29.24 (An explicit observation-reduction error bound). *Define*

$$\begin{aligned} d_K &= \max_z \delta_K(z), \\ d_k &= \max_{z, a} \{\alpha_k(z, a) + \epsilon_k(z, a) \text{span}(\hat{V}_{k+1})\}, \quad B_k = d_K + \sum_{j=k}^{K-1} d_j. \end{aligned} \quad (29.47)$$

For the recursions [\(29.42\)](#),

$$\hat{V}_k - B_k \leq L_k \leq \hat{V}_k \leq U_k \leq \hat{V}_k + B_k, \quad U_0(z_0) - L_0(z_0) \leq 2B_0. \quad (29.48)$$

The envelope recursions are finite linear programs. With rational data they can be solved exactly; certified outward bounds on their extrema also give a valid lower–upper certificate.

Proof. For probability vectors, $|(p - \hat{P}) \cdot v| \leq \epsilon \text{span}(v)$ under the stated total-variation convention. Thus applying either envelope Bellman operator to $\hat{V}_{k+1}$ changes the nominal value by at most d_k . Each operator is monotone and is one-Lipschitz in the sup norm, since its transition vectors have unit mass. Backward induction from the terminal error d_K proves the outer bounds. The nominal pair belongs to every envelope; a second induction gives $L_k \leq \hat{V}_k \leq U_k$. The absolute-value constraints defining a finite total-variation ball have a linear-program representation. Lower bounds on the lower extrema and upper bounds on the selected action’s upper extrema propagate in the required directions. An inexact selected action is evaluated as a fixed controller by [Theorem 29.22](#). $\square$

This bound uses nominal continuation-value spans, not a stability constant for Bayes’ formula. Computing the state-dependent recursions can retain information discarded by the maxima in d_k . Neither refinement substitutes for valid conditional error bounds.

A constructive witness for binary hidden liquidity

Suppose the richer controlled model is hidden Markov with the binary state of [Assumption 29.16](#). A fine report is reduced to a symbol o , and $M_{k,x,a,o}$ is its joint hidden-transition law after summing or integrating the finer reports. The symbol retains the next ledger state $F_k(x, a, o)$ and charge $c_k(x, a, o)$. The summary is $z = (x, \gamma)$ with the fixed update (29.34), driven by these coarse symbols. Assume supplied intervals satisfying, for every admissible full history,

$$\mathcal{I}_k(z) = [b_k^-(z), b_k^+(z)] \subset [0, 1], \quad \mathbb{P}(h_k = 1 \mid H_k) \in \mathcal{I}_k(\Psi_k(H_k)). \quad (29.49)$$

This is a condition on the full posterior, not a claim that it equals the controller's memory γ .

For each $b \in \mathcal{I}_k(z)$ define

$$\begin{aligned} r_b(z, a) &= \sum_o p_b^a(o) c_k(x, a, o), \\ P_b(z' \mid z, a) &= \sum_o p_b^a(o) \mathbf{1}\{z' = (F_k(x, a, o), Q_m \Phi_\gamma^a(o))\}, \\ \mathcal{U}_k(z, a) &= \{(r_b(z, a), P_b(\cdot \mid z, a)) : b \in \mathcal{I}_k(z)\}. \end{aligned} \quad (29.50)$$

The update uses γ , while the physical probabilities use b . The chosen zero-probability update convention is fixed for every symbol, including a symbol impossible under γ but possible under b . Terminal bounds are the minimum and maximum of $(1 - b)c_K(x, 0) + bc_K(x, 1)$ on $\mathcal{I}_K(z)$.

Proposition 29.25 (Endpoint evaluation of posterior envelopes). *Under these hypotheses, (29.50) satisfies [Assumption 29.21](#). Each infimum and supremum in (29.42) is attained at $b_k^-(z)$ or $b_k^+(z)$. For supplied intervals, the backward recursions require $O(K \max_k |\mathcal{Z}_k| |A| |O|)$ arithmetic operations. If $|b - \gamma| \leq \eta_k(z)$ throughout the interval, rectangular envelopes of the form (29.46) are valid with*

$$\alpha_k(z, a) = \eta_k(z) |\bar{c}_1 - \bar{c}_0|, \quad \epsilon_k(z, a) = \min\{1, \eta_k(z) \|P^1 - P^0\|_{\text{TV}}\}, \quad (29.51)$$

where $\bar{c}_h$ and P^h are the conditional mean charge and next-summary law given current hidden state h , z and a . The nominal pair is the mixture at γ . The analogous terminal radius is $\eta_K(z) |c_K(x, 1) - c_K(x, 0)|$.

Proof. The controlled hidden-Markov property makes the conditional coarse outcome law the mixture in (29.31) at the full posterior b . The memory update is already fixed by z, a, o . Therefore both the charge mean and next-summary law in (29.50) are affine in b , proving conditional inclusion and compactness. Pairing with any fixed continuation vector remains affine, so the two endpoints suffice. Each endpoint evaluation sums over at most $|O|$ symbols. Finally, $r_b - r_\gamma = (b - \gamma)(\bar{c}_1 - \bar{c}_0)$ and $P_b - P_\gamma = (b - \gamma)(P^1 - P^0)$, proving the radius formulas. $\square$

The paired envelope preserves the dependence between charge and transition errors through the same posterior. It can be tighter than their rectangular enclosure. On a finite observation tree, intervals in (29.49) can be constructed by exact enumeration of all positive-probability histories under all feasible actions, followed by grouping at each summary. That preparatory enumeration can grow exponentially; the stated arithmetic bound covers the subsequent backward recursions, not the construction of the intervals. Continuous reports require a separate analytic or validated numerical enclosure. The trivial interval $[0, 1]$ is always available but may yield a weak certificate.

In particular, $\eta_k(z)$ cannot generally be set to $1/(2m)$: repeated rounding and discarded reports separate the full posterior from γ . The one-step quantization radius in [Proposition 29.19](#) belongs to

its resampling comparison and does not bound this different error. Given valid posterior intervals, a controller evaluated by (29.35) has the directly computable certificate

$$J_\rho - V_0^{\text{rich}}(H_0) \leq U_m(\rho) - L_0(z_0).$$

Here the physical evaluation may integrate out the discarded reports because the fixed controller does not use them. The new lower bound L_0 is what makes the comparison valid against policies that do use those reports. Section 34.2 gives exact finite-tree checks and an example where information reduction has a strictly positive cost.

Action comparisons can certify more than a cost interval

An interval for the optimal cost can remain wide even when the same action is optimal throughout a summary fiber. The next result uses the existing conditional envelopes to certify decisions directly. The Bellman comparison and upper/lower action screening are standard bounding principles; the statement below keeps the richer-history information class and the exact execution mask of Assumption 29.21.

For the lower and upper functions in (29.42), define

$$\begin{aligned} Q_k^-(z, a) &= \inf_{(r,p) \in \mathcal{U}_k(z,a)} \{r + p \cdot L_{k+1}\}, \\ Q_k^+(z, a) &= \sup_{(r,p) \in \mathcal{U}_k(z,a)} \{r + p \cdot U_{k+1}\}. \end{aligned} \quad (29.52)$$

If a is the only feasible action, set $e_k(z, a) = 0$. Otherwise set

$$e_k(z, a) = \left[Q_k^+(z, a) - \min_{a' \in A_k(z) \setminus \{a\}} Q_k^-(z, a') \right]_+. \quad (29.53)$$

For any feasible summary controller ρ , use the finite recursion

$$\begin{aligned} R_K^\rho(z) &= 0, \\ R_k^\rho(z) &= \sum_{a \in A_k(z)} \rho_k(a | z) \left\{ e_k(z, a) + \sup_{(r,p) \in \mathcal{U}_k(z,a)} p \cdot R_{k+1}^\rho \right\}. \end{aligned} \quad (29.54)$$

The supremum in this last display concerns continuation only; the immediate cost is already accounted for through the action comparison.

Theorem 29.26 (An action-comparison execution certificate). *Under Assumption 29.21, every feasible summary controller satisfies*

$$0 \leq J_k^\rho(H) - V_k^{\text{rich}}(H) \leq R_k^\rho(\Psi_k(H)). \quad (29.55)$$

Consequently an independently certified upper evaluation $\bar{E}_\rho \geq J_\rho$ gives the combined bound

$$J_\rho - V_0^{\text{rich}}(H_0) \leq \min\{\bar{E}_\rho - L_0(z_0), R_0^\rho(z_0)\}. \quad (29.56)$$

If every action used by ρ has $e_k(z, a) = 0$ at every stage and summary, then ρ is optimal among all richer-history policies. This conclusion does not require singleton envelopes or equality of L and U . For the binary posterior envelopes, both action bounds and every step of (29.54) use only their two endpoints.

Proof. Let $Q_k^*(H, a)$ be the true expected charge plus the optimal richer-history continuation after action a . The tower property, conditional inclusion and [Theorem 29.22](#) give

$$Q_k^-(z, a) \leq Q_k^*(H, a) \leq Q_k^+(z, a).$$

With at least two feasible actions, its Bellman advantage satisfies

$$Q_k^*(H, a) - V_k^{\text{rich}}(H) = \left[Q_k^*(H, a) - \min_{a' \neq a} Q_k^*(H, a') \right]_+ \leq e_k(z, a).$$

With one feasible action the advantage is zero. At K the policy and optimal conditional terminal costs agree. If the claimed regret bound holds at $k + 1$, condition on the action selected by ρ and subtract the true Bellman value. The resulting difference is the action advantage plus the conditional expected continuation regret. Bound the first by e_k , the second by $p_k(H, a) \cdot R_{k+1}^\rho$, and then use membership in the conditional envelope. Averaging over ρ_k proves (29.55) by backward induction. The earlier cost-interval certificate gives the other endpoint in (29.56). If all used action excesses vanish, the recursion gives $R^\rho = 0$. Finally, the binary transition law is affine in its posterior, so pairing it with any fixed continuation vector again attains its extrema at the interval endpoints. $\square$

The minimum of the two certificates is useful: neither bound is uniformly sharper. A cost interval measures uncertainty in value, while (29.53) compares the selected action with its alternatives. The same bounds also give safe action screening. If $Q_k^-(z, a) > \min_{a'} Q_k^+(z, a')$, action a is strictly suboptimal at every full history in that fiber and may be removed without changing the richer-history optimum. The inequality is strict so that optimal ties are not removed. This screening concerns optimization; it does not relax a reservation, order-status or completion constraint.

These conclusions remain conditional on valid envelopes for all feasible histories and actions. A small fitted action gap under one set of reports does not establish the required uniform comparison. For model transfer, compute the minimum in (29.56) in the approximate model and add $2B_T$ as in [Corollary 29.23](#).

29.9 Relation to order-flow and signature-control foundations

[Cont et al. \(2025\)](#) provide the order-flow/clearing decomposition and its generator factorization. Equation (29.7) uses that foundation. The specialization here retains ownership, priority, private reports and a parent ledger, and makes the event law depend on a trader's control. These additions formulate a particular execution problem; they do not supersede the original general order-book framework. [Cont and Kukanov \(2017\)](#) already formulate order placement across venues with queue uncertainty, execution costs and a fixed trade slice. Queue-aware placement and venue allocation are therefore foundations, not new objects here. The additional scope is their integration with dynamic quantity choice, action-dependent hidden information, exact parent-order feasibility and an explicit error certificate for the resulting feedback controller.

[Kalsi et al. \(2020\)](#) develop signature execution policies and their approximation against a price signal. The compact signature universality argument in [Theorem 17.1](#) uses the same underlying approximation principle. The kernel augmentation, pathwise completion transformation and cost-unit perturbation certificate identify the specific construction developed in [Section 17](#).

[Bank et al. \(2025\)](#) establish density of signature controls in an admissible progressively measurable class and equality of optimal values for controlled rough differential equations under their driver-filtration, regularity and stability hypotheses; see their Proposition 4.5 and Theorem 4.7. Consequently neither signature-control density nor recovery of an unrestricted optimum is claimed here as a new general principle. [Theorem 29.10](#) treats a different explicit specialization: finite

actions, partially observed marked jump dynamics, history-dependent admissibility masks and an observation law altered by earlier controls. Its proof uses complete observable transcripts, masked score approximation and bounded-loss hybrid coupling. It does not require the hidden driving clock to be observed or the loss to be continuous. The comparison below identifies this scope rather than claiming priority over general stochastic control theory.

Object	Prior foundation	Construction or scope here
Book dynamics	Order flow and clearing (Cont et al., 2025)	Ownership, private reports, conserved parent ledger and controlled rates
Signature execution	Price-signal signature policies (Kalsi et al., 2020)	Exact exponential augmentation, completion and a common cost certificate
Control approximation	Signature-control density and optimal values (Bank et al., 2025)	Masked finite-action feedback on complete endogenous observable transcripts
Primitive model error	Coupling and total-variation contraction	Joint execution-path error, intervention error and bounded-loss regret
Quadratic reference	Exogenous feature law and quadratic rate cost	Explicit autonomy and reduction hypotheses; a charged reference-model defect
Computable partial observation	Belief values, bounding approximations and finite-horizon planning (Smallwood and Sondik, 1973 ; Lovejoy, 1991 ; Walraven and Spaan, 2019)	Feasible belief-memory controller, physical hidden-state evaluation, signature-table realization and model-transfer certificate
Observation reduction	Robust Bellman bounds (Iyengar, 2005 ; Nilim and El Ghaoui, 2005)	Conditional envelopes over full-history fibers, exact feasibility masks, binary endpoint witnesses and a certificate against richer information

The combined statement is [Corollary 29.11](#): under its common-model, feasibility and global-optimization hypotheses, execution regret converges to zero against all policies in the declared observation filtration. [Proposition 29.12](#) supplies an explicit finite-state approximation; [Proposition 29.13](#) separately accounts for using a reference objective. Neither statement converts a local nonconvex fit into a certified global optimum.

The constructive specialization in [Section 29.7](#) supplies a finite algorithm and a posteriori global gap under its additional binary hidden-state and finite-observation assumptions. Its one-hot summary has an explicit, growing dimension. The contribution is the complete execution construction and its checked error accounting, not a new claim to belief concavity, grid planning, or general signature universality. It does not establish signature compression or a sample-complexity advantage: the depth-one construction encodes an already computed table in a growing observable channel. [Theorem 29.22](#) supplies the missing comparison when that controller omits reports. Robust Bellman recursion itself is an established method; the execution-specific statement retains feasibility, conditions its envelopes on full observable histories, and combines a binary posterior witness with physical controller evaluation. Constructing those envelopes is an explicit obligation, not a consequence of a small planning gap or a successful fit on logged trajectories. The action-comparison certificate in [Theorem 29.26](#) uses these same envelopes to bound Bellman advantages, screen strictly suboptimal instructions and, in separated cases, establish exact optimality without singleton conditional laws. This is a refinement of the stated execution comparison, not a new general Bellman principle.

30 Dynamic control of quantity, instruction and destination

30.1 The state and the joint kernel

At decision time k , define

$$X_k = (q_k, s_k, \ell_k, z_k, I_k, r_k, b_k). \quad (30.1)$$

The observable component s_k includes local books, valid external references, recent activity and age-of-information variables. Live orders ℓ_k retain reserved exposure. Queue or parity states z_k may themselves be uncertain and included in b_k . The impact state I_k carries relevant history; r_k records the session and auction phase.

An admissible policy chooses $a_k \in \mathcal{A}_k(X_k)$ from observations available at that time. Section 29 constructs the following joint law from primitive event rates, tagged clearing and the observation filter:

$$\mathcal{K}_k(dF, dP, ds', dy, d\ell', dz', dI', dr' \mid X_k, a_k). \quad (30.2)$$

It preserves dependence between fills, prices and reports. Conditional independence across venues is an optional model restriction, not a consequence of fragmentation. The next belief is obtained from the controlled likelihood.

For bounded costs in a finite state/action approximation, the Bellman recursion is

$$V_k(X) = \min_{a \in \mathcal{A}_k(X)} \mathbb{E} [c_k(X, a, Y) + V_{k+1}(X') \mid X, a], \quad V_N(X) = \Phi(q, s, r). \quad (30.3)$$

The stage cost can include net execution premiums, waiting-price effects, additive inventory risk and message costs, with each component counted once. Finite minima and backward induction establish an optimal policy in this approximation. Continuous-state existence needs additional compactness, measurability and lower-semicontinuity assumptions and is not implied merely by writing the Bellman equation.

30.2 The marginal value of remaining inventory

In a differentiable reduced model, write $V_k(q, s, b)$. An immediate purchase of x shares approximately changes the objective by

$$C(x) - x \partial_q V_{k+1}(q, s, b), \quad (30.4)$$

when x is small and other state effects are negligible. The optimal interior quantity satisfies $C'(x) \approx \partial_q V_{k+1}$. The shadow cost of an unexecuted share therefore connects scheduling urgency to venue-level willingness to pay. With the continuation written exactly, this condition becomes $C'(x) = \partial_q V_{k+1}(q - x, s, b)$.

For a passive order, expected fill size alone is insufficient. Expanding in a random fill F gives

$$\mathbb{E}[V(q - F)] \approx V(q) - V_q(q) \mathbb{E}F + \frac{1}{2} V_{qq}(q) \mathbb{E}[F^2]. \quad (30.5)$$

When V is convex, uncertainty in completion can be costly. The expansion holds only where the derivatives exist and the neglected terms are controlled; the exact discrete recursion needs neither condition.

30.3 A quantitative link from impact to execution decisions

The energy representation also controls sensitivity of the execution policy. This connection is exact under a common response mechanism; it does not rely on a first-order expansion of the continuation value.

Proposition 30.1 (Impact error and execution regret). *Fix a finite decision horizon, a common initial law, the same feasible contingent policies, and the same action-dependent law of fills and observations. For every policy, let $x \in \mathbb{R}^m$ be its realized signed fill vector on a common time/venue grid, with $\|x\|_1 \leq Q$ almost surely. Suppose the two integrable total-loss models differ only through $\frac{1}{2}x^\top Kx$ versus $\frac{1}{2}x^\top \widehat{K}x$, where K and $\widehat{K}$ are symmetric and $\max_{i,j} |K_{ij} - \widehat{K}_{ij}| \leq \epsilon_K$. Then, for every fixed policy,*

$$|J_\pi - \widehat{J}_\pi| \leq \frac{1}{2}Q^2\epsilon_K.$$

If $\widehat{\pi}$ is an optimizer for the approximate cost and an optimizer exists for the true cost, its true regret is at most $Q^2\epsilon_K$. An ϵ_{opt} -optimal approximate policy adds ϵ_{opt} to this regret bound.

Proof. For each realized path,

$$\left| \frac{1}{2}x^\top (K - \widehat{K})x \right| \leq \frac{1}{2}\epsilon_K \sum_{i,j} |x_i||x_j| = \frac{1}{2}\epsilon_K \|x\|_1^2.$$

Take expectations. Add and subtract the approximate costs of the selected policy and a true optimizer; approximate optimality bounds the middle term, and the two fixed-policy errors give the claim. $\square$

For a buy-only parent with authoritative reservations, total fills cannot exceed Q , giving the required volume bound. Strategies that buy and resell need a bound on gross volume instead. Discretized bounded linear kernels from Part II supply K ; the same proof applies to finite-energy discretizations of singular kernels when an entrywise error bound is available. Both models should independently satisfy the relevant admissibility conditions; small coefficient error alone does not preserve absence of manipulation.

Corollary 30.2 (Phantom profit from an estimated kernel). *On a finite single-asset time grid $\{t_1, \dots, t_m\}$, let the estimated symmetric impact matrix of Proposition 30.1 be $\widehat{K}_{ij} = \widehat{G}(|t_i - t_j|)$. Write $\delta_T(\widehat{G})$ for the defect restricted to this grid. For a gross traded-volume cap $Q \geq 0$, the largest estimated gain from the impact component over signed round trips is*

$$\sup_{\substack{x \in \mathbb{R}^m, \\ \sum_i x_i = 0 \\ \|x\|_1 \leq Q}} \left\{ -\frac{1}{2}x^\top \widehat{K}x \right\} = \frac{1}{2}Q^2\delta_T(\widehat{G}). \quad (30.6)$$

If the true matrix K has nonnegative energy on every zero-mass vector and $\max_{i,j} |K_{ij} - \widehat{K}_{ij}| \leq \epsilon_K$, then

$$0 \leq \frac{1}{2}Q^2\delta_T(\widehat{G}) \leq \frac{1}{2}Q^2\epsilon_K. \quad (30.7)$$

Thus every negative estimated round-trip impact cost is an estimation artifact relative to this admissible true impact model.

Proof. For $Q > 0$, write $x = Q\nu$. The zero-mass and volume constraints become those of the fixed-grid defect, while the energy scales by Q^2 . For $Q = 0$, the only feasible trade is zero and the identity is immediate. This is the finite-grid profit identity of Proposition 14.2. For a zero-mass ν with $\|\nu\|_1 \leq 1$, true admissibility gives

$$-\nu^\top \widehat{K}\nu = -\nu^\top K\nu + \nu^\top (K - \widehat{K})\nu \leq \epsilon_K \|\nu\|_1^2 \leq \epsilon_K.$$

Taking the supremum proves (30.7); for sampled scalar kernels this is also (14.2) applied to an admissible kernel. $\square$

Remark 30.3 (Mandate regret and phantom profit). [Proposition 30.1](#) bounds the true regret of a policy selected under a wrong impact model by $Q^2\epsilon_K$. [Corollary 30.2](#) instead measures the negative impact cost the estimate assigns to a closed loop, at $Q^2\delta_T(\hat{G})/2$. Its supremum allows all signed round trips subject only to the gross-volume cap. Additional policy or inventory constraints can reduce the attainable gain; a buy-only class contains no nonzero round trip. The quantity excludes spreads, fees and other costs.

Projecting the estimate onto the cone of matrices with nonnegative energy on zero-mass vectors removes this phantom impact profit on the tested grid. It does not by itself guarantee a smaller regret bound. For a repaired matrix $\tilde{K}$, that bound uses $Q^2 \max_{i,j} |K_{ij} - \tilde{K}_{ij}|$, whose improvement requires a separate error comparison. Reporting the grid, gross-volume cap and defect alongside an estimated kernel therefore records an execution consequence that a fit statistic alone does not identify.

Auction eligibility, allocation, random fills and common non-impact costs may all be included, provided they have the same response law in this comparison. The proposition therefore connects impact-model accuracy to order and venue selection, while identifying exactly what it does not cover: if changed impact also changes fills, prices used by the policy, or observations, one must include response-law error, as in [Corollaries 29.6](#) and [C.1](#). Impact theory constrains modeled costs; it does not derive an exchange's allocation rules.

30.4 A transparent passive-order threshold

Consider one unit and two complete plans. Immediate execution has total expected cost c_M . A passive-first plan fills with probability p , has conditional total cost c_F when filled and c_U when unfilled, and incurs an additional decision cost d . Both conditional costs must include the appropriate continuation under the same benchmark.

Proposition 30.4 (Order-choice threshold). *If $c_U > c_F$ and these conditional costs are fixed in the comparison, the passive-first plan is weakly better exactly when*

$$p \geq \frac{c_U + d - c_M}{c_U - c_F}. \quad (30.8)$$

A threshold below zero makes the passive plan weakly preferable throughout $[0, 1]$; one above one makes it inferior throughout $[0, 1]$.

Proof. Rearrange $pc_F + (1 - p)c_U + d \leq c_M$ using $c_U - c_F > 0$. □

If changing the instruction changes both p and the conditional costs, the complete expression must be recomputed. Increasing a fill rate by exposing the order to more toxic flow can raise c_F . Likewise, a higher failure cost near the deadline often raises the required probability, provided the other terms remain fixed and $c_M > c_F + d$. Without such qualifications, a universal monotonicity claim would be false.

Example 30.5 (Price improvement can be outweighed by waiting). Take dollar-per-share costs $c_M = 0.010$, $c_F = -0.002$, $c_U = 0.025$ and $d = 0.0005$. The threshold is 0.574074. At $p = 0.75$, passive-first expected cost is 0.00525; at $p = 0.25$, it is 0.01875. The second plan is more expensive than immediate execution despite its favorable fill price.

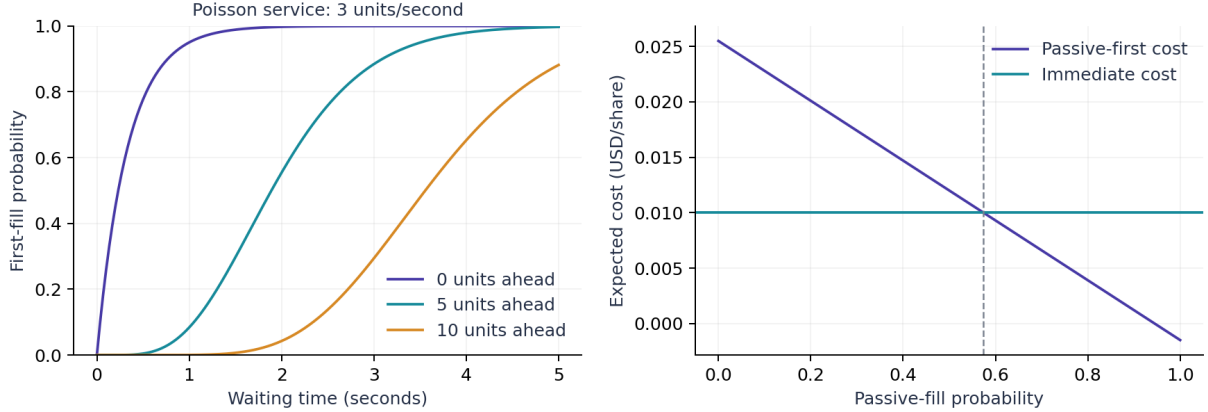


Figure 4: Two synthetic components of order choice. Left: first-fill probability in the Poisson FIFO service model, with no cancellations or repricing. Right: the passive-first cost threshold in Proposition 30.4.

31 Information value, misspecification and robust decisions

31.1 Value of observing a liquidity signal

In a finite-horizon model with hidden initial state h , fix a contingent policy π mapping future observations into actions. Its expected total cost at prior b is linear:

$$J_\pi(b) = \sum_h b(h)g_\pi(h), \quad (31.1)$$

where $g_\pi(h)$ includes all later uncertainty and costs under that fixed contingent plan. Suppose the feasible policy class is the same for every prior in the comparison. Then $V(b) = \inf_\pi J_\pi(b)$ is concave.

Classical result 31.1 (Free information cannot increase optimal expected cost). *Let B be the posterior after an additional signal observed before choosing the contingent plan. If $\mathbb{E}[B] = b$ and the signal does not change the underlying feasible set or impose a cost, then*

$$\mathbb{E}[V(B)] \leq V(b). \quad (31.2)$$

Proof. An infimum of linear functions is concave. Jensen's inequality gives $\mathbb{E}[V(B)] \leq V(\mathbb{E}B) = V(b)$. Equivalently, the informed trader can ignore the signal and implement the prior-optimal plan. $\square$

The result compares correct probabilistic information with the option not to use it. It says nothing about the value of a misspecified hidden-stock feature, a signal that arrives too late, or an information-gathering action with cost. For a signal costing c , acquire it only when $V(b) - \mathbb{E}V(B) > c$. The quantity on the left is specific to the decision loss; it is not a price-prediction score.

31.2 A finite-horizon decision bound

Theorem 31.2 (Regret from a wrong liquidity belief). *Assume the same feasible contingent policies for priors b and $\hat{b}$. For every such policy, suppose its conditional expected total cost has range at most C : $\sup_h g_\pi(h) - \inf_h g_\pi(h) \leq C$. Let $\hat{\pi}$ minimize cost at $\hat{b}$ and π^* minimize it at b . With $\text{TV}(b, \hat{b}) = \frac{1}{2} \sum_h |b(h) - \hat{b}(h)|$,*

$$0 \leq J_{\hat{\pi}}(b) - V(b) \leq 2C \text{TV}(b, \hat{b}). \quad (31.3)$$

Proof. For a function of range at most C , the expectation difference between the two priors is at most $C \text{TV}(b, \hat{b})$. Apply this to each fixed policy and add and subtract costs under $\hat{b}$:

$$\begin{aligned} J_{\hat{\pi}}(b) - J_{\pi^*}(b) &= [J_{\hat{\pi}}(b) - J_{\hat{\pi}}(\hat{b})] + [J_{\hat{\pi}}(\hat{b}) - J_{\pi^*}(\hat{b})] \\ &\quad + [J_{\pi^*}(\hat{b}) - J_{\pi^*}(b)]. \end{aligned}$$

The middle term is nonpositive by optimality. Bound the other terms as stated. $\square$

An ε -optimal optimizer at $\hat{b}$ adds ε to the right-hand side. If each of N stage costs has range at most c and terminal cost range at most c_T , one may take $C = Nc + c_T$. Infinite hard-completion penalties violate boundedness; this theorem then gives no finite guarantee. The bound isolates belief error under an otherwise correct model. Wrong transition kernels, impact laws or feasible sets require additional bounds.

31.3 Robust control and tail objectives

Let $\mathfrak{K}_k(X, a)$ be a set of plausible joint kernels. A robust recursion is

$$V_k^{\text{rob}}(X) = \min_{a \in \mathcal{A}_k(X)} \sup_{K \in \mathfrak{K}_k(X, a)} \mathbb{E}_K[c_k + V_{k+1}^{\text{rob}}(X')]. \quad (31.4)$$

This stagewise form assumes rectangular uncertainty: nature can choose a permitted kernel at each encountered state/action. If one unknown parameter is fixed across the entire trajectory, independently maximizing at each step changes the uncertainty model. A posterior over a fixed parameter or an augmented robust state is then needed.

For a completed-order loss L , a tail criterion can use the representation of [Rockafellar and Uryasev \(2000\)](#),

$$\text{CVaR}_\alpha(L) = \min_{z \in \mathbb{R}} \left[z + \frac{1}{1 - \alpha} \mathbb{E}(L - z)_+ \right], \quad 0 < \alpha < 1. \quad (31.5)$$

Optimizing terminal CVaR is a precommitment problem unless the state and criterion are adapted appropriately. Nesting one-step conditional risk functionals gives a dynamically consistent alternative with a different meaning. Neither CVaR nor a large inventory penalty makes a physically impossible completion constraint feasible.

32 Certifying an improvement over a feasible schedule

A bound on distance from an unknown optimum need not answer the immediate execution question: should a proposed schedule replace the schedule already in use? This section compares both schedules under the *same* uncertain cost law. Common uncertainty then cancels where the schedules agree. Baseline-relative robust improvement is an established control principle ([Ghavamzadeh et al., 2016](#)); convex trading formulations with explicit costs and constraints are developed by [Boyd et al. \(2017\)](#). The result below is an execution specialization: a box of temporary-impact coefficients and an ellipsoid of linear charges give an exact, closed-form acceptance step along a proposed, quantity-conserving schedule change.

32.1 The mandate and the uncertainty set

There are n execution bins. Quantities have one common unit, and the buy schedule belongs to

$$K = \{x \in \mathbb{R}^n : \mathbf{1}^\top x = Q, \ 0 \leq x_i \leq u_i\}, \quad Q > 0, \quad u_i \geq 0, \quad \sum_i u_i \geq Q.$$

The bounds are executable capacities in the declared model, not expected passive fills. Let $b, x \in K$ be a baseline and a proposed schedule, $d = x - b$, and $x(t) = b + td$, $0 \leq t \leq 1$. Every such schedule completes the mandate and respects the same caps. Consider the family

$$C_{z,v}(y) = \frac{1}{2}y^\top (H + \text{diag } z)y + (c + Bv)^\top y, \quad |z_i| \leq \epsilon_i, \quad \|v\|_2 \leq \rho. \quad (32.1)$$

Here H is symmetric, $H - \text{diag } \epsilon \succeq 0$, and the box and ball form a Cartesian product. H may contain a known transient-impact or inventory-risk quadratic; only its diagonal correction is uncertain. The linear charge can include fees or an adverse drift. All costs are in the same units. The family is conditional on the information available when this schedule comparison is made.

Theorem 32.1 (Exact robust schedule step). *Define $\text{sgn}(0) = 0$ and*

$$A_d = d^\top (Hb + c) + \sum_i \epsilon_i b_i |d_i| + \rho \|B^\top d\|_2, \quad (32.2)$$

$$D_d = d^\top H d + \sum_i \epsilon_i \text{sgn}(d_i) d_i^2. \quad (32.3)$$

Then $D_d \geq 0$ and, simultaneously for $0 \leq t \leq 1$,

$$\sup_{z,v} \{C_{z,v}(x(t)) - C_{z,v}(b)\} = A_d t + \frac{1}{2} D_d t^2 =: \Psi_d(t). \quad (32.4)$$

If $A_d \geq 0$, $t_ = 0$ minimizes this expression. If $A_d < 0 < D_d$, its minimizer is $t_* = \min\{1, -A_d/D_d\}$; if $A_d < 0 = D_d$, it is $t_* = 1$. The accepted schedule satisfies, for every law in the family,*

$$C_{z,v}(x(t_*)) \leq C_{z,v}(b) - G_d, \quad G_d = -\Psi_d(t_*) \geq 0.$$

When $0 < -A_d < D_d$, the guaranteed saving is $A_d^2/(2D_d)$.

Proof. Expand the cost difference in (32.1). The uncertain diagonal contribution is $\frac{1}{2} \sum_i z_i ((b_i + td_i)^2 - b_i^2)$. Because b_i and $b_i + td_i$ are nonnegative, its coefficient has the sign of d_i for $t > 0$ whenever it is nonzero. The maximizing corner is therefore $z_i = \epsilon_i \text{sgn}(d_i)$, independent of $t > 0$. Its contribution is $t \sum_i \epsilon_i b_i |d_i| + \frac{1}{2} t^2 \sum_i \epsilon_i \text{sgn}(d_i) d_i^2$. The support function of the ball gives $\sup_{\|v\| \leq \rho} tv^\top B^\top d = t\rho \|B^\top d\|$. Independence of the uncertainty factors proves equality, including $t = 0$. The adverse diagonal matrix is positive semidefinite after addition to H , so $D_d \geq 0$. Minimizing this convex quadratic over $[0, 1]$ proves the remaining assertions. Convexity of K proves feasibility. $\square$

Remark 32.2 (What is exact and what remains conditional). The theorem optimizes the fraction of a supplied proposal; it does not solve the globally optimal robust schedule problem. It is exact for the Cartesian family in (32.1); enlarging a correlated family to this box and ellipsoid gives an upper bound, possibly conservative. Signed trades can change the sign of the squared-quantity difference along the segment, so the displayed single-quadratic formula cannot be used unchanged for round trips. The true coefficients must lie in the supplied uncertainty set. No confidence level follows from an arbitrary choice of radii.

32.2 Why the common model matters

The desired comparison is $\sup_\theta (C_\theta(x) - C_\theta(b))$. It is generally smaller than $\sup_\theta C_\theta(x) - \inf_\theta C_\theta(b)$, which allows two different laws in the same comparison. At $x = b$, the first expression is exactly zero, while the second can be strictly positive. This elementary distinction prevents rejection of a harmless no-change action and can prevent excessive shrinkage of small proposals. The term $\rho \|B^\top d\|$ also shows that uncertainty orthogonal to the schedule change has no effect on its certificate. For example, an uncertain common per-share charge cancels because $\mathbf{1}^\top d = 0$.

Corollary 32.3 (Finite proposal menus and numerical tolerance). *For any finite collection $x^{(j)} \in K$, compute its accepted step and select the largest $G_{d(j)}$, including the unchanged baseline with gain zero. The selected schedule improves on the baseline by at least that gain for every law in the common family, even when the menu was chosen from the same information used to construct the uncertainty set. If only a certified upper bound $\bar{\Psi}_j \geq \Psi_{d(j)}(t_j)$ is available, accept only a proposal with $\bar{\Psi}_j < 0$.*

Proof. Each inequality is uniform over schedules and laws satisfying the stated conditions. Taking a finite, data-dependent selection preserves it. The second assertion follows directly from the upper-bound inequality. $\square$

32.3 Completion must survive every update

Schedule feasibility at planning time is not enough when future capacity can fall. Suppose before bin k the remaining mandate is q_k , current capacity u_k is known, and future executable capacities have guaranteed lower bounds $\underline{u}_{k+1}, \dots, \underline{u}_n$. Assume the lower corner is possible and the trader can execute any quantity up to the actual capacity. The one-step viability condition is explicit.

Proposition 32.4 (Completion-preserving interval). *The quantities preserving completion for all future capacity paths in this model are exactly*

$$\max\{0, q_k - \sum_{j>k} \underline{u}_j\} \leq y_k \leq \min\{q_k, u_k\}. \quad (32.5)$$

This interval is nonempty if and only if $q_k \leq u_k + \sum_{j>k} \underline{u}_j$. Requiring this condition at every update preserves $q_{n+1} = 0$.

Proof. The remaining quantity is $q_k - y_k$. It must be nonnegative and cannot exceed the total guaranteed future capacity, giving the lower bound. Current capacity and no overexecution give the upper bound. The lower corner shows necessity. Sufficiency follows by allocating the residual within the future lower bounds, or by induction using the same interval. At the final bin the lower and upper feasible values both equal q_n . $\square$

Expected fills and a statistical volume forecast are not deterministic capacity guarantees. If the lower-capacity event is only probabilistic, completion is conditional on that event; a market-order fallback with a price limit can itself be infeasible. A partially filled live order also retains its reservation until cancellation is acknowledged. The schedule comparison must therefore be applied to the unreserved mandate with the ledger of Proposition 27.3, not to a quantity already committed elsewhere. Replanning changes the benchmark: a sequence of local comparisons is not automatically a global cost comparison with the original plan.

32.4 An independently reproducible stress experiment

The script `scripts/certified_improvement.py` solves a capped nominal quadratic problem in six bins, computes the exact robust step, and independently enumerates all 2^6 diagonal corners. For each corner it uses the analytic linear-charge support value. A separate scalar minimizer checks the step. It checks quantity conservation, caps, positive curvature, zero-change cancellation, and the viability interval, then writes the following table. The uncertainty multiplier scales both declared radii; the largest displayed multiplier still satisfies the positive-semidefinite lower bound.

For a market study, freeze the baseline, candidate-generation rule, uncertainty calibration and execution horizon before evaluating orders. Report both the realized shortfall difference and how often the certificate accepts a change. Separate failures of coefficient coverage, executable capacity, and ledger reconciliation from optimization error.

Table 8: Synthetic six-bin execution: worst-case cost changes relative to the same feasible baseline. Negative values certify a saving. All numbers are normalized objective units, not measured basis points.

Uncertainty	Full proposal	Accepted fraction	Certified saving
0	-0.026593	1.000000	0.026593
1	-0.009450	0.685922	0.011957
2	0.007694	0.346335	0.003003
4	0.041980	0.000000	0.000000
8	0.110553	0.000000	0.000000

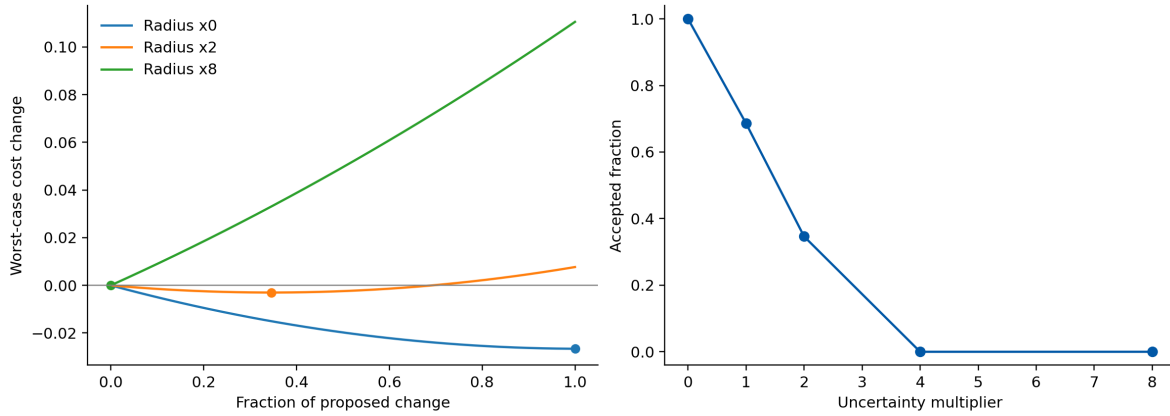


Figure 5: A proposal can have a negative nominal cost change and an adverse worst-case change. The accepted fraction minimizes the exact common-model comparison. The right panel shows how increasing uncertainty contracts the accepted change. Synthetic model; no market observations.

Table 9: Selected closing-auction constraints on a normal 4:00 p.m. close. Times are Eastern; shortened sessions and exceptional events require their own calendar and rules. Sources: [New York Stock Exchange \(2026c,d\)](#); [Nasdaq \(2026d\)](#).

Decision	NYSE	Nasdaq
Ordinary new MOC	Entry cutoff at 3:50; later entry has significant-imbalance conditions	Enter before 3:55
Late LOC	After 3:50, significant-imbalance conditions apply	Before 3:58, with reference-price controls after 3:55
Ordinary cancellation	MOC/LOC and Closing IO cancellations rejected from 3:50	MOC/LOC ordinary cancellation ends at 3:50; specified error corrections only before 3:58
Specialized late access	Established entry/cancel/modify cutoff: 3:59:50; amended cancel/modify cutoff: 3:59, subject to implementation	Imbalance-Only orders follow separate cross rules

33 NYSE and Nasdaq auctions at the deadline

An auction is a distinct matching process with a clearing price and allocation rule. It should not be modeled as a generic Poisson print near the open or close. NYSE and Nasdaq publish distinct auction procedures and imbalance information ([New York Stock Exchange, 2026c](#); [Nasdaq, 2026b](#)). Reopening auctions depend on trading status and event-specific rules. An imbalance observation is available only after its dissemination time, even when the eventual auction price is known in a dataset.

33.1 Exchange phases and commitment deadlines

NYSE’s opening process starts at 9:30 a.m. Eastern, but a DMM may open an individual security later. Its published auction eligibility covers NYSE-listed securities. Nasdaq describes broader cross eligibility, including nationally listed securities ([New York Stock Exchange, 2026c](#); [Nasdaq, 2026b](#)). The optimizer must retain both the execution destination and the listing market. For a mandate tied to a primary-market closing benchmark, a cross on another exchange must be evaluated against that benchmark rather than assumed to reproduce it.

The D Order row distinguishes the established process from the amendment in SR-NYSE-2026-36. NYSE’s published auction timeline states 3:59:50 for entry and modification. The August 2026 filing changes cancellation/modification to one minute before the close, but separately provides for implementation by a Trader Update, anticipated no later than the first quarter of 2027. Its filing effectiveness alone does not establish a production start date. Use the announced implementation date to switch the constraint; do not apply 3:59 retroactively ([New York Stock Exchange, 2026c](#); [U.S. Securities and Exchange Commission, 2026](#)).

These times define changes in feasible actions, not just higher predicted volume. At an irreversible commitment point, the relevant decision is whether the prospective auction allocation is worth giving up the ability to redirect those shares. A receding-horizon algorithm must optimize across the next such boundary; waiting until the final minute can remove an entire order family from its feasible set. Imbalance messages affect beliefs only after receipt, and their indicative prices remain uncertain forecasts of clearing outcomes.

33.2 Eligible interest, matched volume and the clearing problem

Fix an exchange $v \in \{\text{XNYS}, \text{XNAS}\}$, auction $c \in \{o, k\}$ (opening or closing), and a realized pre-auction state ω . All quantities in this subsection are shares. Split each accepted order into accounting pieces only when the pieces have different allocation classes; displayed and reserve pieces must sum to the order's remaining quantity. For a candidate price p , let $n_i(p; \omega)$ be the eligible quantity of piece i after limit, pegging, phase, access and other exchange restrictions have been applied. For an ordinary buy limit L_i , eligibility requires $p \leq L_i$; for a sell, $p \geq L_i$. Define executable demand and supply

$$D_v(p) = \sum_{i \in \mathcal{B}_v} n_i(p), \quad S_v(p) = \sum_{j \in \mathcal{S}_v} n_j(p). \quad (33.1)$$

The letter $S_v(p)$ here denotes a supply curve, not the reference-price process. If every eligible buy can match every eligible sell, the maximum paired volume is

$$V_v(p) = \min\{D_v(p), S_v(p)\}, \quad U_v(p) = D_v(p) - S_v(p). \quad (33.2)$$

Paired volume counts each matched share once, not once on each side. The signed quantity U_v is an aggregate excess-demand statistic; it is not automatically the exchange's published imbalance field.

When the remaining restrictions consist only of pairwise eligibility and share capacities, the exact accounting object is a bipartite matching problem. Write z_{ij} for the shares paired from buy i to sell j , and let $e_{ij}(p) \in \{0, 1\}$ describe permitted matching:

$$\begin{aligned} V_v(p) &= \max_{z \geq 0} \sum_{i,j} z_{ij}, \\ \sum_j z_{ij} &\leq n_i(p), \quad \sum_i z_{ij} \leq n_j(p), \quad z_{ij} = 0 \text{ if } e_{ij}(p) = 0. \end{aligned} \quad (33.3)$$

This formulation prevents an imbalance-offset instruction from being counted as unrestricted two-sided liquidity. Additional priority constraints select individual fills after price selection. Integer share capacities give an integer maximum flow in this restricted bipartite problem. Minimum-fill conditions, all-or-none requirements, state-changing self-trade prevention and other non-pairwise constraints require additional integer or sequential rules; they are not encoded by a fixed eligibility edge alone. With all pairings permitted, (33.2) follows: no flow can exceed either side's total, and a successive pairing of eligible shares attains the smaller total.

The model therefore separates three maps: accepted orders to eligible interest, eligible interest to an auction price, and the price plus allocation state to individual fills. A generative auction model must preserve all three.

33.3 Nasdaq: a lexicographic price-selection operator

For Nasdaq's ordinary opening and closing crosses, Rules 4752(d) and 4754(b) first maximize executed shares, then use unmatched on-open/on-close interest, entered-price and midpoint tie-breaks; price protections and specified special cases also apply (Nasdaq, 2026d). Let Π be the applicable price grid and write $I_c^N(p)$ for the rule-defined unmatched committed interest. In the closing cross this statistic concerns MOC/LOC interest; in the opening cross it also reflects the specified Early Market Hours interest. It need not equal $|U_v(p)|$ when order classes differ.

A mathematical representation of the ordinary selection sequence is

$$\begin{aligned}
\Pi_1 &= \arg \max_{p \in \Pi} V_N(p), \\
\Pi_2 &= \arg \min_{p \in \Pi_1} I_c^N(p), \\
\Pi_3 &= \begin{cases} E_c(\Pi_2), & E_c(\Pi_2) \neq \emptyset, \\ \Pi_2, & E_c(\Pi_2) = \emptyset, \end{cases} \\
p_N^0 &\in \arg \min_{p \in \Pi_3} |p - m_N|.
\end{aligned} \tag{33.4}$$

To define E_c , fix a candidate price p and provisionally allocate the maximum paired volume under the cross's prescribed priority and compatibility rules. Let $f_i^c(p)$ be that provisional fill of a priced, cross-eligible piece i , a_i^c its admitted quantity, and e_i^c its entered price for the ordinary cross, with any prescribed price treatment applied. In the ordinary case the entered-price candidates are

$$E_c(A) = \left\{ p \in A : \sum_{i: e_i^c = p} (a_i^c - f_i^c(p)) > 0 \right\}. \tag{33.5}$$

Market interest without an entered price does not itself supply an entered-price candidate. Provisional allocations are computed at each candidate; they do not assume a previously selected auction price. The definition retains only prices of residual priced interest when such candidates exist; otherwise the criterion does not distinguish the surviving prices. Repricing and special ranked-price adjustments must use the native rule's price fields and branches. In the unrestricted subset, the provisional fills are given by (33.9); restricted matches require the compatibility rules too. Here m_N is the specified Nasdaq/system midpoint, not an arbitrary last trade.

For an example reaching this stage, let one buy LOC contain 600 shares at \$100.02 and one sell LOC contain 500 shares at \$100.00. At the candidate prices \$100.00, \$100.01 and \$100.02, paired volume is 500 and unmatched LOC interest is 100. Only at \$100.02 does residual interest have an entered price equal to the candidate, so $E_k(\Pi_2) = \{100.02\}$. The entered-price criterion selects \$100.02 even if the system midpoint is \$100.01. This example assumes an ordinary unrestricted cross and inactive price protections. Quantities and price conditions are recomputed on the protected candidate set if the initially selected price fails the applicable threshold condition. Opening validation and special ranked-price adjustments remain explicit branches of the operator. A failed opening validation may produce no opening cross; it is not repaired by a fictitious execution at the nearest price.

Within the ordinary sequence, a later criterion cannot sacrifice a share selected by an earlier criterion. The entered-price stage must be retained: a weighted expression containing only volume, unmatched interest and midpoint distance omits a rule even if its weights are small. A scalar encoding of all stages would require bounds and minimum criterion gaps that rigorously preserve the hierarchy. The subsequent protection and special ranked-price branches are separate operations and may change the price selected by the ordinary sequence.

For forecasting, the current reference price and near/far indicative clearing prices are different observables. The near calculation brings in eligible continuous-book interest; the far calculation uses the corresponding auction book (Nasdaq, 2026c). Neither is an already realized future auction price. When market messages arrive, update the distribution of the complete future input to (33.4), not merely a regression on one indicative number.

33.4 NYSE: feasible prices and DMM auction liquidity

NYSE Rules 7.35, 7.35A and 7.35B require a different construction. The DMM selects the auction price; better-priced customer interest and auction-specific conditions constrain that selection. Ordinary DMM Orders are not eligible for the closing auction under Rule 7.35B(a); DMM Auction Liquidity is a separate contribution that can be entered after Core Trading Hours. DMM Auction Liquidity is excluded from published Auction Imbalance Information, and DMM interest is not treated as better-priced customer interest ([New York Stock Exchange, 2026b,h,d](#)). Consequently, a public maximum-volume calculation alone does not determine the actual NYSE auction outcome.

Let $B_{>}(p)$ and $A_{<}(p)$ be eligible better-priced customer buy and sell quantities, with the rule's treatment of market orders and short-sale restrictions. Let $D_C(p), S_C(p)$ include eligible at-priced customer interest, and let $L_D^b(p), L_D^s(p)$ denote DMM liquidity supplied for the auction. Necessary aggregate feasibility conditions for full satisfaction of better-priced customer interest are

$$B_{>}(p) \leq S_C(p) + L_D^s(p), \quad A_{<}(p) \leq D_C(p) + L_D^b(p). \quad (33.6)$$

With restricted matches, these inequalities are replaced by the corresponding flow-feasibility conditions from (33.3). They are necessary accounting constraints, not a claim that a DMM is obliged to supply an unlimited quantity at any proposed price.

For the ordinary closing case with a non-zero published Continuous Book Clearing Price p_{CBC} and Imbalance Reference Price r_k , define

$$\Pi_Y^k(\omega, L_D) = \{p \in \Pi : \min(r_k, p_{CBC}) \leq p \leq \max(r_k, p_{CBC}), \quad (33.6) \text{ and applicable conditions hold}\}. \quad (33.7)$$

The interval comes from Rule 7.35B(g); the listed-ETP NAV exception and temporary relief require separate states. The Continuous Book Clearing Price is itself a rule-defined price for satisfying better-priced imbalance interest, not a synonym for the price maximizing $\min(D_C, S_C)$ ([New York Stock Exchange, 2026b,d](#)).

For an opening, use a separate set Π_Y^o built from the opening reference, applicable indications, timing, price conditions and (33.6). The closing interval in (33.7) must not be copied into the opening model. Opening time is a security-specific random time τ_o , subject to the DMM process, rather than a deterministic fill at exactly 9:30 a.m. ([New York Stock Exchange, 2026h](#)).

A forecasting model can now specify a conditional kernel

$$(p_Y, L_D, \tau_Y) \sim \mathcal{H}_{Y,c}(\cdot \mid \mathcal{I}_t, a), \quad p_Y \in \Pi_Y^c(\omega, L_D), \quad (33.8)$$

where a includes the trader's proposed auction instruction. The DMM decision is an uncertain external response for the execution optimizer. It is not a control the customer can set. One may estimate $\mathcal{H}_{Y,c}$, integrate scenario bounds, or explicitly condition on a stipulated DMM contribution in an illustration. None of these choices licenses calling the DMM's unobserved contribution measured hidden investor inventory.

33.5 Allocation: Nasdaq order ranks and NYSE participant parity

Once an auction price is fixed, total paired volume does not determine a child's fill. In a Nasdaq subset without restricted counterparties, sort eligible pieces on a given side in the cross's prescribed order $i_1, \dots, i_m$. The ordinary hierarchy places market-on-auction interest first, followed by the specified displayed class and then non-displayed interest, with the rule's price/time priorities inside the classes ([Nasdaq, 2026d](#)). If V shares on that side can execute, the rank recursion is

$$R_0 = V, \quad F_{i_j} = \min\{n_{i_j}, R_{j-1}\}, \quad R_j = R_{j-1} - F_{i_j}. \quad (33.9)$$

Table 10: NYSE residual allocation after better-priced customer interest. Sources: [New York Stock Exchange \(2026h,d\)](#).

Stage	Opening	Closing
1	Displayed class, Opening D Orders and LOO: participant parity	Displayed class and Closing D Orders: participant parity
2	Non-displayed class: participant parity	Non-displayed class: participant parity
3	Displayed yielding interest: time	LOC: time
4	Non-displayed yielding interest: time	Closing IO opposite the unpaired side: time
5–6	—	Displayed, then non-displayed yielding interest: time

The restriction on counterparties must still be respected; the recursion is not a substitute for the eligibility edges in (33.3).

For NYSE, allocate eligible better-priced customer interest first, and let R be the residual capacity on one side. At-priced interest then passes through auction-specific classes. The following is the ordinary hierarchy, with DMM positioning and exceptional states supplied by the cited rules.

Thus a NYSE LOC at the auction price does not receive the same treatment as a NYSE LOO at the opening price. A Closing D Order is also not interchangeable with an LOC submitted by the same intermediary. Continuous-market setter priority does not carry into the closing auction ([New York Stock Exchange, 2026e](#)).

Inside a NYSE parity class, the allocation wheel identifies the next eligible participant. With round lot ℓ , residual capacity R and participant capacity n_j , an ordinary wheel visit allocates

$$f = \min\{\ell, n_j, R\}, \quad n_j \leftarrow n_j - f, \quad R \leftarrow R - f, \quad (33.10)$$

and advances according to Rule 7.37. Odd-lot handling and pointer retention are additional branches. The Book Participant's allocation is then assigned by working time; Floor Broker allocation has its own internal parity process ([New York Stock Exchange, 2026a](#)). An upstairs order is part of the Book Participant, not automatically a separate slot in the wheel.

In a divisible-share approximation, after higher classes are satisfied, equal participant sharing with capacity constraints is summarized by

$$F_j = \min\{n_j, \theta\}, \quad \sum_j \min\{n_j, \theta\} = \min\{R, \sum_j n_j\}. \quad (33.11)$$

This is a fluid approximation to repeated visits, not NYSE's exact share and pointer rule. For capacities (600, 300, 300) and $R = 600$, it gives (200, 200, 200), which also follows from two complete 100-share wheel rounds. A global FIFO allocation with the 600-share interest first gives (600, 0, 0). The difference is economically material even though the price and total volume are identical.

33.6 A worked clearing example for the two exchanges

Consider the following artificial closing book, in shares. The limit orders are LOCs; there are no reserves, restricted pairs, short-sale restrictions, yielding instructions or binding price protections. This is a small mechanism example, not a calibration or a venue-performance comparison.

Side	Market-on-close	Limit 100.01	Limit 100.02
Buy	300	400	500
Sell	200	600	300

The resulting executable curves are

Price	Demand	Supply	Paired volume	Demand minus supply
100.00	1200	200	200	1000
100.01	1200	800	800	400
100.02	800	1100	800	-300
100.03	300	1100	300	-800

For the Nasdaq example, \$100.01 and \$100.02 tie on maximum volume at 800 shares. Because all interest is MOC/LOC and unrestricted, the tie-break compares the unmatched imbalance at the two tied prices: 400 shares of excess demand at \$100.01 against 300 shares of excess supply at \$100.02. The smaller imbalance selects \$100.02 and 800 matched shares. Buy fills are 300 MOC plus 500 LOC; sell fills are 200 MOC plus 600 LOC at the more aggressive limit. The remaining 300-share sell LOC is unfilled.

For the NYSE example, stipulate a last-published reference price of \$100.00 and Continuous Book Clearing Price of \$100.01. The latter is consistent with this book: at \$100.01 the 800 better-priced customer buy shares can be satisfied by 800 eligible sell shares. With no DMM contribution, \$100.01 is a feasible selected price with 800 matched shares. If the DMM instead supplies 200 eligible sell shares at that price, 1000 shares can match and 200 of the 400-share at-priced buy LOC can execute. DMM participation and price selection are stated assumptions of this NYSE scenario, not quantities inferred from the public imbalance. The Nasdaq maximum-volume tie-break cannot be used to rule out the stipulated NYSE price.

33.7 Auction impact and the optimal committed quantity

A useful local approximation explains why auction liquidity changes the schedule. Suppose all submitted auction shares fill, and a smooth competitive clearing equation is

$$D_0(p) + y = S_0(p), \quad S'_0(p) - D'_0(p) > 0, \quad (33.12)$$

where y is the trader's buy quantity. Implicit differentiation gives

$$\frac{\partial p}{\partial y} = \frac{1}{S'_0(p) - D'_0(p)}. \quad (33.13)$$

With price measured in dollars per share, the denominator has units shares squared per dollar, and $\partial p / \partial y$ has units dollars per share squared. Deeper marginal auction supply reduces price sensitivity. This is a smooth local model: real price grids, tie-breaks, allocation changes and a DMM's discrete response can make $p(y)$ discontinuous. For a smooth NYSE scenario in which DMM response depends only on price, replace the denominator by $S'_0 + \partial_p L_D^s - D'_0 - \partial_p L_D^b$. If it also depends directly on the submitted quantity, the derivative is

$$\frac{\partial p}{\partial y} = \frac{1 + \partial_y L_D^b - \partial_y L_D^s}{S'_0 + \partial_p L_D^s - D'_0 - \partial_p L_D^b},$$

provided the denominator is nonzero. These response derivatives are model assumptions to estimate or stress, not exchange rules.

For a fully filled buy commitment, dollar execution cost relative to s is $C_A(y) = y[p(y) - s + f_A]$. Its marginal cost is

$$C'_A(y) = p(y) - s + f_A + y p'(y). \quad (33.14)$$

The last term matters: increasing the commitment can change the price of every share in the auction order. Treating the indicative price as an exogenous quote drops that effect.

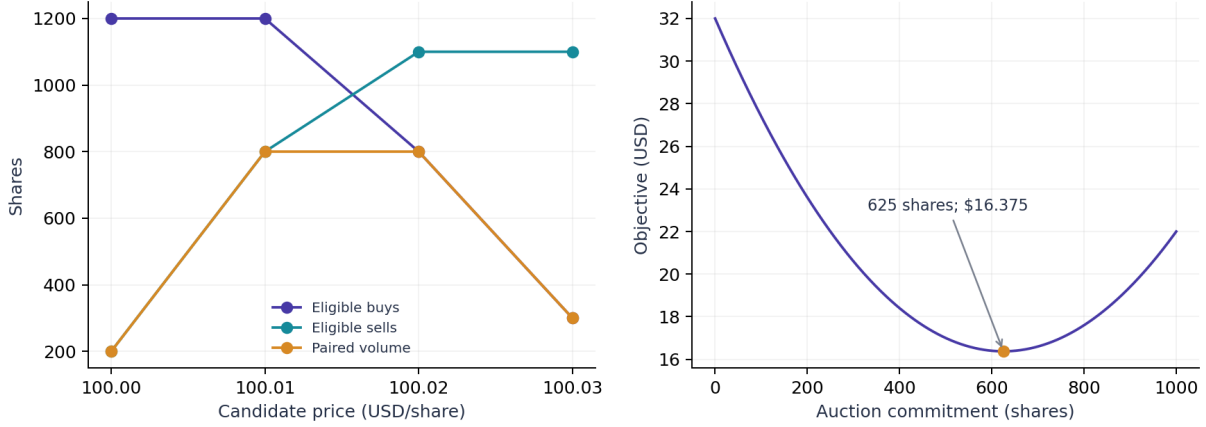


Figure 6: Synthetic auction illustrations: the two candidate Nasdaq prices tie on volume but differ in unmatched interest; the endpoint cost model has an interior optimum at 625 auction shares. Parameters and assumptions are given in the text.

Proposition 33.1 (A schedule with an auction endpoint). *Suppose $Q - y$ shares are completed continuously and y are filled in a final auction. Let their modeled costs be $C_C(x) = a_C x^2 + c_C x$ and $C_A(y) = a_A y^2 + c_A y$, with an additional exposure penalty $r_A y^2$, where $a_C, a_A, r_A \geq 0$ and their sum is positive. Then*

$$y^* = \text{clip}_{[0, Q]} \left(\frac{2a_C Q + c_C - c_A}{2(a_C + a_A + r_A)} \right). \quad (33.15)$$

Proof. The objective is $J(y) = C_C(Q - y) + C_A(y) + r_A y^2$. Its derivative is $2(a_C + a_A + r_A)y - 2a_C Q - c_C + c_A$ and its second derivative is strictly positive. The constrained minimizer is the unconstrained root projected onto $[0, Q]$. $\square$

For $Q = 1000$, $a_C = 2 \times 10^{-5}$, $a_A = r_A = 10^{-5}$, $c_C = 0.012$ and $c_A = 0.002$, the optimum is $y^* = 625$ auction shares and 375 continuous shares. The synthetic objective is \$16.375, compared with \$32 for continuous completion and \$22 for committing all shares to the auction. These amounts combine execution costs and the chosen exposure penalty. Figure 6 and `scripts/auction_examples.py` reproduce the clearing curves, allocation contrast and endpoint calculation.

Full auction fills are a restrictive assumption. With random fill $F(y)$, price $P(y)$ and residual completion cost Φ , a single-auction split becomes

$$J(y) = C_C(Q - y) + \mathbb{E}[F(y)(P(y) - s + f_A) + \Phi(y - F(y))]. \quad (33.16)$$

Assume a common exogenous scenario variable with a distribution independent of y , pathwise differentiable $F(y)$ and $P(y)$, and an integrable bound justifying differentiation under expectation. Then its derivative is

$$J'(y) = -C'_C(Q - y) + \mathbb{E}[F'(y)(P(y) - s + f_A) + F(y)P'(y) + \Phi'(y - F(y))(1 - F'(y))]. \quad (33.17)$$

This expression separates the value of an extra fill, the effect on clearing price, and the cost of an extra unfilled share. In a discrete exchange model, use finite differences of the full action-dependent expected objective instead of differentiating through tick, priority or cutoff discontinuities. If the

scenario distribution itself varies with y , its derivative also contributes a distributional term; a pathwise formula alone does not include that term.

For commitments to both modeled destinations, use their joint law, not independent auction simulations:

$$\min_{x, y_N, y_Y} C_C(x) + \mathbb{E} \left[\sum_{v \in \{N, Y\}} F_v(y_v)(P_v - s + f_v) + \Phi \left(q - x - \sum_v F_v(y_v) \right) \right], \quad (33.18)$$

subject to nonnegative quantities, exchange eligibility and $x + y_N + y_Y \leq q$ after allowing for outstanding commitments. Shared news and order flow couple the outcomes. Auction access may force one commitment to zero; in particular, a non-NYSE-listed security is not eligible for the NYSE primary auction. Deadline constraints and the actual residual opportunity set must make the terminal term attainable. The following valuation discussion applies this accounting to the mandate's benchmark.

33.8 Valuing the auction allocation

Let y be quantity committed to an auction, $F^A \leq y$ its random allocation, and P^A its clearing price. An appropriate terminal opportunity has value

$$\mathbb{E} \left[F^A(P^A - S_0 + f^A) + \Phi(q - F^A, S_T, r_T) \mid \mathcal{I}, y \right], \quad (33.19)$$

where $\mathcal{I}$ includes only known imbalance, reference and phase data. Price and allocation are generally dependent. A limit-on-close order can bound an acceptable price while failing to execute. A market-on-close order does not fix the clearing price.

For intuition only, suppose auction fill is deterministic, its incremental cost is $ay^2 + \mu_A y$, and continuously executing the rest costs $G(Q - y)$. An interior allocation solves

$$2ay + \mu_A = G'(Q - y). \quad (33.20)$$

This is another marginal-value condition, but deterministic fill is an assumption of the example. With uncertain allocation, minimize (33.19) using a joint distribution and a valid residual rule.

Auction commitments consume parent capacity until resolved. After a cutoff, cancellation may be restricted, so the trader loses an option to reallocate quantity. A state variable for cancellability is therefore necessary. If the closing auction is the final admissible opportunity, a simulator cannot repair a missed allocation by assuming an unlimited post-deadline market order. One must reserve earlier completion capacity, accept a specified residual, or change the mandate. Exact cutoffs are instrument-, event- and rule-dependent inputs, not universal constants supplied by the optimization model.

Part VII

Computable certificates under partial observation

34 Certified controllers with hidden liquidity

This part applies [Theorem 29.18](#), [Proposition 29.19](#), and [Corollary 29.20](#) and [Theorems 29.22](#) and [29.26](#), [Corollary 29.23](#), and [Propositions 29.24](#) and [29.25](#) to a finite model in which the

Table 11: Synthetic controlled-liquidity parameters. Transition and fill entries are probabilities; the final column is a per-instruction USD fee. T_{01} is good-to-bad and T_{10} is bad-to-good.

Action	(m_a, ℓ_a)	T_{01}	T_{10}	$p_{a,0}$	$p_{a,1}$	φ_a
Wait	(0, 0)	.05	.18	0	0	0
Lit 1	(0, 1)	.12	.12	.80	.18	.010
Lit 2	(0, 2)	.28	.07	.65	.08	.010
Dark 1	(0, 1)	.04	.15	.70	.03	.025
Market 1	(1, 0)	.35	.06	0	0	0
Market 2	(2, 0)	.60	.03	0	0	0
Split 1+1	(1, 1)	.40	.05	.72	.12	.010

controller does not observe the liquidity regime and its own instructions change that regime. The global certificates are obtained by finite calculation with rational outward enclosures, without an optimization oracle. The remaining numerical checks—the analytic schedule and routing, the liquidity-learning comparison, the exact-completion certificate, the tagged-book rate certificate and signature feedback—are in Appendix D.

34.1 A certified controller with hidden, action-dependent liquidity

This experiment implements [Theorem 29.18](#), [Proposition 29.19](#), and [Corollary 29.20](#). The controller does not observe the liquidity regime, and its actions change the future regime and its reports. Every optimality bound below is a finite-model calculation.

Model, information and feasibility

The parent is 1,000 shares, represented by $q_0 = 10$ units of 100 shares, with $K = 12$ decisions. Hidden liquidity is $H_k \in \{0, 1\}$, good or bad, with initial probability $b_0 = 1/2$ of bad liquidity. At each decision an instruction specifies an immediate market quantity m_a and a passive quantity ℓ_a , in these units. Feasibility requires $m_a + \ell_a \leq q_k$. Unfilled passive interest expires before the next decision; completed parents have only a no-operation action.

After action a , the hidden regime transitions according to T_a . Conditional on $H_{k+1} = h'$, the passive fill is $F_{k+1} \sim \text{Binomial}(\ell_a, p_{a,h'})$ and a public binary signal Y_{k+1} is independent of that fill, with

$$\mathbb{P}(Y_{k+1} = 1 \mid h' = 0) = 0.35, \quad \mathbb{P}(Y_{k+1} = 1 \mid h' = 1) = 0.65.$$

The controller observes its action, own fill, public signal and resulting quantity $q_{k+1} = q_k - m_a - F_{k+1}$. It does not observe H_{k+1} . Table 11 specifies all action-dependent probabilities. The labels *lit* and *dark* name stylized passive alternatives, not an implementation of a venue’s native allocation rules.

For $o = (f, y)$, the primitive finite kernel is

$$M_{q,a,(f,y)}(h, h') = T_a(h, h') \binom{\ell_a}{f} p_{a,h'}^f (1 - p_{a,h'})^{\ell_a - f} z_{h'}^y (1 - z_{h'})^{1-y}, \quad (z_0, z_1) = (.35, .65). \quad (34.1)$$

The stage loss and terminal completion charge, in USD, are

$$\begin{aligned} c(q, a, f) &= m_a + 0.18m_a^2 + 0.08f + \varphi_a + 0.04(q - m_a - f)^2, \\ c_K(q, h) &= 1.8q + 0.15q^2. \end{aligned} \quad (34.2)$$

Table 12: Global certificates under partial observation, in USD. Value endpoints are displayed outward to eight decimals. Gaps are computed from the unrounded enclosures and displayed upward; they need not equal the difference of the displayed value endpoints.

Belief nodes	Optimal-value lower bound	Controller-cost upper bound	Certified gap
9	16.47083929	16.48490582	1.407×10^{-2}
17	16.47985298	16.48561621	5.764×10^{-3}
33	16.48282254	16.48448248	1.660×10^{-3}
65	16.48405766	16.48546396	1.407×10^{-3}
129	16.48416586	16.48439859	2.328×10^{-4}
257	16.48417142	16.48417796	6.530×10^{-6}
513	16.48417251	16.48417296	4.409×10^{-7}
1025	16.48417271	16.48417296	2.413×10^{-7}

The terminal facility fills every remaining share at this stipulated charge. It is part of the synthetic feasibility model. The cost separates immediate execution, passive-fill charges, instruction fees and an inventory penalty.

The hidden law is substantially action dependent. Before observing the next report, a prior of 1/2 leads to bad-state probability 0.435 after waiting and 0.785 after a two-unit market order. Those probabilities are generated by the transition matrices, independently of any belief update. A controller that treats the hidden process as unaffected by trading solves a different decision problem.

Computed global gaps and their interpretation

The implementation solves both finite recursions for $m = 8, 16, \dots, 1024$ subintervals and evaluates each selected table in the physical hidden-state model. It does not simulate by resampling a regime from its rounded belief. The initial prior is exactly a grid node in every case. Table 12 reports a lower bound on the optimum over all policies using the specified observations and an upper enclosure of the deployed controller’s expected cost.

With 1,025 belief nodes, the optimum is enclosed between \$16.48417271 and \$16.48417296, and the controller’s certified gap is at most \$0.0000002413. These small numbers measure planning error within the exact synthetic model; they are not precision claims about market costs. The elementary stage-range bound gives $S_j = 33 + 5.28(12 - j)$ and a uniform mesh guarantee of approximately \$1.09055 at the finest grid. The much smaller computed certificate makes the decision comparison informative.

The deployed policy does not improve monotonically with every refinement: the 17-node policy is slightly more expensive than the 9-node policy, and the 65-node policy is slightly more expensive than the 33-node policy. The certificate, rather than an adjacent-grid difference, determines when the achieved accuracy is adequate. The 257-node case already has a gap below \$0.000007.

Decision comparisons and finite signature implementation

The frozen-prior baseline solves a finite dynamic program which resets its belief to the initial prior after each report, while retaining T_a in each one-step prediction. It is then evaluated under the persistent physical regime. The most-likely-regime baseline filters the reports and applies the fully observed Bellman action for the more probable regime, with a fixed tie rule. The action-independent-transition baseline plans and filters with $T_a = T_{\text{wait}}$ for every action, retaining the action-specific fill probabilities, and is evaluated under the true action-dependent matrices. All three have the same instruction set, loss and completion facility. The first and third are explicitly

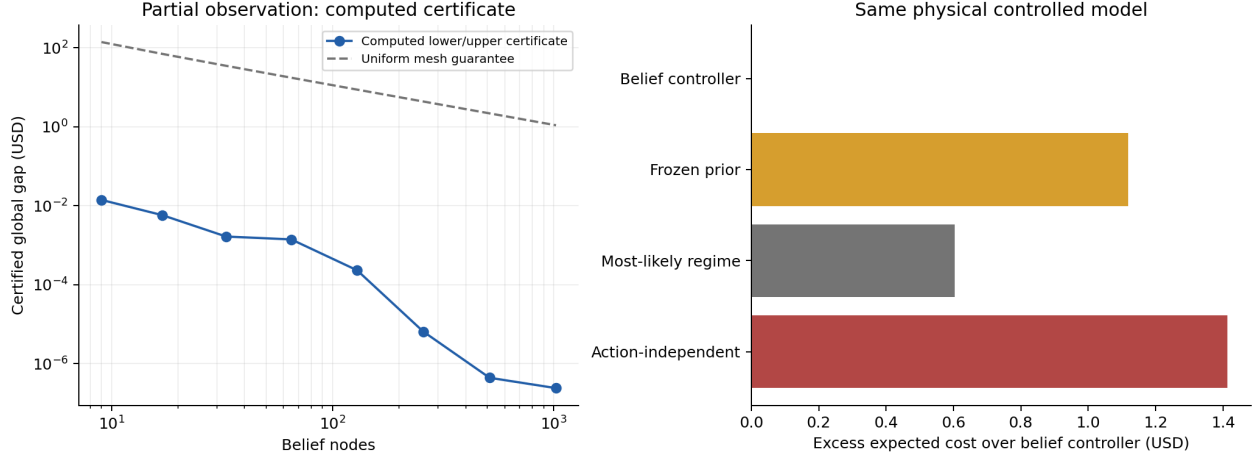


Figure 7: Computed global planning certificates and comparisons in the same controlled hidden-liquidity model. The uniform mesh bound is deliberately conservative. The right panel compares physically evaluated expected costs; no hidden-state observation is supplied to these four controllers.

Table 13: Physical-model policy evaluation, in USD. Costs are rounded upward; excess costs are conservative lower bounds relative to the certified belief controller. Baseline definitions are given in the text.

Policy	Expected-cost upper bound	Excess-cost lower bound
Certified belief controller	16.484173	0
Frozen prior	17.602996	1.118822
Most-likely regime	17.087776	0.603602
Action-independent transition	17.897964	1.413790

specified misspecified-model baselines, not claims to optimality over every controller that omits information.

At the stated initial condition, the belief controller lowers expected objective by approximately 6.36% relative to the frozen-prior baseline and 7.90% relative to the action-independent-transition baseline. A fully observed regime oracle has expected cost \$16.171209. Its lower cost reflects extra information and is not the partially observed optimality benchmark used in the certificate.

The preferred-action table is then converted to the finite-score signature controller of [Corollary 29.20](#). The preferred action has weight w and every other admissible action has weight one. [Table 14](#) evaluates this randomized controller itself, including its effects on future liquidity and beliefs. For $w = 2^{20}$, its certified global gap is at most \$0.00003595. The result therefore covers an implemented finite-score policy rather than only a deterministic limit with infinite logits.

Table 14: The implemented finite-score signature controller, using 1,025 belief nodes. The preferred score is $\log w$; other feasible scores are zero. Costs and gaps are in USD and displayed upward.

Preferred weight w	Expected-cost upper bound	Certified global gap
64	17.04386429	5.597×10^{-1}
1024	16.52062609	3.646×10^{-2}
16384	16.48645748	2.285×10^{-3}
1048576	16.48420866	3.595×10^{-5}

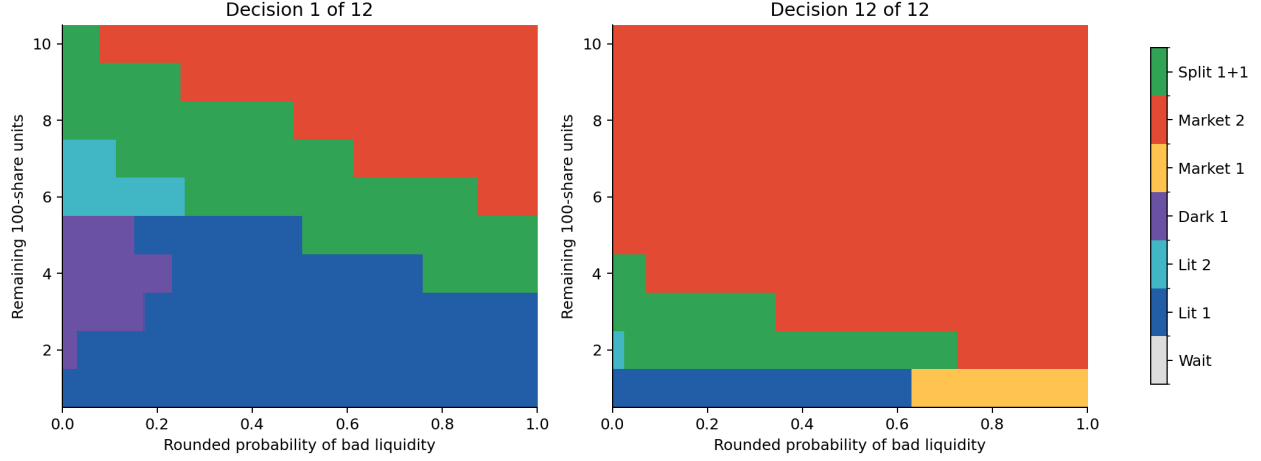


Figure 8: Preferred instructions at the first and last decisions as a function of remaining quantity and observable rounded belief. The hidden regime is not an input. The policy table is encoded by the endpoint of the finite observable summary channel.

Table 15: Prior and deadline sensitivity. All USD columns use outward rounding. The last two columns give lower bounds on baseline excess cost over the certified controller. Every row uses 257 belief nodes.

K	b_0	Controller cost	Global gap	Frozen-prior excess	Transition-blind excess
6	0.25	17.82009	9.353×10^{-7}	0.14683	1.06892
6	0.50	17.83267	5.413×10^{-9}	0.14892	0.80346
6	0.75	17.84515	1.079×10^{-6}	0.01390	0.63533
12	0.25	16.45224	2.106×10^{-6}	2.50259	1.70067
12	0.50	16.48418	6.530×10^{-6}	1.11881	1.42935
12	0.75	16.50961	1.120×10^{-6}	0.25686	1.50526

Sensitivity and independent verification

The same fixed economic parameters are also tested with six or twelve decisions and initial bad-state probabilities $1/4, 1/2, 3/4$, using 257 belief nodes. Table 15 reports all six combinations. The belief controller improves on both specified baselines in every case, although the benefit varies substantially with prior and deadline. This is a sensitivity check of the declared model, not evidence of universal superiority.

All primitive probabilities in (34.1) are integer multiples of 10^{-8} . For grid beliefs, posterior numerators and nearest-node decisions are computed with integer arithmetic, including ties. Elementary nonnegative products, sums and divisions in the bound recursions are enclosed by outward expansion using the adjacent binary64 floating-point numbers. The value enclosures are approximately 1.2×10^{-13} USD wide; the global finite-grid optimization tolerance from (29.38) is below 1.2×10^{-12} USD. The reported policy-approximation gaps are separate from these numerical enclosure widths.

An independent exact-rational exhaustive recursion for three decisions and two units gives

$$V^* = \frac{232172138011}{156250000000} = 1.4859016832704.$$

Its 3,553 distinct recursion states give a value between the grid lower bound and the controller upper bound. Exact-rational evaluation of the deterministic and finite-score controllers is also contained

Table 16: Exact finite-tree observation-reduction checks, in USD. z_0/z_1 denotes the two probabilities of public signal one given next hidden state; m is the number of uniform belief subintervals. The two exact optima and L_0 are displayed downward to six decimals; J_σ and $J_\sigma - L_0$ are displayed upward. The machine-readable results retain every exact rational value.

z_0/z_1	m	V^{rich}	V^{coarse}	J_σ	L_0	Certified gap
0.35/0.65	8	1.485901	1.494181	1.500395	1.314722	0.185673
0.05/0.95	8	1.411837	1.494181	1.552022	0.924095	0.627927
0.05/0.95	16	1.411837	1.494181	1.552022	0.940225	0.611797

in the corresponding outward intervals. A separate forward occupancy calculation agrees with the main backward physical evaluation to better than 10^{-10} USD, conserves total probability to better than 10^{-12} , and visits 9,390 positive-probability stage/memory states. The code checks 24,600 integer posterior-rounding and probability-enclosure cases on the finest grid.

These computations are reproducible without an optimization oracle or a Monte Carlo stopping criterion. They certify global planning accuracy for the stated finite-observation model. Transfer to an estimated market model still requires a justified B_T in (29.40); increasing belief resolution cannot compensate for incorrect transition, reporting or cost assumptions.

34.2 Exact checks of the cost of discarded reports

The finite-observation planning gap and the loss of an omitted observation are different quantities. This experiment checks [Theorem 29.22](#) and [Proposition 29.25](#) on a finite tree where both can be evaluated exactly. It also shows why a fine belief grid cannot recover information that the controller never receives.

Controlled liquidity with a coarser controller

Use the seven instructions, transitions, fills and charges in (34.1)–(34.2), with three decisions, two quantity units and $b_0 = 1/2$. The richer agent observes both the fill and public signal. The implemented summary controller retains the quantity and a rounded filter driven only by fills; it discards the public signal. Its update uses the kernel obtained by summing (34.1) over y . It retains the same exact quantity mask as the richer agent.

All positive-probability fine-report histories under all feasible actions are enumerated with rational arithmetic. Histories are merged only when their stage, quantity, controller memory and exact full posterior agree. At each summary, the smallest and largest of these full posteriors give (29.49). The endpoint recursions then produce L, U and the upper-recursion minimizing policy σ . Independently, full-posterior dynamic programming computes V^{rich} , and fill-history dynamic programming computes the exact coarse-history optimum V^{coarse} . The latter retains the entire fill history and has no rounding error. A separate forward calculation propagates actual hidden-state probabilities under the fixed σ to obtain its physical cost J_σ .

The first row uses the original public-signal probabilities. Its richer optimum is the previously checked $232172138011/156250000000 = 1.4859016832704$. Discarding the signal raises the optimal expected cost by exactly $1293658149/156250000000 = 0.0082794121536$ USD, before any rounding or summary-policy restriction. The remaining rows strengthen only the report, to probabilities 0.05 and 0.95. The coarse model is unchanged because it integrates out that report. Its information loss

risers to exactly

$$V^{\text{coarse}} - V^{\text{rich}} = \frac{205859933299369}{2500000000000000} = 0.0823439733197476 \text{ USD}.$$

Increasing m from eight to sixteen slightly improves the certificate while leaving the selected controller's initial cost unchanged.

These are valid but conservative certificates: the stagewise interval extrema can combine posterior histories that one physical trajectory cannot jointly realize. The upper-recursion policy need not be optimal among summary policies. The reported gap uses the sharper physical evaluation $J_\sigma - L_0$, not $U_0 - L_0$, and is compared with the exact achieved gap $J_\sigma - V^{\text{rich}}$ in the audit output.

There are 1, 24, 324, 3456 distinct full-posterior/memory states at the four stages in each case. Every one of the 349 nonterminal states satisfies $L \leq V^{\text{rich}} \leq J^\sigma \leq U$ in exact arithmetic. The separate forward hidden-state evaluation equals the history-based evaluation exactly and conserves probability at every stage. The maximum discrepancy between the full posterior and rounded memory exceeds 0.43 even in the first case, whereas the one-step rounding radius is $1/16$. This directly checks that those two errors cannot be identified.

A transparent probe example and the sufficiency limit

For a separate two-decision example, one unit remains and good/bad liquidity is static with equal prior probabilities. At the first decision the trader may wait at zero cost or pay 0.02 USD for a report that identifies the state correctly with probability 0.9. At the second decision, a market instruction completes at cost 1 USD. Alternatively, a passive instruction fills at zero cost with probability 0.9 in good liquidity and 0.1 in bad liquidity; any failure is completed at cost 2 USD. Every instruction is feasible and all residuals are completed.

Writing b for the probability of good liquidity in this example, the passive expected cost is $1.8 - 1.6b$. A retained favorable report gives $b = 0.9$ and passive cost 0.36; an unfavorable report gives $b = 0.1$ and selects the market instruction. Thus the richer agent probes and has value

$$V^{\text{rich}} = 0.02 + \frac{1}{2}(0.36 + 1) = 0.70.$$

An agent discarding the report has exact posterior $b = 1/2$, optimally waits, and has value $V^{\text{coarse}} = 1$. Its information loss is 0.30 USD even with an exact coarse filter and no quantization. If the summary remembers whether it probed but discards the report, its post-probe posterior envelope is $[0.1, 0.9]$. The endpoint recursions give

$$L_0 = \min\{1, 0.02 + 0.36\} = 0.38, \quad U_0 = \min\{1, 0.02 + 1\} = 1.$$

The resulting 0.62 USD certificate covers the 0.30 USD actual loss. Retaining the report makes every conditional envelope a singleton; both recursions then equal 0.70, as required by [Corollary 29.23](#). These values are checked as exact fractions.

Action certificates and decision sufficiency

The same three finite trees also check [Theorem 29.26](#). Every one of the 1,237 feasible state-action pairs per case satisfies both action bounds in (29.52) and the Bellman-advantage bound (29.53), using exact fractions. The resulting regret recursion bounds the selected controller at all 349 nonterminal states. Across the three cases, 648, 409 and 417 strict action-screening comparisons identify only truly suboptimal actions. These counts include the different full histories in a common summary fiber.

Table 17: Two certificates for the same implemented controller in the three controlled-liquidity cases, in USD. The cost certificate is $J_\sigma - L_0$ using physical evaluation; the action certificate is R_0^σ . The combined bound takes their minimum. All displayed gaps and bounds are rounded upward to six decimals from exact rational values.

z_0/z_1	m	Actual gap	Cost certificate	Action certificate	Combined
0.35/0.65	8	0.014494	0.185673	0.386751	0.185673
0.05/0.95	8	0.140185	0.627927	1.584820	0.627927
0.05/0.95	16	0.140185	0.611797	1.559358	0.611797

Here the cost certificate is sharper in every row of Table 17; the action recursion is not a uniformly tighter replacement. A separate two-stage case shows the distinction. At the first stage the only feasible action costs 0.02 USD and produces a perfect report of a static, equally likely binary state. The richer agent retains this report and the summary controller discards it. At the second stage both terminal instructions complete the remaining unit by assumption. Instruction L has all-in charges 0.10 and 0.20 USD in the two states; instruction M costs 0.50 USD in either state. These supplied charges define a finite execution example, not an estimated exchange fill law.

The summary’s posterior envelope after the report is $[0, 1]$. Its action bounds are $Q^-(L) = 0.10$, $Q^+(L) = 0.20$ and $Q^-(M) = Q^+(M) = 0.50$. Hence $e(L) = 0$ and $e(M) = 0.40$. The fixed controller choosing L has

$$L_0 = 0.12, \quad J_L = V_0^{\text{rich}} = 0.17, \quad U_0 = 0.22.$$

The physically evaluated cost certificate is 0.05 USD, but the action certificate is exactly zero. It establishes optimality despite the non-singleton envelope and nonzero cost interval. The richer conditional value at the final decision is 0.10 or 0.20, so the summary is *decision sufficient* here without determining that conditional value.

For a randomized controller choosing M with probability $1/4$, physical cost is 0.2575 USD and true regret is 0.0875 USD. The cost certificate is 0.1375 USD while the action certificate is 0.10 USD. The script checks these quantities and the conditional bounds in both hidden states as exact fractions. This also checks the action average in (29.54); randomization is evaluated as implemented.

The script `scripts/observation_summary_checks.py` generates both tables and stores the exact fractions, state counts and conditional checks. The same script checks the Gaussian sign symmetry in Theorem 18.2 and the nonuniform cell created by inserting the initial prior $1/3$ into an eight-subinterval grid. These are finite verification cases supporting the written proofs; they do not constitute a machine-checked proof of the general conditional-envelope theorem.

Part VIII

Implementation and evidence

35 Learning a model suitable for execution

The object to be estimated is the action-dependent joint law

$$p_{\theta}(F, P, Y, B', \ell', I', r' \mid B, \ell, I, r, b, a) \quad (35.1)$$

of fills, prices, reports and next state, given the current state, belief and action. Section 29 supplies one construction: estimate action-dependent primitive intensities and any unidentified mark law, then compose them with the known matching and reporting maps. Three consequences follow from results already proved. First, a generative model of displayed books that is not conditioned on the action cannot evaluate a policy that changes demand, queue composition or disclosure. The relevant accuracy is the uniform primitive error in (29.11), which a good likelihood on the logging distribution does not establish (Theorem 29.5). Second, a belief update takes the trader's own actions as inputs, (28.3); a filter trained only on passive observations can be miscalibrated under an aggressive policy. Third, own impact is identified only by variation in size and timing that is not itself driven by the information the action responds to (Section 11). Historical replay answers the narrower small-intervention question of Section 39.4.

Recurrent or attention-based encoders can parameterize (35.1). A latent coordinate is not measured hidden inventory without an identification argument, and positivity-preserving decoders of spread, gaps and depth enforce valid book geometry, not the parent ledger or exchange priority. For transient impact a simple additional state is

$$I_{k+1} = \rho I_k + \zeta x_k, \quad 0 \leq \rho < 1, \zeta \geq 0, \quad (35.2)$$

with ζ the impact-state loading per executed share; the chosen structure must be checked for profitable round trips as in Sections 6 and 14.

36 From optimization to executable messages

A practical controller uses receding-horizon optimization. At each reliable decision point it updates its state, simulates or integrates candidate continuations, sends the first admissible instruction, and then incorporates the resulting reports before replanning. Optimization frequency should reflect the event rate, computation budget and value of changing a decision. Faster replanning is not useful if it repeatedly forfeits queue position or acts on stale observations.

Execution controller at one decision point

1. Incorporate sequenced market events and authoritative order reports; reject duplicated fills and retain unresolved exposure.
2. Update reference prices, book coverage, session/auction phase, remaining quantity and each child’s reservation.
3. Update liquidity and queue beliefs using the actions actually sent and the observations that have become available.
4. Form the admissible actions from quantity, price, NYSE/Nasdaq destination, native order instruction, participant access, lifetime and mandate constraints.
5. Evaluate joint fill/price scenarios and continuation values, including terminal feasibility, risk and model uncertainty.
6. Choose a first action, reserve its maximum possible exposure, and send the instruction with a unique identifier.
7. Replan after new information or a scheduled decision time. Release capacity only after authoritative execution or terminal cancellation.

The optimization can use convex scheduling, dynamic programming, mixed-integer programming, scenario trees, or a learned policy. The choice depends on which state dimensions and constraints matter. Reinforcement learning is a possible solver for the same declared objective, not a substitute for defining it. Hard quantity and instruction restrictions should be enforced by construction or an admissibility layer; a reward penalty alone does not guarantee them.

The planner must distinguish a requested cancel from a confirmed cancel, a route instruction from an execution, and a venue acknowledgment from a fill. Message arrival order can differ from event time. Market data and private execution reports also have different clocks and sequencing conventions. Nasdaq ITCH and OUCH describe public market data and the participant’s own order lifecycle, respectively; NYSE Pillar FIX describes the NYSE order-entry and execution-report channel ([Nasdaq, 2026a,e](#); [New York Stock Exchange, 2026f](#)). A feed decoder should not invent private lifecycle events absent from public data.

36.1 A two-exchange implementation

The practical interface has four components: a parent scheduler, a NYSE adapter, a Nasdaq adapter, and an authoritative commitment ledger. Each adapter validates native instructions and returns a joint law for acceptance, fills, prices and reports. The scheduler requests execution subject to the mandate; it does not assume that a requested rate is delivered by a passive order. Child orders at both destinations draw from the same parent reservation budget.

For example, a patient buy can compare resting displayed interest with a supported hidden or midpoint instruction on either exchange. When the conditional cost of missing the deadline rises, marketable limits or IOC instructions may become preferable. Approaching the primary closing auction, the controller compares a valid auction commitment with continuous completion. This is a policy comparison within the specified state and access constraints, not a fixed ranking of NYSE versus Nasdaq. Correlated fills, fees, adverse selection and lost cancellation flexibility belong in that comparison.

36.2 Causality and latency

Let τ_e be matching-engine event time and τ_r the time an event becomes available to the controller. The policy filtration uses τ_r , together with completion/sequence rules needed to interpret the event. It cannot condition on an event merely because τ_e precedes the decision timestamp. Similarly, a new

order sees the book at its arrival, not necessarily the book observed when it was sent. A latency model should couple arrival-time depth, price movement and cancellations rather than attach an independent constant slippage to every route.

Latency can be economically asymmetric. It changes the chance of receiving an attractive displayed quote, the chance of canceling before an adverse execution, and the value of waiting for more information. These effects can change the optimal order type even when expected price drift is unchanged.

37 From benchmark gaps to a joint execution law

The usefulness of AC and OW is that their assumptions make a decision transparent. Their limitations should be attached to those assumptions, not attributed to every member of an entire model family. In particular, AC uses linear permanent impact with a constant *coefficient*; the price displacement is not constant. Under the continuous, fixed-quantity, single-asset convention of Proposition 5.3, its permanent cost is schedule-independent. The original AC paper also considers portfolios and a liquidity-adjusted VaR criterion (Almgren and Chriss, 2001). OW explicitly retains transient impact and book resilience (Obizhaeva and Wang, 2013); its block-shaped benchmark is not a model without liquidity recovery.

Likewise, an empirical square-root relation between aggregate metaorder size and price change is not an instruction to replace instantaneous trading cost by a square root. The relevant input, horizon and conditioning must match. A concave price response can produce a convex total charge, while a nonlinear transient specification can admit profitable round trips unless response and decay satisfy additional restrictions (Gatheral, 2010; Curato et al., 2017). Sections 8 and 10 retain those distinctions.

37.1 Six requirements and their present status

Fill law. Sections 24, 25, 29 and 36 already distinguish instructions, venues, reservations, tagged orders and executable messages. A schedule is a requested control, not proof that every requested share fills. A market application still needs venue-specific priority rules, size-dependent partial fills, tick and lot admissibility, and the joint timing of messages, acknowledgements and executions. Replacing an order resets its priority according to those rules; a pending cancel does not release a reservation merely because the trader sent it.

Impact and changing liquidity. Nonlinearity and cross-impact are represented in the structural analysis; they are not empirically calibrated here. An intraday clock, a volatility state and a signal can enter the controlled book law. A U-shaped seasonal profile is a candidate fit to test by venue, asset and day type, not a universal law to impose. Own impact must enter the transition law as well as the current transaction price if it changes subsequent opportunities. A signal-dependent drift must also be separated from price movement caused by the trader.

Information and adverse selection. The belief and observation certificates address discarded information and supplied uncertainty sets. They do not make parameters known. A marked jump can join a fill, a news shock and a deterioration of depth in one event. Calibration must preserve their dependence. Hidden toxicity calls for a belief state or another sufficient observation summary; enlarging a state vector with unavailable information creates an infeasible benchmark. A survival estimator for fill time also needs a stated censoring target: policy-driven cancellation

can be informative, and cancellation and fill can instead be modeled as competing events. Treating cancellation as independent censoring requires justification.

Competition. A controlled book can encode a reduced-form response to displayed size, quote changes and schedule predictability. It does not thereby solve a game among strategic agents. A Nash claim needs opponents' information, action sets and best responses, and a joint equilibrium calculation. The model-risk family should include changes in the response to one's own policy; perturbing only exogenous order-arrival rates misses this channel.

Objectives and constraints. The mandate ledger and terminal condition are part of the problem. Fees, rebates and hedge costs belong in net cash flows. Capital and inventory limits belong in feasibility; drawdown needs wealth and its running peak if modeled as a path constraint. A quadratic risk charge is one declared preference, not a substitute for every such requirement. Adaptive mean-variance and ratio objectives do not in general satisfy the ordinary additive Bellman recursion. Precommitment or an equilibrium interpretation must be specified before claiming time consistency.

Evaluation. The monograph already supplies model-error and robustness certificates. What remains is evidence that their uncertainty family covers the relevant market law. A fixed historical book cannot identify the market response to an unobserved intervention merely by inserting simulated orders. Counterfactual support, dependence stress and impact-aware evaluation are separate from solving the chosen control problem accurately.

37.2 What is shared by the Bellman step

At decision time t , let x retain remaining quantity, live-order reservations, available queue information, impact memory and any observed signal or hidden-state belief. A joint kernel $P_t(dz | x, a)$ describes executed sizes and prices, fees, price/depth changes and message acknowledgements. With a state update F_t and a one-step shortfall charge ℓ_t , a finite-horizon expected-cost problem reads

$$V_t(x) = \inf_{a \in \mathcal{A}_t(x)} \int \{\ell_t(x, a, z) + V_{t+1}(F_t(x, a, z))\} P_t(dz | x, a). \quad (37.1)$$

Every allowed event must preserve the physical ledger. The terminal value specifies completion, an executable fallback, or an explicit failure penalty. Assigning $+\infty$ to noncompletion is a hard constraint, not a claim that completion is always feasible.

This recursion is shared by many models because it is conditional optimization. It does not justify independent fill and price kernels. Nor does inserting a worst-case kernel separately at each node necessarily evaluate one unknown, fixed model: a nodewise adversary permits different choices after different histories. Such rectangular uncertainty is a stronger problem unless the family is stable under this pasting, or the state retains the common latent parameter. The baseline-relative certificate in Section 32 instead compares the proposal and baseline under the same model.

37.3 A dependence term that reverses the routing decision

A two-date completion problem isolates one missing interaction without claiming to describe a calibrated exchange. There are Q shares left. An immediate market action costs C_M . Alternatively, a passive order fills $F \in [0, Q]$, earns a rebate r per share, and the residual is completed at the next date at an incremental cost K per share. Both choices complete the mandate; the fallback is assumed available for the full residual. All prices are relative to one initial benchmark, with K including whatever terminal slippage this supplied law specifies.

Proposition 37.1 (The fill–completion dependence charge). *For square-integrable F, K , the expected passive cost is*

$$C_P = \mathbb{E}K(Q - \mathbb{E}F) - r\mathbb{E}F - \text{Cov}(K, F). \quad (37.2)$$

An independent coupling with the same marginals underestimates this cost by $-\text{Cov}(K, F)$ when the covariance is negative. Its absolute error is at most $\sqrt{\text{Var}(K)\text{Var}(F)}$. If only discrete marginals $p_i = \Pr(F = f_i)$, $q_j = \Pr(K = k_j)$ are supplied, the exact smallest and largest expected costs are the minimum and maximum of

$$\sum_{ij} \pi_{ij} \{k_j(Q - f_i) - rf_i\}, \quad \pi_{ij} \geq 0, \quad \sum_j \pi_{ij} = p_i, \quad \sum_i \pi_{ij} = q_j. \quad (37.3)$$

Proof. Expand $\mathbb{E}[K(Q - F) - rF]$ and use $\mathbb{E}[KF] = \mathbb{E}K\mathbb{E}F + \text{Cov}(K, F)$. Cauchy–Schwarz gives the bound. Every joint law with the stated finite marginals is a feasible matrix π , and every such matrix defines a joint law. The feasible set is nonempty and compact, so the linear objective attains both extrema. $\square$

Take $Q = 1$, $r = 0.001$, $C_M = 0.06$. A paired three-event law has

$$(F, K) = (1, 0.02), (0.5, 0.08), (0, 0.30), \quad p = (0.45, 0.35, 0.20).$$

Both this law and the independent coupling have $\mathbb{E}F = 0.625$ and $\mathbb{E}K = 0.097$. Nevertheless, the independent cost is 0.035750, whereas the paired cost is 0.073375: the preferred choice changes from passive to immediate execution. The loss from using the independently selected action under the paired law is 0.013375 per unit parent quantity. Under a convex mixture with paired-law weight θ , passive cost is $0.035750 + 0.037625\theta$, crossing the market cost at approximately $\theta = 0.64452$. These are normalized synthetic costs, not market bps.

Table 18: Same marginal fills and completion prices; different joint laws. The marginal-only worst case is an extremum of (37.3). All values are synthetic.

Law	Passive expected cost	Selected action
Independent	0.035750	Passive
Paired	0.073375	Market
Marginal-only worst case	0.073375	Market

The calculation in `scripts/realism_stress.py` compares exact rational expectations with the marginal-coupling linear program and checks the routing switch on a mixture grid. It is a dependence stress, not an identification of endogenous impact. If K also changes with submitted size or timing, that response requires its own controlled law.

37.4 A falsifiable evaluation contract

Section 40.5 turns these requirements into a preregistered evaluation contract.

38 Completion-preserving execution on discrete, delayed venues

The preceding impact and execution certificates do not by themselves specify an order gateway. This section completes the interface between a parent mandate, discrete child decisions, delayed private reports and a finite decision model. The required ingredients are a pathwise commitment

ledger, a viability test and an explicit separation between optimization and message delivery. Their guarantees concern the declared event maps; exchange conformance remains a separate validation task.

38.1 Normalized reports and a conserved parent ledger

Consider a buy parent of Q integer shares. Let F be cumulative confirmed fills, $r = Q - F$, and o_i the upper bound on additional executable shares of child i , including sent but unresolved messages.

Remark 38.1 (Delayed cancellation and replacement). In the notation of this section, with F cumulative confirmed fills and o_i the reservation of child i , the invariant of Proposition 27.3 reads

$$0 \leq F \leq Q, \quad o_i \geq 0, \quad \sum_i o_i \leq Q - F. \quad (38.1)$$

That proposition already covers messages in flight. Two of its clauses carry the weight at a gateway. First, sending a cancel changes no reservation; only an authoritative terminal report, after its cumulative executions are reconciled, releases the remainder. Second, a replacement that can coexist with the old order is two possibly executable children and must reserve both quantities until the old order is resolved. A smaller joint reservation is admissible only when the venue's replacement is atomic and its treatment of in-flight fills is established.

The normalized report contract is part of the proof. A terminal report may arrive through a different channel from execution messages; receipt of a public delete alone does not establish that all private fills were booked. The adapter needs sequence recovery and reconciliation, including duplicate reports and session-scoped identifiers. A trade correction is a new ledger event, not a duplicate. During an unresolved gap, keep conservative exposure and prevent new commitments that rely on the missing information. The reference code verifies this normalized contract; it does not implement the entire Nasdaq OUCH or NYSE Pillar protocol (Nasdaq, 2026e).

For $Q = 10$, a child reserving six shares leaves room for only four more. If a cancel request incorrectly releases those six shares, a replacement of ten can coexist with the original six and expose the parent to sixteen. Correctly retaining the reservation prevents this overcommitment before any distributional assumptions about fills are used.

38.2 Feasibility is stronger than an expected fill forecast

Let $\mathcal{S}_k(x, a)$ be the union of possible successor states over the admitted event laws, including delayed-message outcomes. The state contains both r and all o_i . Define a terminal set $\mathcal{V}_N = \{x : r(x) = 0, \sum_i o_i(x) = 0\}$ and recursively

$$\mathcal{V}_k = \{x : \exists a \in A_k(x), \mathcal{S}_k(x, a) \subseteq \mathcal{V}_{k+1}\}. \quad (38.2)$$

Theorem 38.2 (Robust completion by an admissibility mask). *In a finite-horizon finite-state model with nonempty outcome supports, $x \in \mathcal{V}_k$ if and only if there is a policy that reaches $\mathcal{V}_N$ for every allowed sequence of outcomes from x . Choosing only actions whose successors lie in $\mathcal{V}_{k+1}$ preserves this guarantee under arbitrary reoptimization within that mask.*

Proof. At N the assertion is the definition. If an action has every successor in $\mathcal{V}_{k+1}$, the induction hypothesis supplies a completing continuation after each successor. Conversely, a policy completing under every outcome must choose a first action all of whose successors admit a completing continuation, hence satisfy (38.2). The argument applies after each realized transition and proves the reoptimization statement. $\square$

This is the finite-horizon target-tube recursion of Bertsekas and Rhodes (1971); Aubin (1991) develops the general viability theory. Its use here is the explicit inclusion of unresolved messages in the successor supports.

This is a worst-case support guarantee, stronger than high expected fills. It can be empty if a halt, failed connectivity or unavailable liquidity makes completion impossible. The optimizer must then report infeasibility or a separately authorized mandate relaxation; it cannot manufacture liquidity through a terminal penalty.

A useful terminal specialization has no unresolved orders after the current stage, a remaining requirement r , and a guaranteed next-stage immediate capacity C . If the current action produces fills $f \in [f_{\min}, f_{\max}]$, with integer bounds and $0 \leq f_{\max} \leq r$, completion for every outcome is possible exactly when

$$r - f_{\min} \leq C. \quad (38.3)$$

Necessity follows from the worst fill; sufficiency follows by purchasing $r - f$ at the next stage. With $r = 5$, $C = 3$, an action guaranteeing at least two fills is feasible. A passive order with mean fill four but a possible zero fill is infeasible for this guarantee. When live children remain, (38.3) alone is insufficient: cancellation resolution and joint outstanding exposure belong in the full viability recursion.

38.3 Integer allocation rather than independent rounding

Ticks constrain prices, and share or lot grids constrain quantities. The continuous allocation problem is a benchmark; its components cannot be rounded independently without checking total quantity and capacity. Consider m destinations, integer capacities C_i , integer parent Q , and separable costs $c_i(n)$ for $n = 0, \dots, C_i$, with $c_i(0) = 0$. Assume discrete convexity: the marginal costs $d_i(n) = c_i(n) - c_i(n-1)$ are nondecreasing in n .

Classical result 38.3 (Exact integer marginal allocation). *If $Q \leq \sum_i C_i$, repeatedly assigning the next unit to a destination with the smallest available next marginal cost gives a feasible minimizer of $\sum_i c_i(n_i)$ subject to $\sum_i n_i = Q$ and $0 \leq n_i \leq C_i$. Ties can be resolved by any fixed deterministic rule.*

Proof. Every feasible allocation selects Q marginal costs and, for each destination, a prefix of its nondecreasing sequence. The greedy selection always takes a smallest unselected available marginal. Any unavailable later marginal is at least its available predecessor, so no cheaper unselected marginal is missed. Thus it selects Q globally smallest marginals in a prefix-feasible order. No other sum of Q marginals is lower. $\square$

Greedy marginal allocation for separable discrete-convex costs is classical (Gross, 1956; Ibaraki and Katoh, 1988); it is recorded here because the separable convex routing cells of Section 27 satisfy its hypotheses once lot and capacity grids are fixed.

For $c_1(n) = n^2/2$, $c_2(n) = n^2$, $Q = 5$ and capacities at least five, the continuous optimum is $(10/3, 5/3)$ with cost $25/3$. The integer optimum is $(3, 2)$ with cost $17/2$ and an integrality gap $1/6$. This example concerns guaranteed quantities with separable costs. Random fills, coupled impact or shared fee tiers generally destroy separability; then the finite joint action set must be optimized directly. Likewise, rounding a price can change queue rank and fill law discontinuously, so a smooth rounding-error estimate is not automatic.

38.4 Joint scenarios and a runnable reference controller

At each decision, use outcomes of the form

(fills, execution prices, fees, elapsed time, reports, next book state).

The planner first applies the reservation and viability masks, then compares negative implementation shortfall plus continuation reward. Report each cost component in cash or a clearly stated per-parent normalization; a fill-conditioned spread is not the same unit as full-parent shortfall. A marketable limit can control price while failing to complete; a completion assumption must therefore include the available liquidity and allowable prices, not just an order label.

The package includes a deterministic discrete-event engine with separate exchange-event and local-receipt times, a tagged FIFO queue cell and a normalized commitment ledger. The FIFO cell is a declared mathematical mechanism. It is not a representation of NYSE’s participant parity rules (New York Stock Exchange, 2026e). The simulator retains fills between a cancel request and its effective exchange cancellation. Receipt-time callbacks cannot read future exchange events. The example records both clocks and tests a fill that precedes effective cancellation even though the cancel was sent first. Transport gap recovery and exchange-specific binary decoding remain adapter obligations, not implied capabilities of the reference engine.

A production loop has four ordered operations: reconcile available reports; construct an admissible joint action; reserve and send its actual child messages; then evaluate the emitted action and its outcomes. The calibration and implementation theorem in Section 39 compares this emitted policy to the optimum in the same feasible model. Hardware timeouts and stale observations must be represented in its state or error allowances. They are not free perturbations of a schedule.

38.5 Synthetic end-to-end checks and interpretation

Table 19: Execution implementation checks. Share counts and costs are synthetic; the latency trace uses arbitrary clock units.

Quantity	Value
Parent size (shares)	10
Reservation retained on cancel request	6
Fill before cancellation is effective	3
Integer allocation cost	8.500000
Continuous allocation cost	8.333333
Integrality gap	0.166667

The new suite exhaustively compares greedy integer allocation with all feasible small allocations, checks the worst-fill capacity condition, and exercises partial fills, duplicate normalized reports, rejection, non-atomic replacement, cancel races and conservative recovery states. Random finite decision models independently check the policy-transfer bound by exact backward evaluation. These checks support the stated algorithms and identities.

38.6 Persistent flow and endogenous conditioning

Under a fixed stationary Markov policy, a finite irreducible continuous-time book chain is exponentially mixing. Bounded state observables then have exponentially decaying stationary covariances. This does not follow for arbitrary adaptive controls from the controlled-book assumptions alone.

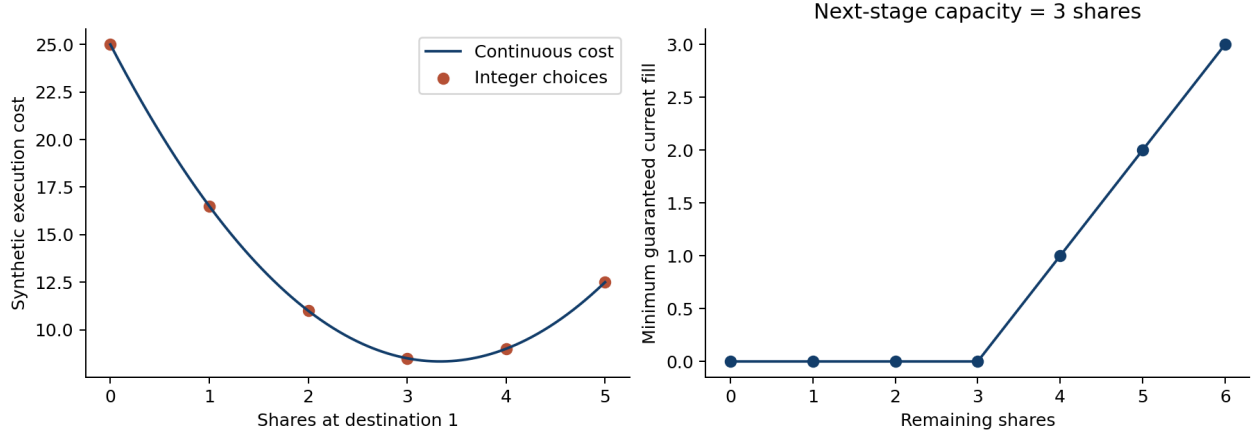


Figure 9: Discrete implementation diagnostics generated by the accompanying reference suite. All axes use synthetic model units; the curves illustrate the declared mechanisms rather than estimated market behavior.

Rodriguez Dominguez (2026b) derives long-memory flow from switching frictions and heterogeneous horizons. A finite-state approximation to that asymptotic persistence requires an explicit truncation or model-error allowance, charged via the primitive budget of Theorem 29.5 or the reference defect of Proposition 29.13; a finite state fitted to autocorrelations alone carries no such charge. Rodriguez Dominguez (2026a) makes the conditioning representation itself an equilibrium object. For the present results this bears on one hypothesis: the reference law in Proposition 29.13 and the uncertainty family in Section 32 are fixed, whereas widespread adoption of a common representation can move the law those bounds must contain.

39 Calibration, implementation error, and the evidence boundary

This section supplies a finite, explicit interface between a fitted event model and an implemented policy. The inequalities are elementary finite-horizon perturbation and concentration results; their purpose is to make the assumptions and units of a deployment claim auditable, not to claim new concentration theory.

39.1 An observed decision process and its clocks

Let $k = 0, \dots, N$ index decision opportunities. A state x records the available public history or a declared sufficient summary, private order reports, outstanding commitments, clock time and remaining horizon. A report belongs to the controller’s information only after receipt and sequence recovery. An exchange timestamp is not a receipt timestamp. For genuinely hidden state, x must be a posterior or an observation history; replacing it by a point estimate needs an additional reduction allowance. An event-time model must retain holding times or calendar time whenever costs, signal decay or the terminal deadline depend on them. If the number of events before a deadline is unbounded, the finite N result below applies to a declared truncation or batching scheme, with an additional tail error.

A feasible action $a \in A_k(x)$ is a message or a deliberately empty message. Write $P_k^a(x, \cdot)$ for the next-state kernel and $r_k(x, a)$ for the conditional mean net reward accrued before the next decision. Partial fills, fees, simultaneous price changes and elapsed-time charges are jointly accounted before

taking this mean. For execution, reward is negative cost. The additive objective is

$$J(\pi) = \mathbb{E}^\pi \left[\sum_{k=0}^{N-1} r_k(X_k, A_k) + g(X_N) \right].$$

This definition does not identify risk-penalized reward with cash profit. A realized random reward can replace r_k inside the expectation.

All comparisons start from the same fixed initial state or initial law. The model and deployment use the same observation class and feasible sets. A support error that admits overfills or a missing action that changes feasibility is not repaired by a small reward-error bound. A conservative common feasible mask is permitted; the resulting benchmark is the optimum within that mask. Comparison with a larger action class needs a separate restriction gap.

39.2 A finite-horizon transfer with a compiled policy

Let hats denote the fitted model. Total variation uses the convention $\text{TV}(p, q) = \frac{1}{2} \sum_y |p_y - q_y|$. Suppose, uniformly over feasible states and actions,

$$|r_k - \hat{r}_k| \leq \eta_k, \quad \text{TV}(P_k^a, \hat{P}_k^a) \leq \epsilon_k, \quad |g - \hat{g}| \leq \eta_N.$$

Let S_k bound the span of every fitted-model continuation value of a policy in the comparison class, including its optimal value. Bounded stage rewards give the simple choice $S_k \leq \sum_{j=k}^{N-1} R_j + R_N$, where R_j is a common range width across all states and actions of fitted stage reward and R_N is the terminal range width. Set

$$B_N = \eta_N, \quad B_k = B_{k+1} + \eta_k + \epsilon_k S_{k+1}. \quad (39.1)$$

Theorem 39.1 (Statistical and implementation errors in one objective). *Under these assumptions, $|J(\pi) - \hat{J}(\pi)| \leq B_0$ for each common feasible policy, and $|J^* - \hat{J}^*| \leq B_0$. Suppose the actual implemented policy $\tilde{\pi}$ satisfies*

$$\hat{V}_k^*(x) - \hat{Q}_k^*(x, \tilde{\pi}_k(x)) \leq \zeta_k(x), \quad \zeta_k(x) \geq 0,$$

where $\hat{Q}_k^* = \hat{r}_k + \hat{P}_k \hat{V}_{k+1}^*$. Then

$$J^* - J(\tilde{\pi}) \leq 2B_0 + \hat{\mathbb{E}}^{\tilde{\pi}} \sum_{k=0}^{N-1} \zeta_k(X_k). \quad (39.2)$$

The expectation on the right is under the fitted deployed-policy law. It may be bounded by $\sum_k \sup_x \zeta_k(x)$.

Proof. For a fixed action and continuation v , $|(P - \hat{P})v| \leq \text{TV}(P, \hat{P})(\max v - \min v)$. Insert and subtract the fitted continuation in the one-step recursion. Backward induction gives (39.1) for a fixed policy and, since the action sets agree, for the optimal Bellman operators. Telescoping the fitted optimal value along the implemented policy gives $\hat{J}^* - \hat{J}(\tilde{\pi}) = \hat{\mathbb{E}}^{\tilde{\pi}} \sum_k [\hat{V}_k^*(X_k) - \hat{Q}_k^*(X_k, \tilde{\pi}_k(X_k))]$. Insert the two model-comparison bounds to obtain (39.2). The same proof applies to history states. $\square$

Remark (Relation to the path-coupling certificate). When x is the full structural state of Section 29 and the decision-time maps agree, Theorem 29.5 applied to one decision interval supplies the transition budget: if the one-clock mark laws differ by at most η in total variation, then $\epsilon_k \leq$

$1 - e^{-\Lambda\eta(t_{k+1}-t_k)}$ in (39.1). Neither certificate dominates the other. Corollary 29.6 charges one loss range L against the whole-path mismatch $1 - e^{-\Lambda\eta T}$, whereas (39.1) weights each interval's mismatch by the span S_{k+1} of its own remaining continuation and additionally accepts reward-mean errors η_k and history states.

If a fixed pair of baseline and candidate policies has fitted improvement $\hat{\Delta}$, its true improvement is at least $\hat{\Delta} - B_0^{\text{candidate}} - B_0^{\text{baseline}}$; $2B_0$ is a common-envelope choice. This certificate may be conservative. A shared-law advantage comparison can be tighter because common errors cancel. The implementation allowance must evaluate the action actually emitted after rounding, risk checks and timeouts. An optimizer tolerance before those transformations is not an implementation guarantee.

39.3 A transparent finite-sample construction

Proposition 39.2 (Simultaneous coverage of a finite calibration table). *Consider L prespecified state-action-time rows, each with at most M next-state categories. In row ℓ , collect a fixed number $n_\ell \geq 1$ of independent draws from that row's joint reward/next-state law. The reward has a known range width R_ℓ . Independence between reward and next state, or between different rows, is not needed. Let empirical means and frequencies define $\hat{r}_\ell, \hat{P}_\ell$. For $0 < \delta < 1$, define*

$$e_\ell = \min \left\{ 1, \sqrt{\frac{(M+2) \log 2 + \log(L/\delta)}{2n_\ell}} \right\}, \quad (39.3)$$

$$d_\ell = R_\ell \min \left\{ 1, \sqrt{\frac{\log(4L/\delta)}{2n_\ell}} \right\}. \quad (39.4)$$

With probability at least $1 - \delta$, all rows satisfy $\text{TV}(P_\ell, \hat{P}_\ell) \leq e_\ell$ and $|r_\ell - \hat{r}_\ell| \leq d_\ell$ simultaneously. Consequently Theorem 39.1 holds on that same event, even for a policy selected using the table.

Proof. For a fixed subset C of next-state categories, the empirical indicator mean differs from $P_\ell(C)$ by more than u with probability at most $2e^{-2n_\ell u^2}$. There are at most 2^M subsets; total variation is the maximum of these absolute differences. A union bound over rows gives $L2^{M+1}e^{-2n_\ell u^2} \leq \delta/2$ with the unclipped rowwise choice. The bounded-variable inequality gives a further failure probability at most $\delta/2$ for the reward means. The deterministic bounds 1 and R_ℓ justify clipping. On their simultaneous event the preceding theorem applies uniformly, so subsequent policy selection does not require another union bound over policies. $\square$

This deliberately conservative construction is a complete positive result for a stated sampling experiment. It is not an assertion that adjacent market events are independent. A single dependent book path, data-dependent bins, adaptive sample counts or hidden confounding do not satisfy it merely because many messages were recorded. Such settings require their own martingale, mixing or block-sampling argument. A row with no supported observations receives the trivial bounds $e_\ell = 1$ and $d_\ell = R_\ell$, or its action is excluded; it does not receive zero error.

If the deployment row is within total variation v_ℓ of its calibration law and its mean reward has changed by at most w_ℓ , use $\epsilon_\ell = \min(1, e_\ell + v_\ell)$ and $\eta_\ell = d_\ell + w_\ell$. This follows by the triangle inequality. The drift budgets must be supplied or validated; a regime detector alone does not prove them. A latency-tail, unmodeled-event or state-reduction allowance can enter the same budgets only after its corresponding coupling or mean-error argument is stated. There is no blanket independent-delay assumption in this finite-kernel result.

39.4 Replay, interventions and supported policy evaluation

Public order-by-order data support displayed-book reconstruction. They do not reveal all latent liquidity: the Nasdaq ITCH specification distinguishes non-displayed executions from displayed-order lifecycle events (Nasdaq, 2026a). A simulation of an inserted order additionally chooses its arrival time, priority and the market’s response. Replaying unchanged external messages is a small-intervention modeling assumption, not an observed response to the inserted order. Raw packet replay helps reproduce transport and decoder behavior; a correctly ordered event stream can be sufficient for an economic experiment that explicitly supplies transport delays. Neither format creates missing counterfactual data.

A simple positive off-policy statement makes the missing assumptions precise. Consider independent one-decision records (X_i, A_i, Y_i) from a logging policy β , a candidate policy π fixed independently of these evaluation records, and $|Y_i| \leq M_Y$. Suppose the conditional outcome law given (X, A) is invariant under the intervention, actions are conditionally randomized or unconfounded given X , and $\beta(a | x) \geq b_0 > 0$ whenever $\pi(a | x) > 0$. The importance-weighted estimate

$$\hat{J}_{\text{IS}} = \frac{1}{n} \sum_i \frac{\pi(A_i | X_i)}{\beta(A_i | X_i)} Y_i$$

is unbiased for the candidate value on the logging context distribution. Conditioning on X , summing over A cancels β and proves this claim. Each summand lies in $[-M_Y/b_0, M_Y/b_0]$, so the bounded-variable inequality gives the fixed-sample confidence radius

$$\frac{M_Y}{b_0} \sqrt{\frac{2 \log(2/\delta)}{n}}.$$

This elementary instance illustrates the established high-confidence off-policy evaluation programme (Thomas et al., 2015). For sequential policies one generally needs trajectory or sequential density ratios and support for the induced histories; one-step weights are insufficient. An omitted confounder or an opponent response that changes with deployment invalidates the invariance assumption. At an unlogged action, two laws can agree on every observed record and assign opposite candidate rewards. No amount of additional data from the same unsupported policy identifies that comparison.

40 What would constitute market evidence?

40.1 Data requirements matched to the claim

Data selection should follow the intended claim. Market-by-price data provide aggregate price-level states. Market-by-order data preserve more of the displayed order lifecycle, but does not reveal all hidden orders or investor inventory. Auction statistics, imbalance and status supply different channels. Even a full displayed book at one exchange is not the national market. A faithful decoder must respect native message types, execution identifiers and printability flags to avoid counting one execution more than once (Nasdaq, 2026a).

40.2 Evaluation of forecasts and decisions

Forecast diagnostics should cover fill-size distributions, time-to-fill, non-fill probability, conditional markouts and dependence across venues. Book energy scores or price CRPS remain useful for their own targets but cannot establish execution accuracy. Queue depletion, cancellation survival, replenishment and auction allocations require separate checks.

Table 20: The evidence required grows with the strength of the execution claim.

Claim	Required evidence
Displayed-book prediction	Causal book states, quality flags, chronological splits and calibrated distributional scores
Passive fill prediction	Private order lifetimes/fills or defensible queue replay, cancellations, censoring and latency
Reserve inference	A validated observation model, native identities or private order evidence where available, and sensitivity to non-identification
Auction execution	Imbalance/statistics/status feeds, accepted order commitments, allocation and effective cutoff rules
Cross-venue routing	Synchronized relevant books/references, route and execution logs, fee eligibility and joint fill dependence
Large-order impact	Data supporting counterfactual response estimates, with action selection and market conditions addressed
Policy improvement	Comparable mandates, costs, residual treatment, tail outcomes and an evaluation design supporting the policy comparison

Policy diagnostics should include arrival-price shortfall or the specified benchmark loss, total charges, completion probability, residual size, tail cost, participation, market impact, and message intensity. Report conditional results by spread, volatility, time of day, order size, venue and liquidity regime. A lower mean cost can conceal larger non-completion risk or a worse tail, so the same mandate must be applied to every comparator.

Useful baselines include rounded TWAP, causal volume schedules, participation with a completion rule, immediate execution, a calibrated parametric scheduler, and a simple queue-aware order-placement policy. A neural controller should earn its complexity against these controls. A reserve-input ablation should hold targets and training procedures fixed while comparing observed-book inputs, observable execution flow, and the same inputs with inferred stock.

40.3 Selection bias and counterfactual policies

Logged fills are conditional on a historical policy. Orders that were never submitted have no observed counterfactual fill. Training and evaluating an alternative policy on those logs therefore require assumptions about support, state sufficiency and action-dependent response. A replay simulator supplies one such model; it does not erase selection bias by replaying faster.

For a one-decision problem, a logged-action importance weight is $w = \pi(a | s) / \mu(a | s)$, where μ is the behavior policy. A zero denominator on actions selected by π destroys identification by that method. Sequential execution compounds the problem because earlier actions change inventory and observations. Model-based evaluations should state their counterfactual assumptions; controlled prospective comparisons should retain mandate constraints and monitor meaningful outcome differences. A random train/test split of nearby events is not a substitute for chronological validation across independent dates.

The intended sequence is mechanical verification, distributional calibration, decision evaluation under stated assumptions, and then appropriately designed prospective evidence. None of these steps certifies the others. The synthetic study in Section 34 completes numerical checks within its

declared model; the data-dependent stages remain an empirical research task.

40.4 Two conditional restrictions and their empirical tests

Most evidence requirements in Table 20 are demanding. Two results nevertheless supply specific restrictions that can be tested without deploying a counterfactual execution policy. Their data requirements are identification requirements: public price and trade records alone need not identify either the strategic trader count or complete metaorders.

(i) *The temporal-skew coefficient.* Under the symmetric game of Section 19, Theorem 19.2 gives

$$\Gamma + \frac{1}{N}\Gamma^* = a_N\Sigma + b_N\Lambda, \quad a_N = \frac{N+1}{N}, \quad b_N = \frac{N-1}{N}.$$

The null restriction is $b_N/a_N = (N-1)/(N+1)$, after separating the temporary-cost term $2\eta I/N$ and fixing the normalization of Σ and Λ . It concerns the equilibrium first-order operator, rather than the price-flow response kernel. A test therefore needs an identified common response operator, observed or otherwise identified strategic participation N , and data supporting the symmetric-mandate specification. Remark 19.5 lists the main confounds. With those quantities identified, fit the skew coefficient freely and test the stated equality; across comparable participation groups, the null also pins down its monotone dependence on N . A raw transaction count is not the strategic trader count in this model.

(ii) *The joint size-tail restriction.* Corollary 15.4 links the impact exponent δ , the metaorder tail index α , and the permanent fraction c :

$$\delta = \alpha - 1, \quad c = \frac{1}{\alpha}.$$

With identified metaorder sizes and impact paths, estimating α and δ freely gives one scalar equality to test. If permanent displacement is also identified, estimating c adds the second equality. A comparison using average execution displacement at finite Q must retain the cutoff correction from Theorem 15.3,

$$\frac{1}{Q} \int_0^Q I(u) du - cYQ^\delta = \frac{Y\delta q_0^{1+\delta}}{(1+\delta)Q},$$

or explicitly specify a different boundary model. The test must distinguish metaorder size from individual prints and address selection in size, horizon and liquidity. Inference must account for estimating the exponents, cutoff and nuisance parameters from dependent observations; the number of scalar equalities alone does not establish a reference distribution for a test statistic.

These are tests of combined structural assumptions, and a rejection does not identify which assumption failed. Neither requires a counterfactual policy deployment or a replay simulator. Both require suitable trader-level or metaorder information, or a justified reconstruction and identification model. They are empirical targets, not tests already performed by the synthetic experiments in Section 34.

40.5 A preregistered evaluation contract

Fix the parent mandates, causal information, fee convention, baseline, candidate generation rule and uncertainty calibration before any policy is selected. Split by trading day into chronological training, calibration and test periods, and keep overlapping outcome windows within one period. Compare a simple cell with successive additions of queue and partial-fill state, message delay, joint fill–mark laws, time-varying liquidity, own impact and opponent response, and stress their interactions jointly: marginal-only ablations cannot detect the covariance term in Proposition 37.1. Report completion probability and residual-size tails alongside shortfall, fees, fallback cost and violations of the declared uncertainty bounds; a lower mean cost obtained by leaving difficult shares unfinished is not improved execution.

Retain the exchange and receipt clocks, data and protocol versions, private acknowledgements and fills, the fee schedule, position and capital limits, model and executable hashes, and the calibration split. Report separately (i) matching and accounting correctness, (ii) conditional coverage and drift, (iii) realized order outcomes, and (iv) processing and network-delay distributions under load. Synthetic verification, historical replay and an identified intervention study answer different questions, and none should be labelled as another. Operational recovery, exchange certification and hardware timings cannot be inferred from the mathematics or the synthetic evidence.

Part IX

Outlook

41 Research questions beyond the benchmarks

1. **Constrained nonlinear transient execution.** Determine existence, global optima and quantitative approximation error for a specified nonhomogeneous response, rate bounds, spread and risk objective. The inequality $\beta + \delta \geq 1$ alone does not settle any of these questions.
2. **Mechanisms that break scale invariance.** Test whether observed size-dependent schedules are explained by rate caps, endogenous horizons, changing liquidity or nonhomogeneous impact. The homogeneous unconstrained power-response model itself predicts invariant normalized shape.
3. **Causal cross-impact estimation.** Estimate matrix kernels under explicit temporal and round-trip constraints and quantify the effect of the constraints on out-of-sample execution loss. PSD projection is a known tool; choosing and identifying the correct causal model is a separate problem.
4. **Liquidity learning under intervention.** Identify how the trader’s own orders change fills, observations and later depth. Compare a full posterior with simpler observable state and report decision loss as well as fit scores. Section 29.8 supplies the certificate side of that comparison; the open task is constructing validated posterior envelopes from realistic reports and timestamps.
5. **Auction allocation under incomplete observation.** Estimate a joint law for eligible interest, price, DMM participation and own fills, with effective NYSE/Nasdaq rules. Public imbalance data alone do not determine all allocation state or identify investor inventory.

42 Conclusion

Execution is the joint choice of quantity over time, instruction and destination under uncertain completion. Impact benchmarks are useful when their response, clock, cost convention and feasible

strategies are explicit. Linear impact gives quadratic energy and exact exponential or power-law examples. Nonlinear response requires separate admissibility and existence analysis; power homogeneity still fixes size scaling, while a power kernel also changes horizon scaling. A necessary round-trip restriction is not a certificate for a complete model.

The marginal continuation cost of an unexecuted share connects a schedule to order placement. Under separable convex costs it yields marginal-cost routing; with uncertain fills, queue persistence or information effects the choices belong in a joint dynamic model. Reservations retain exposure until authoritative reports resolve it. Hidden-liquidity estimates should be judged by their incremental value for that decision, without equating execution value with price predictability or latent coordinates with measured investor inventory.

Nasdaq and NYSE auctions require distinct price and allocation representations. The same public demand/supply totals need not imply the same child-order outcome. DMM response, order classes, counterpart restrictions, deadlines and residual opportunities enter the control problem. The worked examples establish their arithmetic under stated assumptions; the quadratic endpoint model provides an explicit scheduling benchmark with full fills.

The tagged order-book construction makes the dynamic execution kernel an output of primitive flow and clearing. It preserves private fills, priority and the cash ledger; its stationary linear-response limit gives a conditional propagator representation. Primitive event-rate error then reaches the execution objective through a common path-coupling bound, alongside the separate impact-cost error. The resulting certificate does not require an endogenous book to have the signature model's exogenous quadratic objective.

Within a finite controlled book, complete observable histories and exact admissibility masks support a vanishing signature policy-class gap. Combined with uniform model errors and certified global optimization tolerances, this gives conditional consistency. The explicit finite-state softmax experiment separates approximation error from persistent model misspecification. A reference objective has a transferable guarantee only when its intervention differences and objective-reduction error are accounted for; an exogenous quadratic identity requires its additional feature-law and rate-cost assumptions.

For a finite-observation binary hidden model, lower and upper backward recursions turn that conditional programme into a computable global certificate. If the deployed controller discards reports, conditional action-law envelopes give a separate lower bound for the richer-history optimum. The endpoint construction and exact finite-tree examples show how to certify this comparison, while the singleton case identifies a sufficient summary. Coarser information can have a positive cost even with an exact filter; increasing a belief grid does not remove that loss. Conversely, an action-separation certificate can establish an optimal instruction even when the summary does not determine the optimal conditional value. Obtaining tight, validated envelopes from realistic reports remains a separate task.

The additional finite-sample construction makes one route to uncertainty coverage explicit: independent observations in every declared state-action row give simultaneous reward and transition bounds, which transfer through the Bellman recursion to the implemented policy. Exact reservation and integer-allocation results address feasibility; the completion viability recursion identifies when a guaranteed finish is possible at all. These results include their observation, support and boundedness assumptions.

The contribution is a theoretical framework linking impact assumptions to feasible execution decisions. Its derivations state the conditions under which schedules, routing rules, information bounds and auction allocations follow. The numerical comparison illustrates learning within one specified liquidity model. Empirical calibration and policy evaluation are separate extensions; they are not needed for the conditional mathematical statements proved here.

A A continuous-time characterization

The discrete controller has a continuous-time analogue, with the marked generator in (29.5) retaining joint fills and price changes. The following algebra also permits additional continuous dynamics; its characterization remains subject to the regularity conditions below. Consider a translation-invariant model whose expected future dollar outlay is $U(t, q, m, z) = qm + v(t, q, z)$. The state z includes observable market features and a sufficient liquidity belief. Let the reference midpoint have local drift $\alpha(z)$, and let $\mathcal{L}^a$ be the between-event generator for z under passive action a . It includes any belief drift generated by observing no execution.

A passive event has intensity $\lambda_a(z)$ and mark (f, g, j, z') , where f shares execute at premium g per share over the pre-event midpoint, the midpoint changes by j , and the other state changes to z' . After subtracting qm , the event's incremental continuation expression is

$$fg + (q - f)j + v(t, q - f, z') - v(t, q, z). \quad (\text{A.1})$$

This follows by substituting $U = qm + v$ and the recorded cash payment $f(m + g)$ into the cost-plus-continuation generator. A primitive event can have $f = 0$ and still change the price or another state. If such events are included in the marked sum, they must be excluded from the separately specified between-event generator $\mathcal{L}^a$. The price-jump term is multiplied by the residual, not by the initial parent quantity. Conditioning only on f while discarding its dependence with j loses the adverse-selection mechanism.

For simplicity, suppose an aggressive impulse of size x has premium cost $C_M(x, z)$, changes z to z^M , and has no separate instantaneous midpoint jump outside that convention. Define

$$\mathcal{M}v(t, q, z) = \inf_{0 < x \leq q} \{C_M(x, z) + v(t, q - x, z^M)\}. \quad (\text{A.2})$$

If an impulse separately moves the midpoint by j_M , the expression must also include $(q - x)j_M$, consistently with (A.1); it should not also be hidden in another cost component.

A formal dynamic-programming characterization is

$$\begin{aligned} \min \left\{ \mathcal{M}v - v, \quad \partial_t v + \inf_{a \in \mathcal{A}(q, z)} \left[\mathcal{L}^a v + \alpha(z)q + \lambda_R \sigma^2(z)q^2 + c_a(z) \right. \right. \\ \left. \left. + \lambda_a(z) \mathbb{E}_a \{ fg + (q - f)j + v(t, q - f, z') - v(t, q, z) \} \right] \right\} = 0. \end{aligned} \quad (\text{A.3})$$

Here λ_R is the inventory-risk weight and c_a a running decision cost. This expression is a characterization under suitable regularity and admissibility assumptions, not an existence theorem. Impulse costs or a frequency restriction are needed to exclude infinitely many interventions in finite time. Auctions impose separate boundary/transition conditions; they are not encoded by a smooth event intensity near the closing time.

The familiar limit/market-order control structure is developed in the execution literature (Cartea and Jaimungal, 2015). Equation (A.3) states the joint event term explicitly so that a fill premium and a correlated midpoint jump are not accidentally counted as independent advantages.

B Portfolio execution and a consistent impact kernel

For d assets let $\mathbf{q}_k \in \mathbb{R}^d$ denote signed residual positions and $\mathbf{x}_k = \mathbf{q}_k - \mathbf{q}_{k+1}$ the trade vector. A quadratic portfolio program is

$$\min_{\mathbf{x}} \frac{1}{2} \sum_{i,j} \mathbf{x}_i^\top G_{ij} \mathbf{x}_j + \lambda \sum_k \mathbf{q}_{k+1}^\top \Sigma_k \mathbf{q}_{k+1} \Delta t_k + \sum_k \boldsymbol{\alpha}_k^\top \mathbf{q}_{k+1} \Delta t_k, \quad (\text{B.1})$$

subject to inventory, participation and admissibility constraints. Correlation matters because independently minimizing each asset's exposure can destroy a hedge. The off-diagonal blocks of G represent cross-impact and must be estimated and constrained consistently. Dark-pool portfolio execution has related stochastic-control structure (Kratz and Schöneborn, 2015); it does not follow by independently applying a one-asset policy to every security.

Proposition B.1 (A positive quadratic impact construction). *Let Γ be a symmetric positive semidefinite $d \times d$ matrix and $0 \leq \rho \leq 1$. Set $G_{ij} = \rho^{|i-j|}\Gamma$ on a finite time grid. Then the quadratic impact term in (B.1) is nonnegative for every signed trade sequence, including every round trip.*

Proof. For $0 \leq \rho < 1$, the matrix $R_{ij} = \rho^{|i-j|}$ is the covariance matrix of a stationary, variance-one AR(1) process and is positive semidefinite. For $\rho = 1$, R is the all-ones matrix and is also positive semidefinite. The full block matrix is $R \otimes \Gamma$, whose eigenvalues are products of the nonnegative eigenvalues of its factors. Its quadratic form is therefore nonnegative. $\square$

Remark B.2 (Scope of the separable construction). For $0 < \rho < 1$, the temporal matrix is an exponential kernel sampled on a regular grid. Its PSD property is sufficient for nonnegative round-trip cost; necessity concerns only the zero-mass subspace. The continuous-time exponential schedule in Theorem 7.2 is a useful limiting benchmark, not the exact finite-grid optimum. For $N \geq 2$ grid points with endpoint trades, unit scalar impact coefficient and no other costs, the exact normalized weights are

$$w_0 = w_{N-1} = \frac{1}{N - (N-2)\rho}, \quad w_i = \frac{1 - \rho}{N - (N-2)\rho} \quad (0 < i < N-1).$$

They sum to one and give a constant vector Rw ; the cost is $Q^2(1 + \rho)/[2\{N - (N-2)\rho\}]$. As the grid becomes fine at fixed horizon, $\rho = e^{-\varrho\Delta t}$ yields the continuous endpoint-block formula. At $\rho = 0$ the finite-grid matrix is the identity; it is a limiting exponential case, not the exponential of a finite decay parameter. At $\rho = 1$ every fixed-mass allocation has the same cost.

The symmetry of Γ is part of this model. It does not establish that antisymmetric causal cross-impact is cost-free: Proposition 10.1 provides a counterexample. A numerical portfolio benchmark must retain the same discretization, coefficient normalization and constraint set.

This proposition concerns the declared impact component only. It does not prove that a complete market with rebates, other signals, constraints or additional transient effects admits no arbitrage. It does provide a simple way to avoid introducing a negative-cost trading loop solely through an unconstrained quadratic impact fit. More general no-dynamic-arbitrage conditions are the subject of Part II; see also Gatheral (2010) and Schneider and Lillo (2016).

C A bound for transition-model error

The belief bound in Theorem 31.2 holds the transition mechanism fixed. A uniform one-step comparison of the complete transition law is a special case of Theorem 39.1.

Corollary C.1 (Finite-horizon simulation bound). *Consider two finite-horizon models with the same state, observation, action and cost-outcome spaces, the same admissible policies, the same initial state/history distribution, and N stages, terminal completion included. Suppose each stage loss lies in $[0, c]$ and, uniformly over admissible histories and actions, the conditional joint laws of the next observation, next state and stage-loss outcome differ by at most ε in total variation. Then every fixed policy satisfies*

$$|J_\pi - \hat{J}_\pi| \leq \frac{1}{2}c\varepsilon N(N+1),$$

and a policy optimal in the approximate model has true regret at most $\min\{Nc, c\varepsilon N(N+1)\}$.

Proof. Apply Theorem 39.1 to the reward $-(\text{stage loss})$, with $g = \hat{g} = 0$ since completion is a stage. A joint total-variation bound ε bounds both marginals: the next-state kernels differ by at most ε , and because the stage loss has range c its conditional means differ by at most $\eta_k = c\varepsilon$. After stage k there remain $N - k - 1$ stages, so every fitted continuation value has span at most $S_{k+1} = (N - k - 1)c$. Recursion (39.1) gives

$$B_0 = \sum_{k=0}^{N-1} [c\varepsilon + \varepsilon(N - k - 1)c] = c\varepsilon \sum_{j=1}^N j = \frac{1}{2}c\varepsilon N(N+1).$$

An approximate optimizer has $\zeta \equiv 0$, so its regret is at most $2B_0$. Total loss lies in $[0, Nc]$, which gives the trivial bound. $\square$

If initial state/history laws instead differ by δ_0 in total variation, the fixed-policy bound becomes

$$|J_\pi - \hat{J}_\pi| \leq Nc\delta_0 + \frac{1}{2}c\varepsilon N(N+1),$$

and the regret bound becomes $\min\{Nc, 2Nc\delta_0 + c\varepsilon N(N+1)\}$. Indeed, the conditional total cost of a fixed policy lies in $[0, Nc]$, so changing only its initial law changes its expectation by at most $Nc\delta_0$; the previous comparison handles the kernels. The shared initial law is essential for the original fixed-policy statement: with one stage, observed state $z \in \{0, 1\}$, identical kernels and loss z , initial probabilities $\mathbb{P}(z = 1) = 1/4$ and $3/4$ give costs differing by $1/2$ even when $\varepsilon = 0$.

The uniform condition includes histories that a new policy may reach but the logging policy rarely visited. Fitting one-step likelihood well on the logging distribution does not establish it. Nor does a small displayed-book forecast error imply a small total-variation error for the joint execution kernel. With negative net charges, a common finite stage shift can place costs in a bounded interval without changing policy comparisons, provided the horizon is fixed. An unbounded terminal penalty again prevents a finite bound of this form.

D Reproduction and scope of the calculations

D.1 Reproduction instructions and scope

Reference implementation. The Python reference implementation checks exact reservation invariants, supported discrete control, compiler action gaps, robust joint-moment decisions and sequential noise bounds in synthetic cells, using independent enumeration, linear programming and Bellman evaluation where applicable. It is a normalized research model, not an ITCH/OUCH decoder or an exchange-certified gateway, and none of its tables is a market backtest or a hardware benchmark.

The source archive contains the LaTeX manuscript with an embedded author–year bibliography, generated PNG figures, Python scripts, parameters, scenario arrays, machine-readable results, table inputs, and the Lean 4 sources of [Appendix E](#) under `lean/`. From the project root, install the pinned dependencies and run

```
python -m pip install -r requirements.txt
python scripts/reproduce.py
python scripts/certified_improvement.py
python scripts/realism_stress.py
python deployment/verify_production.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

Table 21: Common abbreviations used in execution research.

Abbreviation	Meaning
ATS	Alternative trading system
BBO / NBBO	Best bid and offer / national best bid and offer
CRPS	Continuous ranked probability score
CVaR	Conditional value at risk
FIFO	First in, first out, within the applicable priority class
FOK / IOC	Fill-or-kill / immediate-or-cancel
IS	Implementation shortfall
MBP / MBO	Market by price / market by order
MOO / MOC	Market-on-open / market-on-close
LOO / LOC	Limit-on-open / limit-on-close
NOII	Net Order Imbalance Indicator
POMDP	Partially observable Markov decision process
TWAP / VWAP	Time-weighted / volume-weighted average price

The reproduction command solves both belief grids, simulates all six policies, regenerates the figures and numerical inputs, solves the completion-feature quadratic programmes and runs the mathematical and accounting checks. The cross-impact checks compare exact cell integrals with independent state-ODE integration for the energy–area and lift-endpoint identities. The three-point defect checks minimize the original quadratic form over the total-variation polygon in exact rational arithmetic and verify the Lipschitz and phantom-profit constants at explicit attaining examples. The tagged-book script constructs and solves the finite controlled process, compares its transition matrices with uniformization and event simulation, and checks the primitive-rate and stationary-response calculations. The random-number construction, action tie-breaking and nearest-grid rule are explicit in the code. The reported stochastic estimates and table inputs are generated together by this implementation. The observation-summary script also enumerates all feasible histories in three finite controlled-liquidity cases, constructs posterior-fiber bounds, and checks the resulting conditional certificates against exact rational optima and an independent forward hidden-state evaluation. Its separate probe example checks both positive information loss and exact sufficiency. The same exact trees verify action bounds and strict screening against every feasible full-history state–action pair. A non-singleton example separates decision sufficiency from a zero-width value interval and checks both deterministic and randomized controller certificates.

The empirical content is deliberately synthetic. Its economic implications are conditional: persistent liquidity can make learning valuable; non-fill cost can outweigh price improvement; the cheapest initial fee need not give the cheapest allocation; and quantity reservations prevent overfills under their stated acknowledgment assumptions. The completion-feature experiment additionally tests a deterministic error certificate under a known law. These checks establish reproducibility of the stated examples and do not certify an exchange simulator or validate estimated market parameters.

The mathematical proofs are given in the text. The computational checks test selected identities, counterexamples and implementations.

D.2 Analytic schedule and immediate routing

Set $Q = T = \eta = 1$ in Proposition 7.1. Define the unweighted impact term $I = \int_0^1 v_t^2 dt$ and exposure $R = \int_0^1 q_t^2 dt$. Table 22 reports numerical quadrature on 2,001 equally spaced points. Each row uses

Table 22: Normalized analytic schedule calculations.

κ	Impact I	Exposure R	$I + \kappa^2 R$
0	1.0000	0.3333	1.0000
1	1.0185	0.2945	1.3130
3	1.5523	0.1625	3.0149

Table 23: Synthetic routing of 1,000 shares; continuous relaxation.

Venue	Shares	Cost (USD)	Marginal cost (cents/share)
A	120.00	1.7760	1.9600
B	339.46	5.5139	2.0486
C	540.54	9.3207	2.0486

a different objective weight κ^2 ; its last column should not be compared as if all rows solved the same objective. At $\kappa = 3$, the greater impact cost purchases a substantial reduction in unweighted exposure.

For routing, set $C_v(x) = a_v x + b_v x^2/2$ with

$$a = (0.010, 0.012, 0.014), \quad b = (0.000080, 0.000025, 0.000012), \quad c = (120, 550, 900). \quad (\text{D.1})$$

Here quantities are shares, a is dollars per share and b is dollars per share squared. The 1,000-share allocation in Table 23 costs \$16.6106 in this model. Venue A has the lowest initial cost but reaches its 120-share cap. Venues B and C equalize marginal cost at about 2.04865 cents per share; A's lower marginal cost at capacity is consistent with (27.2). Decimal allocations solve a continuous relaxation. Integer shares can be obtained by solving the corresponding discrete marginal-cost allocation; the decimals are not submitted orders.

D.3 A finite-horizon liquidity-learning problem

The dynamic example buys $Q = 20$ units, with one unit defined as 100 shares, over $K = 12$ waiting intervals followed by terminal completion. This unit is a numerical scale, not an assertion about a security's effective round lot. A hidden regime $H \in \{0, 1\}$ is drawn once per trajectory and remains fixed. Its initial probability is $\mathbb{P}(H = 1) = 1/2$. Prices carry no information about H ; only the controlled fill observations update its posterior.

At each step choose $m \in \{0, \dots, q\}$ immediate units, a passive size $\ell \in \{0, \min(1, q-m), \min(3, q-m), \min(5, q-m)\}$, and one of two passive channels when $\ell > 0$. Require $m + \ell \leq q$. All immediate units fill at charge

$$C_M(m) = m + 0.08m^2 \quad \text{USD}. \quad (\text{D.2})$$

Passive fills satisfy

$$F \mid H, v, \ell \sim \text{Binomial}(\ell, p_{v,H}), \quad (p_{v,H}) = \begin{pmatrix} 0.15 & 0.60 \\ 0.05 & 0.75 \end{pmatrix}. \quad (\text{D.3})$$

The rows are a stylized displayed passive channel and a midpoint channel; the columns are the less and more liquid hidden regimes. The displayed channel has net charge $-\$0.30$ per filled unit and the midpoint channel $\$0.05$. These hypothetical net charges are not an actual fee schedule. All

passive orders expire before the next decision, so there are no live orders across intervals in this example.

After the interval, $q' = q - m - F$ and the inventory penalty is $0.008(q')^2$ dollars. At the terminal time all remaining units execute at $C_M(q)$. The objective is

$$\mathbb{E} \left[\sum_{k=0}^{K-1} \{C_M(m_k) + c_{v_k} F_k + 0.008 q_{k+1}^2\} + C_M(q_K) \right]. \quad (\text{D.4})$$

The penalty is a specified risk preference. Independent zero-mean reference innovations can be integrated out of expected execution cost; they are not sampled in the reported Monte Carlo calculation. Consequently, the reported standard errors describe regime and fill randomness, not the distribution of full realized dollar shortfall under a sampled price process.

The unit fills in (D.3) are independent conditional on regime, size and venue. Across intervals they are dependent through the persistent regime. This abstraction omits queue position, impact resilience, price changes conditional on fills, and venue-specific order protocols. It is chosen to isolate the scheduling value of learning execution availability. The general framework permits those omitted states; this small experiment does not validate them.

D.4 The six policies

The *belief policy* solves the Bellman recursion on 101 equally spaced belief values. It updates the posterior from the exact binomial likelihood after both fills and non-fills. Continuation values are linearly interpolated in belief; simulation selects the action at the nearest belief grid point. This is an approximate solution of the finite-action partially observed problem.

The *frozen-prior* comparator is separately optimized under a model that keeps the regime mixture fixed at its initial value at each future step. Its Bellman recursion suppresses posterior updates. It therefore describes a misspecified non-learning model of persistent liquidity, rather than simply reusing the learning controller and withholding its state. It is evaluated on the same persistent-regime trajectories as every other policy.

The *regime oracle* observes H before trading and uses the corresponding endpoint-belief policy. It has more information but the same order-size and venue choices. It is an information benchmark, not an available trading strategy. *TWAP market* divides residual inventory across remaining opportunities, rounding the next quantity upward to integer units. *Market now* buys all 20 units at the first decision. *Midpoint first* posts up to five units at the midpoint at each step and completes the remainder at the terminal time.

D.5 Results and their interpretation

All six policies use the same 16,000 regime draws and per-step uniform random numbers. With NumPy generator seed 5257, the implementation draws the regime uniforms first and then a path-major 16000×12 array of fill uniforms. Binomial inverse distribution functions map those common uniforms to fills. Common random numbers support paired comparisons; their role is variance reduction, not a source of independent market evidence. Table 24 decomposes the objective into execution charges and the inventory penalty.

The belief policy has mean objective \$10.1493 versus \$12.6297 for the optimized frozen-prior model, a reduction of 19.6%. The paired difference is \$2.4803 with standard error \$0.0212; a normal Monte Carlo 95% interval is [\$2.439, \$2.522]. This interval describes numerical sampling under the specified model, not parameter-estimation error or market uncertainty.

Table 24: Synthetic policy comparison for a 2,000-share parent. Charges, penalties, objectives and Monte Carlo standard errors are in USD. Terminal quantity is the mean number of 100-share units before forced completion.

Policy	Charge	Penalty	Objective	MC SE	Terminal units
Belief policy	3.935	6.214	10.149	0.078	0.90
Frozen prior	3.847	8.783	12.630	0.096	2.08
Regime oracle	3.305	5.650	8.954	0.075	0.85
TWAP market	22.720	9.400	32.120	0.000	1.00
Market now	52.000	0.000	52.000	0.000	0.00
Midpoint first	20.792	18.512	39.304	0.273	8.52

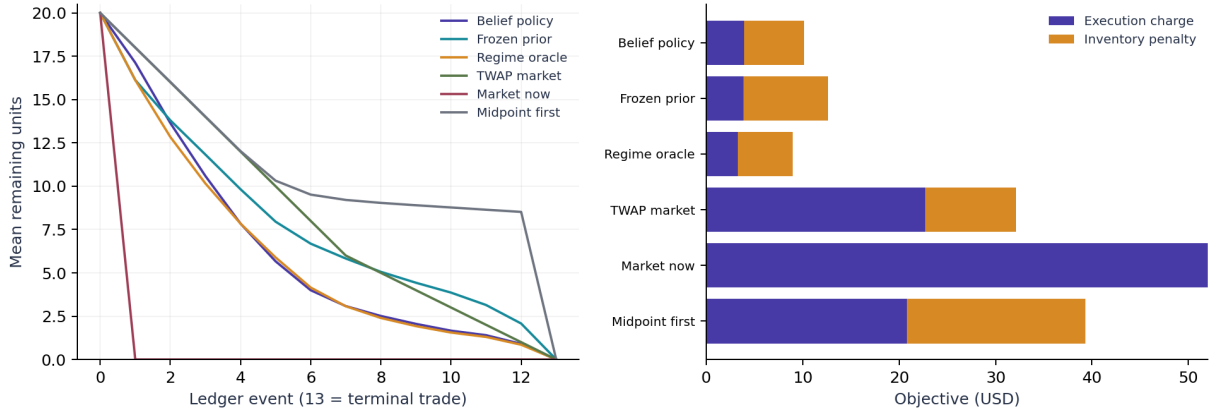


Figure 10: Synthetic execution paths and objective decomposition. The left panel plots mean residual quantity after successive ledger events; its final point is the terminal completion trade, not an extra waiting interval. The right panel separates execution charges from the chosen inventory penalty.

Execution charges alone are \$3.9349 for the belief policy and \$3.8470 for the frozen-prior comparator. The improvement therefore comes principally from reducing the inventory penalty, from \$8.7826 to \$6.2145, while reducing mean residual quantity before terminal completion. The oracle’s mean objective is \$8.9543; its advantage over the belief policy is \$1.1950 with paired standard error \$0.0076. This is a cost of incomplete regime information within the stipulated model.

Midpoint-first leaves substantial quantity for expensive terminal completion in the less liquid regime. Market-now avoids the inventory penalty but pays the largest quadratic immediate charge. Changing fill probabilities, impact coefficients or risk preferences can change these rankings.

Every simulated parent completes, because unlimited terminal market execution is a declared assumption. The simulation also enforces $m + \ell \leq q$. These checks establish quantity consistency in the toy model; they do not establish that a live exchange would provide terminal liquidity or that asynchronous cancellation is safe. Proposition 27.3 addresses the separate accounting condition for live orders.

D.6 Numerical checks and limits

The interpolated initial value is \$10.1091. Increasing the belief grid from 101 to 201 points changes that value by \$0.00032349. This is a targeted resolution check at the stated initial condition, not a uniform policy-error bound. The simulated mean differs from the coarser grid value by \$0.0402, i.e.

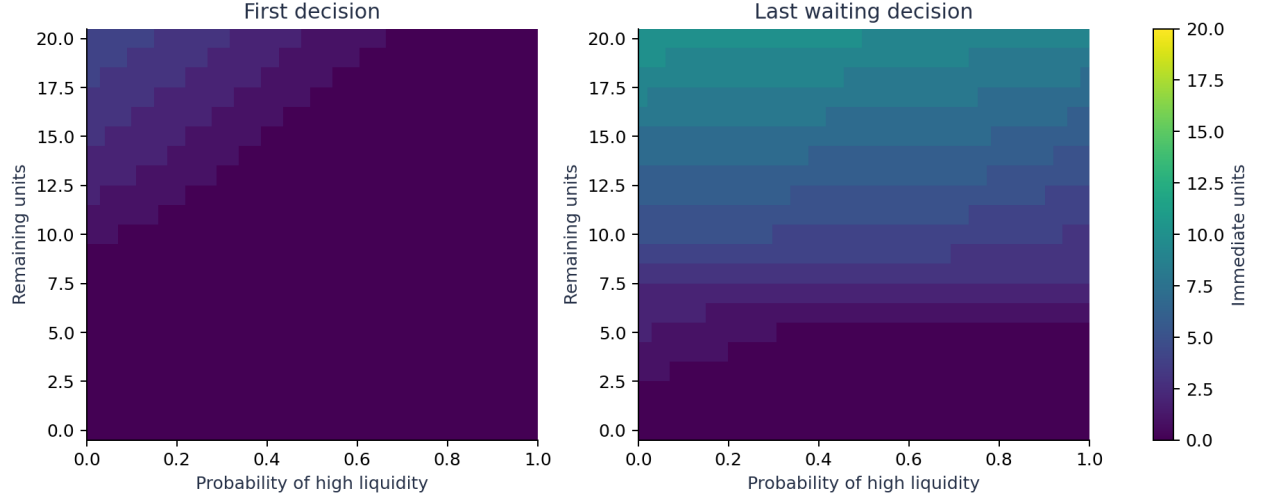


Figure 11: Immediate quantities selected by the approximate belief policy at the first and last decisions, over remaining inventory and belief. The state includes both urgency and execution availability. Color scales are shared.

0.51 of the belief policy’s reported Monte Carlo standard error of \$0.078 (Table 24); nearest-grid action selection and simulation are separate from interpolation in the Bellman recursion.

The source also checks the pathwise shortfall identity on 500 independently generated completed ledgers, the posterior-mean identity for binomial observations, the immediate-allocation quantity constraint, and the reserve transfer example. All figures and numerical tables are generated from the same Python script and JSON results. Code and parameter files permit the reader to change probabilities, capacities, impact, horizon or risk preference without substituting undocumented assumptions.

The tests deliberately do not claim superiority of LSTMs, transformers or reinforcement learning. A small partially observed example can be solved by dynamic programming and provides a useful benchmark for more flexible methods.

D.7 Testing the exact-completion certificate

The additional experiment tests Theorems 17.7 and 17.8 directly. Set $X = T = 1$, $\bar{V} = 2$, $\eta = 0.15$, and let U be uniform on $\{0, 0.2, 0.4, 0.6, 0.8, 1\}$. The exogenous path is $Y_t = (t, (t - U)_+)$, the true kernel is e^{-t} , and an observed linear execution charge is $a_t = -0.3(t - U)_+$. Use the features $\phi = (1, t, (t - U)_+)$ in (17.5) with $M = (1, 1/2, 1/2)$ and coefficient budget $r = 1$. Thus every tested policy completes exactly and stays in $[0, 2]$ for all $U \in [0, 1]$, not just at the sampled support points. An Euclidean coefficient bound is $R = 2$.

For each approximate decay rate in $\{0.9, 1, 1.1\}$, fit the objective using a multinomial sample of 100, 1000 or 10000 exogenous paths, with a separate fixed seed 525717. State-ODE integration computes the quadratic and linear coefficients independently of the signature shuffle implementation. The true finite law supplies the matrix and linear coefficient errors; the kernel error is bounded in the uniform norm and the optimizer reports its convex duality gap. Table 25 compares actual regret within this coefficient polytope with (17.7).

All nine cases satisfy the bound. The certificate can be conservative because it is uniform over the coefficient class and combines worst-case matrix and linear errors. Larger samples need not improve every realized estimate monotonically. No claim about an unknown market distribution

Table 25: Synthetic verification of the execution certificate. Errors are measured against the known finite law; the bounds are deterministic for these realized estimates, not statistical confidence intervals.

Decay rate	Paths	Actual class regret	Certificate
0.9	100	0.000019	0.069293
0.9	1,000	0.000004	0.043048
0.9	10,000	0.000002	0.038963
1.0	100	0.000008	0.006747
1.0	1,000	0.000002	0.003264
1.0	10,000	0.000000	0.000608
1.1	100	0.000017	0.042059
1.1	1,000	0.000004	0.040242
1.1	10,000	0.000002	0.035514

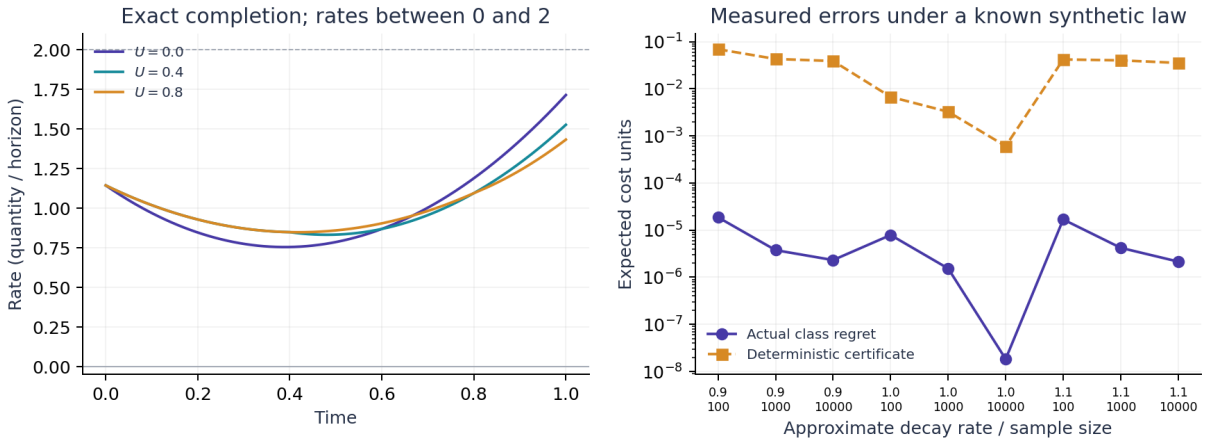


Figure 12: Left: feasible rates from the true-law optimum for three signal paths. Right: actual class regret and its deterministic certificate in the nine kernel-and-sampling scenarios. Exact completion follows analytically from the feature construction; the numerical checks also integrate every rate.

follows from measuring errors against this known synthetic law.

The algebra is checked separately on six kinked paths with a depth-one rate using time, the signal and both exponential coordinates. Integrating the required signature coordinates and an independent impact-state ODE gives costs agreeing within 10^{-10} . This verifies the numerical implementation of the exact finite-degree identity; the proof remains [Theorem 17.3](#). Further checks reproduce the finite-signature counterexample, the cutoff correction, the sharp binary information constant, and the cell-averaged Riesz energy.

D.8 A tagged book, a learned controller and a rate certificate

The final illustration implements the construction in [Section 29](#). A parent buys one share on a synthetic two-price book with bid 100 and ask 101. Up to two external bid units and one own unit can rest at the bid; one external ask unit can rest at the ask. External bid arrivals, ask arrivals and unit buys have rates 0.4, 0.6 and 0.5 per second. Each existing external bid cancels at rate 0.15. Unit sells arrive at true rate $\lambda = 1.2$. Buy and sell requests are immediate-or-cancel against the opposite quote. At capacity an external limit arrival is rejected; if an opposite queue is empty an

Table 26: Synthetic order-flow misspecification. A policy is optimized at each estimated sell rate and then evaluated unchanged at the true rate 1.2. The true optimum costs 1 and crosses immediately. The bound is (D.5); it is conservative and becomes uninformative relative to the loss range for the larger rate errors.

$\hat{\lambda}$	Initial action	Model value	True value	Regret	Bound
1.0	Cross	1.000000	1.000000	0.000000	0.725077
1.2	Cross	1.000000	1.000000	0.000000	0.000000
1.4	Post	0.998043	1.025897	0.025897	0.725077
1.6	Post	0.966724	1.025897	0.025897	1.318720
2.0	Post	0.888503	1.079066	0.079066	2.000000
2.4	Post	0.782390	1.079066	0.079066	2.000000

immediate-or-cancel request expires. All matching uses the actual ordered queues, with executions at resting prices.

The nine possible bid queues are three unposted queues with zero, one or two external units, and six queues with one own unit inserted at every possible position among them. Combined with two ask occupancies and a stopped state after parent completion, this gives 19 states. The initial book has two external bid units and one ask unit, with no own order. There are four decisions at 0, 0.25, 0.5, 0.75 seconds and a deadline at $T = 1$. An action posts or retains the own bid, cancels it and waits, or cancels it and crosses an available ask. Cancellation is atomic in this example. A retained order keeps its priority. A separately stipulated terminal facility buys any remaining unit at 102, so completion is guaranteed even when the local ask is empty. This facility is an explicit model assumption, not inferred liquidity in the book.

Costs are measured relative to 100: a passive fill costs 0, an aggressive fill costs 1, and terminal completion costs 2. No additional message, risk or impact charge is imposed. Consequently every complete path loss lies in $[0, 2]$. Posting is free, so the cancel-and-wait action is weakly dominated here; the nontrivial decision is how long to retain a queue position before crossing. This specialization isolates priority, completion risk and primitive-rate misspecification.

The script builds the generator by pushing six raw mark types through the matching routine, computes $e^{h\mathcal{L}}$, and solves the finite Bellman recursion. If only the sell rate is changed to $\hat{\lambda}$, (29.11) gives $d = |\hat{\lambda} - \lambda|$: both the sell-mark mass and the compensating dummy mass must be included. Thus the policy-regret bound is

$$\min\{2, 4(1 - e^{-|\hat{\lambda} - 1.2|})\}. \quad (\text{D.5})$$

Here $D_{\text{cost}} = 0$, $L = 2$, the policy class is the full finite feedback class, and exact backward induction has $\tau = 0$ mathematically. Matrix calculations are implemented in floating-point arithmetic. The displayed certificate tests response-law error; it does not test a nonzero additional impact-kernel error.

At estimated rate 1.4, the selected policy posts initially. Its model value is 0.998043, whereas its true value is 1.025897: the realized expected regret is 0.025897, below the rate certificate 0.725077. A separate event-time simulation draws exponential waiting times and individual raw marks without using the transition matrix. Using 40,000 paths in each environment gives 0.994425 with standard error 0.002118 in the estimated environment, and 1.025125 with standard error 0.001977 in the true environment. Both estimates are within 1.71 standard errors of their respective matrix-based values. Seeds are 4304211 and 4304212.

Verification covers 3,888 small variable-size matching cases, share conservation, price priority, an explicit FIFO check, and the equal-depth counterexample with different own fills. A Poisson-series calculation agrees with the independently computed matrix exponential to within 10^{-13} in maximum

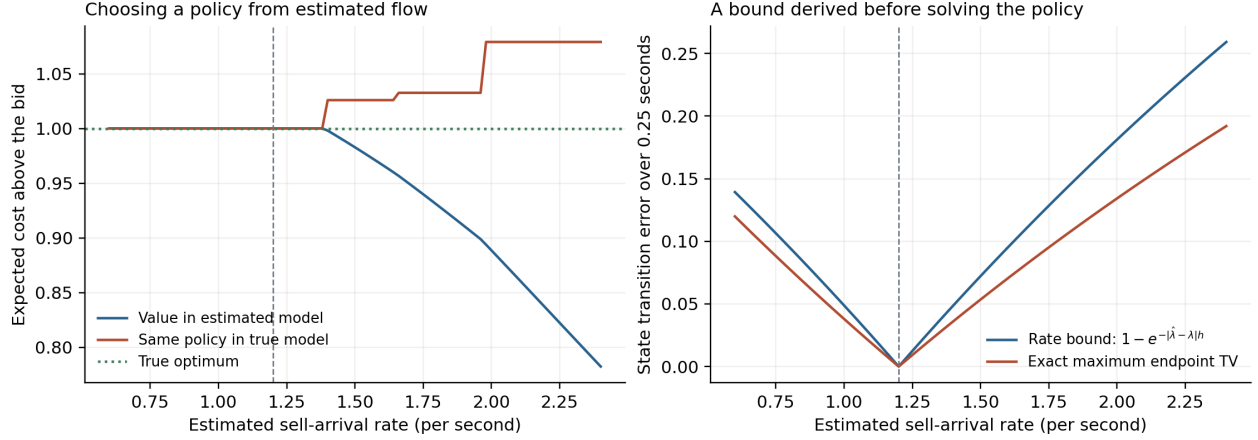


Figure 13: The order-flow construction and its consequences. Left: the value of each learned policy in its fitted environment and its cost when deployed unchanged in the true synthetic model. Right: exact maximum state-endpoint total variation over 0.25 seconds and its primitive-rate bound. The right panel measures the state marginal; the theorem also bounds the richer joint path of states, fills and reports. The dashed vertical line marks the true sell rate.

row total variation, with its omitted tail recorded separately. The fixed-policy and regret bounds are checked at every state and remaining decision time for seven estimated rates, giving 532 state-stage-model comparisons. Separate matrix calculations verify the stationary response remainder for constant and piecewise signed inputs and the exact two-state exponential response. The proofs establish the general statements; these checks establish the stated numerical implementation and expose its finite state, atomic-cancellation and guaranteed-terminal-liquidity assumptions.

D.9 Feasible signature feedback with a computed optimization bound

A further experiment implements [Proposition 29.12](#) in the same 19-state book. The state is fully observed in this specialization. The report alphabet may therefore include the post-event state, and a based state-indicator channel has first-level signature $e_{S_{t_k}}$ at each decision. The stage-specific logit coefficients assign score r to the Bellman-optimal action in each observed state and score zero to the other admissible actions. Crossing an empty ask receives exactly zero probability through the mask; after completion only a no-operation action is retained. This is an implemented signature-softmax policy, rather than an unexecuted proposal for a policy class.

For $K = 4$, $L = 2$ and at most three actions, [Proposition 29.12](#) gives the global approximation bound

$$\tau(r) = \min\{2, 16/(e^r + 2)\}. \quad (\text{D.6})$$

Here the Bellman oracle solves the fully observed finite model. Depth-one state logits can approximate every deterministic Markov optimizer as $r \rightarrow \infty$, so this class has the same infimum as the full feedback class. The computed objective gap is checked against (D.6) at every state and remaining decision time. No local training residual is used as a global optimality certificate.

Table 27 gives two sets of coefficients: one from the true sell rate 1.2, and one from the estimated rate 1.4. Both policies are evaluated at the true rate 1.2. The latter satisfy the combined bound

$$\min\{2, 4(1 - e^{-0.2}) + \tau(r)\}.$$

Increasing r removes the softmax approximation error. Under the correct model, regret tends to zero;

Table 27: Depth-one signature feedback in the fully observed synthetic book. Regrets use the true rate 1.2 and initial state of [Appendix D.8](#). The two coefficient sets are constructed by finite backward induction at rates 1.2 and 1.4. Bounds hold before rounding.

r	Correct-model regret	$\tau(r)$	Misspecified regret	Combined bound
0	0.223352	2.000000	0.223352	2.000000
2	0.016095	1.704112	0.065887	2.000000
4	0.001220	0.282695	0.031569	1.007772
8	0.000019	0.005364	0.026002	0.730441
12	3.49×10^{-7}	0.000098	0.025899	0.725175
20	1.17×10^{-10}	3.30×10^{-8}	0.025897	0.725077

under rate misspecification it approaches the previous deterministic-policy regret of approximately 0.025897. Model error and policy approximation have distinct effects even in this small example.

The verification script reconstructs 31 finite transcripts, distinguishes initial observations and co-timestamped report order, and checks 1,260 flooring, finite-score and exact-mask cases. It evaluates 14 signature policies, checks 1,064 state-stage optimization comparisons and 1,064 consecutive-hybrid comparisons under the laws of earlier replacement policies. It also verifies the frozen-mode counterexample and 256 finite-policy comparisons with an autonomous reference model. The latter checks concern the terminal state marginal; the full-path estimate is established by the coupling proof. All these are finite synthetic checks, not a numerical proof of the general Lusin approximation, a generic depth complexity result, or an empirical execution study.

E Three machine-checked lemmas

Three results of this monograph are formalised in Lean 4 against Mathlib and checked by the Lean kernel. The artifact is the `lean/` directory of the source archive, pinned to Lean 4.19.0 and Mathlib v4.19.0. The reported axioms for each theorem are contained in the standard set `{propext, Classical.choice, Quot.sound}`; the development contains no `sorry`. The build and axiom logs are supplied with `lean/COMPILE.md`.

1. [Theorem 13.1 on a finite grid](#) (`loop_cost_eq_area_pairing`). For a closed inventory loop sampled on a finite grid under linear permanent impact $M = S + A$ with the trapezoidal cost convention, the arrival benchmark and the symmetric part telescope to zero and the total cost equals the trace pairing of A with the discrete Lévy area, $\text{tr}(A\mathcal{A}(\gamma))$. [Corollary 13.2](#) follows in the formalisation as the case $d = 1$.
2. [Corollary 12.8, ray form](#) (`ray_sign_conditions`). Along a fixed zero-mass direction, with the remainder of the graded expansion bounded at order ϵ^6 , nonnegativity of the round-trip cost forces $D^2\mathcal{C}[0][\nu, \nu] \geq 0$, and forces $D^4\mathcal{C}[0][\nu, \nu, \nu, \nu] \geq 0$ on the null set of the quadratic layer.
3. [Theorem 27.1](#) (`value_convex`). With venue costs convex on their capacity intervals and box capacity constraints, the aggregate value function is convex on feasible parent quantities.

The first item covers a finite grid with the prescribed trapezoidal convention; the continuum statement of [Theorem 13.1](#) is proved in the text and is not formalised. The formal pairing is the trace $\text{tr}(A\mathcal{A}) = \sum_{j,k} A_{jk}\mathcal{A}_{kj}$, which by antisymmetry of both factors equals $-2\sum_{i<j} A_{ij}\mathcal{A}_{ij}$, the form in [\(13.2\)](#); the two statements agree with no change of orientation. The second item isolates the algebraic step of the proof of [Corollary 12.8](#): the scalar remainder bound is assumed, and its derivation from [Assumption 12.1](#) is not formalised. The third assumes a finite greatest lower bound at feasible quantities and nonempty endpoint allocation sets, proves feasibility of their convex

combinations, and imposes no conditions at infeasible quantities. A zero-cost example verifies consistency of the greatest-lower-bound hypothesis. Attainment and the KKT conditions are not formalised.

The general certificates of [Sections 17](#) and [29](#), including the partial-observation and action-comparison results, are not formalised. The pinned sources expose the precise assumptions and axiom dependencies of the three stated formalizations.

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